



PUBLIC

2023-05-05

Solution Guide for SAP Service Cloud Version 2

Content

1	Solution Guide for SAP Service Cloud Version 2.	9
2	Minimum System and Software Requirements	10
2.1	Web Applications.	10
	Minimum Hardware Requirements.	10
	Minimum Network Requirements.	11
	Display Resolution.	11
	Additional Software.	11
	Browser Settings.	11
	Browser Versions.	12
	Domain Settings.	12
	Desktops and Laptops.	13
2.2	Language Settings.	13
2.3	Integration and Add-Ins.	14
3	Help and Support.	15
4	Global Search.	17
4.1	Configure Global Search.	18
4.2	Basic Search.	18
4.3	Syntax Search.	19
4.4	Advanced Search for One Entity.	20
4.5	Advanced Search for Multiple Entities.	20
4.6	Edit Saved Searches.	20
4.7	Delete Search.	21
5	User Interface.	22
5.1	Shell Bar.	22
	Favorites.	23
	Notifications.	24
	User Menu.	24
5.2	Home Page.	30
	Use Cards.	30
	Add Background Image.	32
6	Company.	34
6.1	Org Structure.	34
	Create Your Org Structure.	34

	Create an Org Unit.	35
	Add Employees and Managers to Org Units.	38
6.2	Employees.	38
	Create Employees.	39
	Manage Attachments.	40
	Working Hours.	41
	Create Sales Responsibility.	42
6.3	Distribution Channel.	43
	Create Distribution Channels.	43
6.4	Division.	43
	Create Divisions.	43
7	General.	45
7.1	Language and Region.	45
7.2	Change User Interface Texts for Applications.	46
7.3	Company Settings.	47
	Auto Sign Out.	47
7.4	Branding.	47
	Set Company Logo.	48
	Set Background Screen.	48
	Set Login Image.	49
7.5	Adaptation.	49
	Customize Your Solution.	49
	Manage Fields.	50
	Add Extension Fields.	50
	Add and Manage Sections.	51
	Add Mashups to Custom Sections.	52
	Create Custom Buttons.	52
	Create and Manage Custom Tabs.	53
7.6	Extensibility.	54
	Create Extension Fields.	54
	Create Validations.	55
	Create Determinations.	56
7.7	Extensibility Using SAP Build Apps.	58
8	Users and Control.	60
8.1	Users.	60
	Business User Details.	61
	Create Business Users and Assign a Business Role.	61
8.2	Business Roles.	62
	Business Role Details.	63
	Create Business Roles and Assign a Business User.	63

8.3	Security Policies.	64
	Security Policy Features.	65
	Create Security Policies.	65
	Edit Security Policy.	66
8.4	Identity Provider Configuration.	67
	Configure Identity Provider.	67
9	Accounts and Individual Customers.	69
9.1	Configure Accounts.	69
9.2	Create Accounts.	71
9.3	Assign Status.	72
9.4	Assign Industry.	73
9.5	Create or Add Contacts.	74
9.6	Add Accounts to Account Team.	75
9.7	Create Addresses.	75
9.8	Maintain Attachments.	76
9.9	Create Sales Data.	77
9.10	View Customer Insights.	77
10	Contacts.	80
10.1	Create Contacts.	80
10.2	Maintain Attachments.	81
	Upload Attachments From Your Computer.	81
	Add Attachments From SAP System.	81
11	Agent Desktop.	83
11.1	Configure Agent Desktop.	83
11.2	Customer Search.	90
11.3	Communication Channels.	90
	Configure Communication Provider.	90
	Phone.	91
	Chat.	91
	SMS.	92
	Email.	92
11.4	Customer Hub.	96
	Timeline.	96
	Objects.	96
11.5	Knowledge Base.	96
	Configure Knowledge Base Mashup Integration.	97
	Configure Knowledge Base Provider.	98
11.6	Session Information.	98
11.7	Widget Integration.	99

	Phone.	100
	Chat.	103
	SMS.	104
11.8	Integrate External Systems into Agent Desktop with Mashups.	106
	Configure SAP Event Mesh.	107
	Configure SAP S/4HANA Cloud.	109
	Configure SAP S/4HANA.	110
	Configure SAP Cloud Integration.	111
	Configure SAP Service Cloud.	114
	Configure Service Agent Desktop.	115
	Configure Mashup: Create Scenario.	115
	Configure Mashup: Display Scenario.	117
	Adapt Agent Desktop for Mashup.	118
12	Cases.	124
12.1	Configure Cases.	125
12.2	Create Cases.	135
12.3	Manage Cases.	136
	Case Worklist.	137
	Use Case Types.	137
	Work with Case Status.	139
	Analyze Case Data with Qualtrics XM Discover.	142
13	Service Categories.	144
13.1	Create Service Catalogs.	144
14	Registered Products.	146
14.1	Configure Registered Products.	147
14.2	Create Registered Products.	147
14.3	Use Registered Products.	147
14.4	Search Registered Products.	148
	Basic Search in Registered Products.	148
	Enterprise Search for Registered Products.	149
15	Installation Points.	150
15.1	Create Installation Points.	150
15.2	Use Installation Points.	151
15.3	Search Installation Points.	151
16	Installed Base.	153
16.1	Configure Installed Bases.	154
16.2	Use Installed Base.	154
17	Warranty.	155

18	Utilities.	156
18.1	Contract Account.	156
	Contract Account Workspace.	157
18.2	Premise.	161
	Premise Workspace.	161
	Move-In Process.	165
18.3	Agent Desktop.	165
19	Intelligent Services.	167
19.1	Case Categorization.	167
19.2	Similar Case Recommendation.	171
19.3	NLP Classification.	173
19.4	Machine Translation.	175
19.5	Case Summarization.	178
19.6	Case Topic Analyzer.	180
19.7	Profanity Check.	182
19.8	Intelligent Services FAQ.	183
20	Mashups.	185
20.1	Create HTML Mashups.	186
20.2	Copy Existing Mashups.	187
20.3	Delete or Deactivate Mashups.	187
20.4	Add Mashups as Tabs.	188
21	Notifications.	189
22	Approval.	190
22.1	Configure Approvals.	190
23	Autoflow.	193
23.1	Configure Autoflows.	193
	Configure Conditions.	193
	Configure Actions.	195
23.2	Monitor Autoflow Logs.	199
24	Products.	200
24.1	Configure Products.	200
24.2	Create Products.	202
24.3	Manage Products.	203
	Products Worklist.	203
	Adaptation of Products View.	205
25	Output Management.	207
25.1	Maintain Main Templates.	207

25.2	Maintain Form Templates.	209
25.3	Select Form Templates.	213
26	Data Import and Export.	214
26.1	Download Template.	215
26.2	Prepare Data.	215
26.3	Import.	217
26.4	Export.	218
26.5	Monitor.	218
26.6	Available Entities.	219
26.7	Entity-Specific Documentation.	219
	Account.	219
	Contact Person.	228
	Employee.	232
	Organizational Unit.	236
	Individual Customer.	242
	Product.	249
27	Data Administration.	253
27.1	Manage Attachment Types.	254
27.2	Assign Attachment Types.	254
27.3	Change History.	255
27.4	Define MIME Types for Attachments.	255
	Predelivered MIME Types for Attachments.	256
27.5	Attachment Configuration.	260
28	Analytics.	261
28.1	Configure Single Sign-On for SAP Analytics Cloud.	262
28.2	Design Stories.	264
28.3	View Stories.	266
28.4	SAP Analytics Cloud FAQ.	266
29	Microsoft Teams Integration.	268
29.1	Configure Microsoft Teams Integration.	268
29.2	Use Microsoft Teams Integration.	269
	Create Appointments.	269
	Share Workspaces.	270
	Make Outbound Calls.	272
29.3	Microsoft Teams Integration FAQ.	272
30	Surveys.	273
30.1	Configure Surveys.	274
	Configure Survey Categories.	274

	Configure Question and Answer Categories.	275
30.2	Create Surveys.	275
	Edit Surveys.	276
	Add Questions to Question Bank.	276
	Derive Calculations for Numeric Questions.	277
	Add Scores to Questions.	278
	Survey Score Calculations.	279
	Add Logic.	283
	Add Exceptions.	285
	Add Translations to Surveys.	286
30.3	Answer Surveys.	287
30.4	Survey Reporting and Analysis.	287
31	Library.	289
31.1	Configure Library.	289
31.2	Organize Files and Folders.	291
31.3	Manage Access to Files and Folders.	296
31.4	Share Links to Files and Folders.	297
32	Integration.	298
32.1	Configure User Roles for Integration.	298
32.2	Monitor messages.	299
32.3	Event Monitoring.	300
32.4	Communication Systems.	300
32.5	Configure Communication.	301
32.6	View Inbound Configuration.	302
32.7	View Outbound Configuration.	302
32.8	Create Value Mapping.	303

1 Solution Guide for SAP Service Cloud Version 2

This guide aims to inform you about the capabilities of SAP Service Cloud Version 2, help you set up your solution, and accomplish your daily tasks.

With SAP Service Cloud Version 2, your service agents have customer information at their fingertips. Using the collaboration tools and knowledge base, you can quickly address customer issues.

Create new cases and update existing cases automatically. With native integration with Microsoft Teams collaboration, you can share content from within agent teams and stay up to date on the latest service requests, comments, discussions, and decisions online. SAP Service Cloud Version 2 automatically assigns tasks to a case based on relevant attributes to help guide agents through complex processes. Create workflow rules that generate notifications, update fields, and trigger requests for multilevel approvals based on context and time.

Create serialized registered products with customer, location, and warranty information, and add to a case, a maintenance plan, or replicate equipment from SAP ERP as registered products.

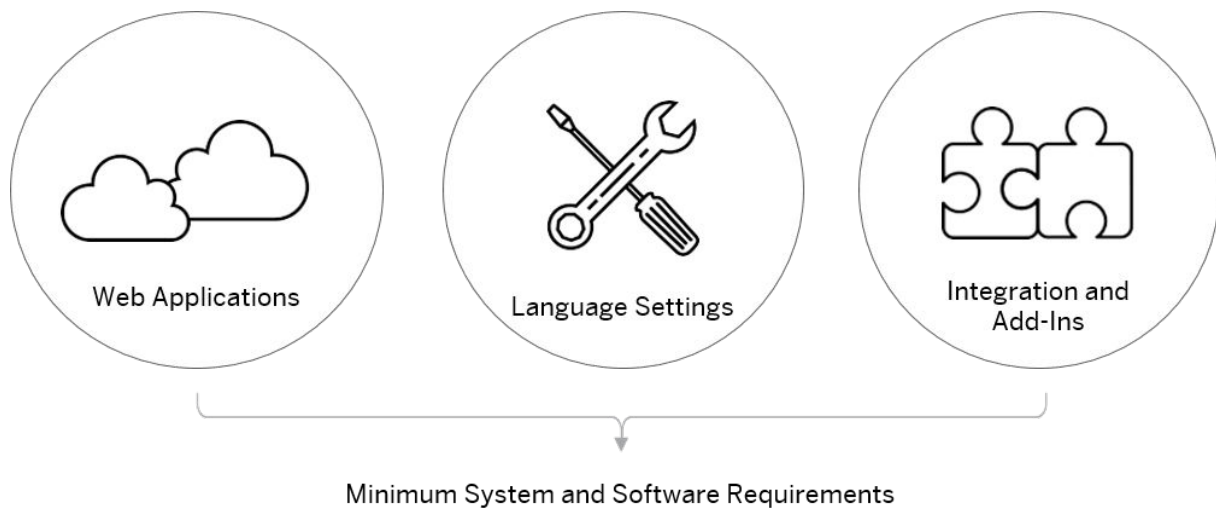
For more information about initial setup of your system, setting up the solution on mobile devices and customizing the solution to your needs, see the appropriate guide on the SAP Help portal.

Related Information

[SAP Service Cloud Version 2 on the SAP Help Portal](#)

2 Minimum System and Software Requirements

This section provides an overview of minimum system and software requirements.



2.1 Web Applications

You will learn about the minimum hardware and network requirements, display resolution, additional software, browser settings and versions and domain settings for web applications.

2.1.1 Minimum Hardware Requirements

Minimum desktop and laptop hardware requirements

- Processor: Intel Core 2 Duo (2.4 GHz with a 1066 megahertz {MHz} front-side bus) or equivalent
- Memory: 6 gigabytes (GB)

2.1.2 Minimum Network Requirements

Minimum bandwidth and latency requirements for

Number of Users		20	100	500	1000
Bandwidth Required (Mbps)	Sales	0.25	1.24	6.2	12.4
	Service	0.6	3	15	30

Network latency should be 200ms.

i Note

The calculations consider only bandwidth used by the solution and don't include any other internet or internal network traffic.

SAP Support may request wired networking during investigation of incidents.

For detailed calculations and more information about bandwidth requirements, refer to the [Network Sizing](#) blog.

2.1.3 Display Resolution

The solution is best displayed at a screen resolution of 1920 x 1024 and 100% scale.

2.1.4 Additional Software

You will require Adobe Reader 8.1.3 or higher as an additional software to work on certain functions.

2.1.5 Browser Settings

Download settings and formats, pop-up settings and advanced settings.

Downloads

Downloading files should be enabled in your browser, since the solution can export files to your local device. If your browser supports restricting downloads by file type, you should enable at least the XML and CSV file types.

If your browser supports trusted sites, we recommend maintaining the download settings only for the Trusted Sites security zone and adding the following URLs to this zone:

- Your system URL, or alternatively the general pattern to which the URL adheres, such as `http://*.crm.cloud.sap`
- Your system URL, or alternatively the general pattern to which the URL adheres, such as `http://*.crm.cloud.sap`
- The pattern `https://*.crm.cloud.sap`
- The pattern `https://*.cxm-salescloud.com`

Pop-ups

The pop-up blocker should either be disabled, or the system URL should be registered as an exceptional site for the pop-up blocker. The Trusted Sites security zone settings listed in the **Downloads** section applies to pop-ups as well.

2.1.6 Browser Versions

Versions tested and handling browser updates.

SAP performs manual and automated testing on the supported and recommended browsers. Recommended browsers provide better performance and usability. Later versions of the recommended browser should be compatible, but have not yet been tested.

Use the latest, stable version of the recommended browsers. Please be aware that some browsers download and apply updates automatically. SAP makes every effort to test and support the most recent, stable versions of all recommended browsers.

2.1.7 Domain Settings

Allow applications to access resources from the ***.crm.cloud.sap** domains.

If you use a URL filter to blocklist a category of websites, you must allow ***.crm.cloud.sap** domains as an exception. Static resources based on the Content Delivery Network (CDN) should be in allow list for the domain ***.cxm-salescloud.com**.

2.1.8 Desktops and Laptops

Browser support by operating system for desktops and laptops.

Platform	OS	Recommended Browser	Supported
Microsoft Windows	Windows 10 & above	Google Chrome	Google Chrome
			Microsoft Edge (Chromium)
Apple Mac OS X	10.10 "Yosemite"	Google Chrome	Google Chrome
			Microsoft Edge (Chromium)

2.2 Language Settings

Browser settings for language support.

To ensure that texts appear consistently in the same logon language, review the following three language settings:

- Before you log on to the solution, choose the preferred language from the dropdown list.
- Make sure that the browser locale is correct, the locale is used to determine the date, time zone, or amount formats.
- In the operating system itself, ensure that the preferred language is defined as the display language.

2.2.1 Languages Supported

Supported languages and restrictions for specific languages.

The user interface is available in the following languages.

- AR — Arabic (Functional limitations for this language are described in the note below)
- CS — Czech
- DA — Danish
- DE — German
- EN — English
- ES — Spanish
- FI — Finnish
- FR — French
- HE — Hebrew (Functional limitations for this language are described in the note below)
- HU — Hungarian
- IT — Italian

- JA — Japanese
- KO — Korean
- NL — Dutch
- NO — Norwegian
- PL — Polish
- PT — Portuguese
- RU — Russian
- SK — Slovak
- SV — Swedish
- TH — Thai
- TR — Turkish
- ZH — Simplified Chinese
- ZF — Traditional Chinese

i Note

Right-to-left languages (Arabic script and the Hebrew alphabet are written from right to left) are **NOT SUPPORTED** for email integration, analytics, or maps.

2.3 Integration and Add-Ins

Additional software requirements for integration scenarios and add-ins.

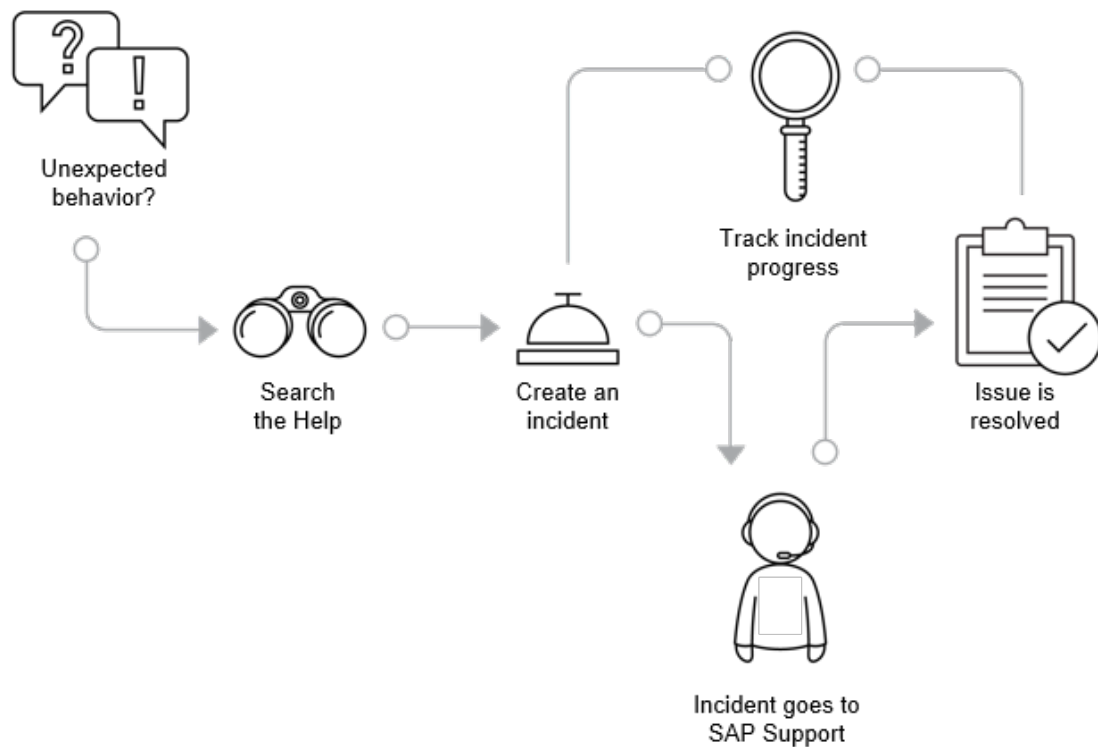
You can enable additional functionalities by integrating with external systems or software that can be installed, via the [Download](#) screen, for specific business purposes and users.

2.3.1 Requirements for Software Integration

To increase customer service, large enterprises can integrate existing SAP S/4 HANA systems with the solution.

3 Help and Support

Learn how to find answers to your questions about the solution and how to report issues and problems.



As you're working, if you encounter unexpected behavior in the solution that interrupts your work or reduces the quality of the service, you can search the [Built-In Support](#) (🔍) for more information or create an incident that is sent to SAP Support. You can search the entire SAP knowledge database, as well as additional resources, for information that can solve your problem.

If you want to review the product documentation that can help you solve the issue yourself, you can find links to the SAP Help Center and this solution guide in the [Built-In Support](#) panel.

Your administrator is the first line of support but if they can't solve the issue for you, they can send the incident to SAP for further investigation.

Prerequisite: One time registration is required to create an SAP Universal ID and use the real-time chat and submit an incident. We recommend you register for your SAP Universal ID with your S-User ID. Once registered, incident reporting and expert chat are available in the [Built-In Support](#) panel.

When creating an incident, be sure to create it from the screen where you're experiencing the issue. The system identifies the screen automatically and includes this information in the incident. To help support process your incident more quickly, always include step-by-step instructions on how to re-create the behavior you're reporting.

As an administrator, after you submit an incident, you can track it for updates.

You can chat with product support specialists as well as with our customer interaction teams for updates on existing incidents. Each chat generates an incident that you can view and edit. The incident contains the chat transcript and can be used to follow up on the reported issue.

Get help within the *Built-In Support* tool by selecting the (ⓘ) icon at the top of the panel.

! Restriction

Built-in support is available only in English.

Related Information

[Full documentation for Built-In Support](#)

[About SAP Universal ID](#) 

4 Global Search

Search for entity details across the solution.



Currently, the available entities for search are:

- Cases
- Individual Customers
- Installed Bases
- Registered Products
- Installation Points
- Library
- Accounts

Global search allows you to search through the entire solution. After clicking the search bar, you can choose to do a simple or advanced search. You can search for a term without specifying any details, or you can search in all categories, or you can specify a specific category.

The search bar is on top of the screen.

i Note

The system displays a list of all the previously searched terms.

4.1 Configure Global Search

Administrators can configure Global Search in [Settings](#).

Find [Settings](#) in the user menu at the top right of the screen.

4.1.1 Filter Configuration

You can define filters and make them visible for users in the advanced searches for the currently available entities.

Procedure

1. Go to your user profile, navigate to ► [Settings](#) ► [All Settings](#) ► [Filter Configuration](#) ►.
2. From the left pane, select the entity you want to configure.
3. Use the [Add](#) (+) button to add a new field.
4. Optional: Click [Reset](#) to erase all the fields in the predefined configuration.
5. Click [Save](#) once you have created a new filter configuration.

4.1.2 Search Admin Settings

You can activate the search bar popover to open for the user when they hover over it.

Procedure

1. Go to your user profile, navigate to ► [Settings](#) ► [All Settings](#) ► [Search Admin Settings](#) ►.
2. Switch on the [Display on Hover](#) toggle to activate the search bar.
3. Click [Save](#).

4.2 Basic Search

Learn about performing a basic search in the solution.

Enter the text in the search field. You can enter a text, number, or date. Any record matching the text shows up. You can view the record in detail by clicking the desired record. The system opens a quick view with the details

of the record. Here you can view details or exit the quick view and navigate to the associated entity details screen.

i Note

In case there are more than three records matching your search criteria, you can choose the [Show More](#) option and view all the matching records in the [Search Results](#) screen.

4.3 Syntax Search

Learn about using the syntax search in the solution.

You can perform a more specific search by using either a query or command line, or by entering the required details in a search form. Also, you can build a command line by selecting the filter icon on the search bar and adding the filter fields that are known.

A query or a command line can fetch you more specific results for your search. Search command line provides suggestions while you are typing. When you press enter, the search result screen appears. This screen has the record details listed in the right pane, and the search form displayed on the left pane.

You can change the search terms and filters on the left pane. As you change the search criteria, notice that the command line field value also changes accordingly. Once you make the changes, click [Apply](#) at the bottom of the form to view the updated records in the right pane of the screen.

On the search form, you have the following additional options:

- Add rule – Adds a new search filter.
- Apply – Applies the changes to the specifics and carry out the search.
- Reset – Erases the search criteria that you have entered previously and start a fresh search.
- Save search – Saves the search query.

i Note

The search is not case sensitive

Additionally, you can also use the asterisk (*) symbol to perform a wildcard search.

When you enter a search term with multiple words that are separated with spaces, the system considers each space as an asterisk or wild card for the previous word. Multiple words are also combined with a logical AND operator. The only exception to this is when the search term is enclosed within double quotes ("").

For instance,

- sap is queried as sap* - all matches starting with the word sap, ignoring the capitalization of the letters
- sap se is queried as sap* AND se*
- "john smith" is queried for an exact match

Use the asterisks appropriately to expand your search results as in the following examples:

- sap* matches any text starting with sap
- *sap* matches any text containing the word sap

- *sap matches any text finishing with the word sap

4.4 Advanced Search for One Entity

Find out how to use the advanced search for one entity in the solution.

You can access the search form directly by clicking the filter icon at the right corner of the search bar. Once you enter the search term and click the filter icon, the system displays the search form with an option to select the entities from the drop-down list in the show section. Upon selecting the entities, the system displays a list that will help you further narrow down your search to get more accurate results.

Example: To search for cases containing the word SAP, enter the term SAP in the search bar and click the filter. In the search form that appears, choose cases from the drop-down list available in [Show](#) section. The system displays the list with options to filter the results further. The solution is configured to fetch the most appropriate filter options corresponding to the entities that you selected in the previous step.

The filter option appears only when you select a single entity in the show section.

i Note

If you wish to change the default filters, your administrator can modify them for each entity using [Filter Configuration \[page 18\]](#).

4.5 Advanced Search for Multiple Entities

Find out how to use the advanced search for multiple entities in the solution.

Once you enter the search term and click the filter icon, the system displays the search form with an option to select the entities from the drop-down list in the show section. Upon selecting multiple entities in the show section, click [Apply](#) to search.

On the search form, you have the following additional options:

- [Reset](#) – Erases the search criteria that you have entered previously and start a fresh search.
- [Save Search](#) – Saves the search query.

4.6 Edit Saved Searches

Here is how you can edit a saved search in the solution.

You can edit recent saved searches in the popover by clicking on the edit icon.

To edit searches that do not appear in the recent saved searches, click [Show All Saved Searches](#) and select the search you wish to edit. Here, you can edit the title, syntax, and add new rules.

4.7 Delete Search

This function deletes the saved search template that is currently stored in the solution.

You can delete recent saved searches in the popover by clicking on the delete icon. Click [OK](#) if you are sure to delete the saved search.

To delete searches that do not appear in the recent saved searches, click [Show All Saved Searches](#) and delete the search.

5 User Interface

Familiarize yourself with the User Interface of the solution.

Knowing how to navigate around the solution with the visible elements that you see and interact with is critical in managing your data in the solution.

[Shell Bar \[page 22\]](#)

The horizontal header that contains User Menu, Plugins, and Logo.

[Home Page \[page 30\]](#)

The home page is the default start page, typically the first thing you see after logging in.

5.1 Shell Bar

The horizontal header that contains User Menu, Plugins, and Logo.

This static shell bar is always visible and therefore you can access the options such as navigation menu, search, user menu and so on, anytime.

The following table lists the icons available in the shell bar:

Icon	Name	Description
	Navigation Menu	Provides easy access to the core functionalities and makes browsing easier by providing a main menu of options. To search for the functionalities in the navigation menu, you can either scroll down the list or start typing in the search field until you get the match.
	Company or Brand Logo	Displays the default SAP logo that you can replace with your own logo. If you're not on the home page, click the logo to return to the home page.
	Global Search	Allows you to search and filter the entity details across the solution.
	CTI Widget	Helps you to support your customers. Computer Telephony Integration (CTI) is identified as the technology of linking of computers and telecom instrument to respond calls, to pull out details from the database based on the phone numbers and even route calls to different people in some cases.
	Built-In Support	Provides digital services to find help, create tickets, and contact support right in your cloud application.

Icon	Name	Description
	Favorites	Allows you to create a list of favorite entities for your easy access.
	Notifications	Lets you know that something new has happened so you don't miss anything that might be worth your attention.
	Switch Application	Allows you to quickly switch between applications. Currently, we support only AppGyver.
	User Menu	Usually shows the initials of your user name. If you click the User Menu, you can see your user name and the email address along with the following options: • Adaptation [page 24]: Customize and extend your solution using various tools. • Settings [page 29]: A central area where you can view and maintain the settings for your solution. • Change Password [page 29]: Allows you to change your existing password. • Sign Out: Use this option to end your session.

Related Information

[Set Company Logo \[page 48\]](#)

[Global Search \[page 17\]](#)

[Favorites \[page 23\]](#)

5.1.1 Favorites

Add entities such as [Accounts](#), [Contacts](#), and [Individual Customers](#) to your favorites list for quick access.

You can use  icon to add your entities to the favorites list and use  icon to remove. To view the list, select  icon available on the top right corner of the shell bar. You can search or filter your items and open them in detail within a click.

The [View All](#) option available under the list gives you the same list in a separate tab where you can select multiple items and then you can remove them from your favorites list at once.

5.1.2 Notifications

Notifications are a way to let you know that something new has happened so you don't miss anything that might be worth your attention.

The Notifications feature is located on the top right corner of the shell bar, next to the Favorites icon. The list of notifications contains maximum five items with the details such as category of actions, time occurred, and so on.

When you click [Show All](#), a separate tab opens where you can see the entire list of notifications. You can filter the items from this list based on date, type and status.

5.1.3 User Menu

Provides access to the user profile, settings, and additional settings for account features.

The User Menu usually shows the initials of your name and available on the top right corner of the shell bar.

5.1.3.1 Adaptation

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

This document describes all functions that you can access as an administrator using the [Start Adaptation](#) button that you can find under the user profile dropdown list.

5.1.3.1.1 Customize Your Solution

As an administrator, you can customize the UI look and feel for all users.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#).

The system displays an orange line in the header to indicate that you are in adaptation mode.

2. Make your changes. Note that any change you make applies to only the main layout.
3. When your changes are complete, click [End Session](#) in the header area.

5.1.3.1.2 Manage Fields

Add, hide, and reorder fields.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange outline.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
 - a. To change the sequence of fields, drag the **Move** icon (☰) next to a field and move the fields around.
 - b. To hide fields, click the **Hide Field** icon (✕).
 - c. To add fields, select the **Add from list** icon (≡+) to bring up the fields available in the entity. You can select one or multiple fields at a time.
4. Click [Apply](#).

5.1.3.1.3 Add Extension Fields

Add extension fields for your objects.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
4. To add fields, select the **Add from list** icon (≡+) to bring up the fields available in the entity.
5. Find the [Extensions](#) node in the list and drill down inside the structure to view the list of available extension fields.
6. Select the **Add Field** icon (+) to add the desired field.

Note: You can add only part of structure fields, for example, you can add only the amount value and not the currency.
7. Click [Apply](#).

5.1.3.1.4 Add and Manage Sections

Add, hide, show, and reorder sections.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
 2. Hover over the desired section to view the editing icons.
 - a. To move the section up or down, click the **Sort** icons ().
 - b. To hide the section from the screen, click the **Hide Section** icon ().
 - c. To add sections, follow these steps:
 1. Click the **Add Section** icon () on the orange/yellow outline.
 2. Select [Create Section](#). The system creates an empty section above the selected spot and a [Create Section](#) pop-up box opens up.
 3. Enter the section name.
 4. Optional: Select the [Show Section Header](#) checkbox to make the section name visible as header.
 5. Click [Apply](#).
 - d. To manage multiple sections together, follow these steps:
 1. Click the orange icon ().
 2. Select [Manage Sections](#). The system displays the list of visible sections in the [All Sections](#) pop-up box.
 3. You can perform the following actions:
 - To change the sequence of sections, drag the **Move Section** icon () next to sections and move them around.
 - To hide sections from the screen, click ().
 - To make hidden sections visible, select the **Add from list** icon () to bring up the [Add Section](#) pop-up box. Search and select the desired sections and click [Apply](#) to make those sections visible again in the UI.
 - To delete custom sections from the screen, select the **Delete Section** icon ().
4. Click [Apply](#) to make changes to all these explicit actions at one go.

Note

You can't delete standard sections; you can only hide them from the screen.

5.1.3.1.5 Add Mashups to Custom Sections

After creating a custom section, you can add mashups to that section.

Procedure

1. Open a custom section.
2. Click the **Drill Down** icon (▾) next to [Add Mashup](#) to drill down to the list of available mashups.
3. Search for the desired mashup and click the **Drill Down** icon (▾) to view the mashup properties.
4. If necessary, you can adjust the appearance of the mashup by editing the following options in the [Edit Properties](#) pop-up box.

Note: The options available to you'll vary depending on the mashup.

- Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
- Define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of [Height\(%\)](#) to 100, the newly added mashup occupies the full height of the screen.
- Configure the parameters to define which screens can be used in the mashup.

5. Click [Apply](#).

5.1.3.1.6 Create Custom Buttons

As an administrator, you can create custom buttons to trigger end user action such as release sales orders, print invoices, and so on. You can create them at designated sections on the screen, for example, near an action group.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange outline.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit** icon (✎) to open the [Section Items](#) pop-up box.
4. Select the **Add** icon (+) to open the [Add Buttons](#) pop-up box.

i Note

You can create a custom button under a standard button group or create a stand-alone button. To create a custom button under a standard button group, drill down inside the group.

5. Add a button name.

6. Add an event name. We recommend that you add a name that explains what the button is used for. For example, Create a Sales Order.

User actions, such as click of a button, or clicking a link triggers certain events. These events are associated to a button.

7. Optional: Add parameters.

Parameters are data that is passed when an event is triggered. You can use these data points as input for mashups.

For example, when you capture the current address and send it to third-party system during a button click, the system will pick up the address that you mention in the parameter section during a button creation.

8. Click [Apply](#).

5.1.3.1.7 Create and Manage Custom Tabs

As an administrator, you can create custom tabs to embed mashups.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
2. Hover over the tab header section to view the editing icons.
3. Select the **Edit** icon (✎) to open the [Edit Tabs](#) pop-up box.
4. Select the **Add** icon (+) to create a custom tab.
5. Add a name for your tab and press the **Return** key.
6. Click [Apply](#).

A custom tab is now created with the same title.

7. Optional: To hide the tab and launch it based on a button click, select the drill down icon (▾) next to the tab and select the [Show on Event Trigger](#) checkbox. The system now hides the tab by default and makes it visible only after you click the button you associate with an event. Ensure that the desired parameters of the event are associated to the mashup present in custom tab.
8. Optional: To hide a custom tab, select the **Hide Tab**(✕).
9. Optional: To delete a custom tab, select the **Delete Tab** icon (🗑).
10. Optional: To change the sequence of tabs, drag the **Move** icon (☰) next to the tabs and move them around.

Click [Apply](#) after performing all the optional steps for the changes to reflect.

5.1.3.1.7.1 Add Mashups to Custom Tabs

After creating a custom tab, you can add mashups to that tab.

Procedure

1. Open a custom tab.
2. Click the **Drill Down** icon (>) next to [Add Mashup](#) to drill down to the list of available mashups.
3. Search for the desired mashup and click the **Drill Down** icon (>) to view the mashup properties.
4. Optional: If necessary, you can adjust the appearance of the mashup by editing the following options in the [Edit Properties](#) pop-up box.

i Note

The options available to you'll vary depending on the mashup.

- Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
 - Define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of [Height\(%\)](#) to 100, the newly added mashup occupies the full height of the screen.
 - Configure the parameters to define which screens can be used in the mashup.
5. Click [Apply](#).

5.1.3.2 Settings

Access the Settings page from the user menu to navigate to different settings on the system.

Go to ► [User Menu](#) ► [Settings](#) ». On the [Settings](#) page, you can view both [All Settings](#) and [Dashboard](#) tabs.

The [All Settings](#) tab lists all the settings on your system in a card view. The settings are categorized into different categories such as Company, General, Users and Control, and so on. Each of these categories further contains four subcategories of settings. If any of the settings category contains more than four subcategories, a link showing +1 More, +2 More, and so on, is displayed. From the [All Settings](#) page, you can navigate to different apps on your system. You can also search for settings in the instant search bar. The [Dashboard](#) tab lists [Pinned](#) and [Recently Visited](#) sections. You can view the settings that you've pinned in the [Pinned](#) section, and all the settings that you visited recently appear in the [Recently Visited](#) section.

5.1.3.3 Change Password

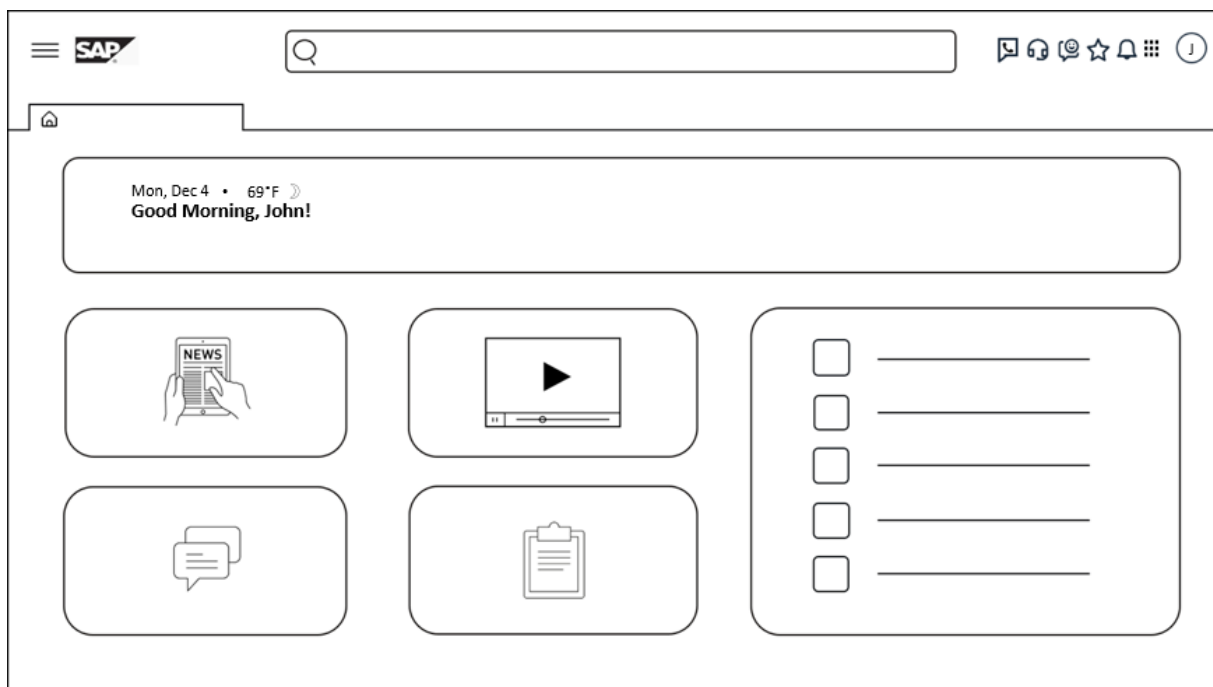
Change the password for users from the user interface.

Change the password for your user by choosing the ► [User Menu](#) ► [Change Password](#) » option. Enter your current password. If you're unsure if you've entered in the correct current password, you can view the password

by clicking the  (show) icon. Next, enter a new password and also re-enter the new password, and click [Confirm](#) to save your new password.

5.2 Home Page

The home page is the default start page, typically the first thing you see after logging in.



The home page displays different types of cards that help you focus on important tasks and performance indicators for your business.

5.2.1 Use Cards

Home page cards are the blocks of content in the solution.

You can see different types of cards on the home page such as News card, Message card, Quick links card and so on. Home page cards highlight important information and enable you to take action or view more information.

You can see the following cards on the home page.

- [Greeting Card](#): Shows a greeting message to the user along with the current date, and the weather condition based on your browser settings.
- [Message Card](#): Shows messages for the business users.
- [News Card](#): Displays news one after the other based on the RSS feeds.
- [Video Thumbnail Card](#): Appears on the home page with the pre-delivered video thumbnail. When you click the video thumbnail, it launches in a separate window.

- *Quick Links Card*: Serves as a launch pad for different quick create pages within the solution. Quick links card on home page shows the object icons and the object names that help you to navigate.
- *Learning Content Card*: Shows a list of five useful articles that help you understand the solution.
- *Video Dashboard Card*: Shows a list of five useful videos that help you understand the solution.

5.2.1.1 Create Cards

Create your personalized cards on the home page.

i Note

We support only Message cards.

As a part of the configuration, you can set the title, description, button type, and background color of the message card.

As an administrator, you can publish messages for your business users by following these steps:

1. Navigate to ► *User Menu* ► *Start Adaptation* ►.
The system displays an orange line in the header to indicate that you are in adaptation mode.
2. Hover over the cards and click the plus (**+**) icon.
3. Select *Create Section*.
4. To create a message card, select the arrow (**>**) icon next to *Add Message Card*.
A new pop-up opens.
5. Enter the details such as title, message, button type, and background color.
6. Click *Add*.
7. Select *End Session* when your changes are complete.

5.2.1.2 Manage Cards

Create, delete, hide, and rearrange your cards on the home page via adaptation.

As an administrator, you can manage your cards by following these steps:

1. Navigate to ► *User Menu* ► *Start Adaptation* ►.
The system displays an orange line in the header to indicate that you are in adaptation mode.
2. Hover over the desired sections to use the following options:
 - **↑** : To move the card towards left
 - **↓** : To move the card towards right
 - **×** : To hide the card
 - **✎** : To edit the card

i Note

The edit functionality is only available for custom Message cards and Greeting card.

3. Click the plus () icon to view the available options.
 1. Select [Manage Sections](#) and use the following options:
 -  : To hide the cards.
 -  : To remove the custom cards permanently.
 -  : To open the [Add Section](#) pop-up and add the cards using the plus () icon.
 2. Select [Create Section](#) to create your own cards by entering the details.
For more information, see [Create Cards \[page 31\]](#).
4. Select [End Session](#) when your changes are complete.

5.2.1.3 Edit Greeting Card

Change your greeting card features on the home page.

As an administrator, you can follow these steps to edit your greeting card:

1. Navigate to ► [User Menu](#) ► [Start Adaptation](#) ►.
2. Hover over the greeting card and select the  (Edit) icon.
A new window opens for editing.
3. To view the weather information on the card, enable the [Weather](#) toggle button to [Show](#).
4. To update the background image of the card, choose any of the following:
 - Select [Default](#) and choose one of the images from [Default Image](#).
 - Select [Custom](#) and upload your image.

i Note

The image format should be JPG or PNG without exceeding 1 MB size limit.

5. Click [Save](#) and select [End Session](#).

5.2.2 Add Background Image

Add background image on home page.

As an administrator, you can follow these steps to add your own background image on home page:

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ►.
2. Under [General](#), select [Home Page Settings](#) to open [Home Page Image](#) tab.
3. You can use one of the following options:

- Select *Default* and choose one of the images.
- Select *Custom* and upload your own image.

i Note

The image format should be JPEG or PNG without exceeding 1 MB size limit.

4. Click *Save*.

6 Company

Learn to manage employees, create divisions and distribution channels, and create your company's structure.

6.1 Org Structure

An organization structure is a set of organizational units, and their hierarchical relationships, that together form the company.

As an administrator, learn to set up your organizational structure including business organizations and sub-units. The organizational structure establishes reporting lines and allows automatic work distribution.

6.1.1 Create Your Org Structure

Setting up an organizational structure enables automated routing of work. The organizational structure is composed of units and provides a unified representation of your company's organizational data.

To create your org structure, you must first plan your org structure. Use your existing organizational structure plan as a basis for creating the org structure in the system. When you start creating the structure, we recommend that you build from the top down; that is, start with companies and then add the sales organizations or departments, including your sales teams. Obtain all data relevant to organizational management, including managerial and functional data.

→ Tip

When editing your org structure, always enter the valid-from date for your changes.

6.1.1.1 Recommendations

Under [Settings](#) > [Company](#) > [Org Structures](#), you establish your organizational structure as a prerequisite to going live. This is also where you maintain and update your organizational structure. As you set up your org structure, we recommend that you observe the following principles:

- The uppermost node of your org structure should be an org unit defined with the [Company](#) function.

→ Tip

You must create and then edit the org unit details to select the [Company](#) function.

- The node that you create directly below the [Company](#) node should be either a [Sales Organization](#) or a [Service Organization](#). If your company uses both the sales and service functions of the solution, then your org structure will have both types of nodes below the company.

→ Tip

It's possible to setup multiple sales/service organizations in parallel organizational structures but it isn't possible to stack sales/service organizations. Because a sales organization is required to process all corresponding sales documents (for example, leads, opportunities), and a service organization is required to process service requests. We recommend that you define only one sales organization and one service organization per org structure.

- The nodes that you create below the [Sales Organization](#) node can be for example, [Sales Units](#), and the nodes that you create below the [Service Organization](#) node can be for example, be [Service Units](#). It's also possible to create an organization, in which all units underneath a sales/service organization are not flagged as a sales/service unit.

However, it is not recommended to interrupt the setup of the sales or service function along the organizational hierarchy as application logic may not consider functional units underneath a node that is not flagged as service or sales unit.

→ Tip

Within the org structure, multiple levels of sales or service units may be defined. Service units represent the service organization and teams that process service requests. Sales units represent the sales departments or offices where sales documents are initiated. Nodes for [Sales Units](#) or [Service Units](#) can be defined below the Sales Organization, or Service Organization node, or below the nodes of other sales or service units.

- When editing your org structure, always enter the valid-from date for your changes. There's no need to explicitly activate the organizational units. They are activated based on their validity period.

⚠ Caution

During maintenance of the org structure, always set the validity of org units within the validity period of the parent org unit.

6.1.2 Create an Org Unit

Learn how to create an [Org Unit](#).

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Org Structure](#) ►.
2. Create a new org unit by selecting [Create](#) (+).
3. Enter the [Valid From](#) date (today's date, or a future date, fixed date) and the [Valid To](#) date. You can edit the [Valid From](#) and [Valid To](#) fields in the detailed view of the org unit.
4. Enter the [Address](#) details for the unit. You can edit the [Address](#) field in the detailed view of the org unit under the [General](#) tab.
5. Select the appropriate functions for the unit.

Function	Description
Sales	The Sales function indicates that the org unit is responsible for the processing of documents, such as opportunities or sales quotes. In sales reporting, sales data is aggregated at the sales unit level. The sales organization is a specific type of sales unit, which is used to head complete areas or countries/regions. Relevance Select this function if employees assigned to this org unit will perform tasks in the areas described above or if sales data is to be aggregated at the level of this org unit.
Sales Organization	The Sales Organization function indicates that the org unit represents the top level of a sales hierarchy. Note Example For your domestic sales, you create a sales hierarchy and define the top-level org unit as a sales organization. Then below that org unit, you add additional units, defined as sales units, that represent regional sales groups and sales departments. The sales organization is also used for defining sales data in master data. • Sales data in the product and account master can be defined in relation to the sales organization in combination with the distribution channel and division. • Internal price lists can be defined based on the sales organization and distribution channels. • The combination of the sales organization, distribution channel, and division is also included in sales documents, such as sales orders or service orders. Relevance Select this function if the org unit represents the top unit of a sales hierarchy. All sales-relevant master data and pricing must include a sales organization.

Function	Description
Service	The Service function indicates that the org unit is responsible for the managing, monitoring, and execution of services, such as the handling of product inquiries, providing recommendations, and pre- and post- sales support. Customer service can be provided by a service desk or by field or in-house service teams. Relevance Select this function if employees assigned to this org unit will perform tasks in the areas described above.
Service Organization	The Service Organization function indicates that the org unit represents the overall organization responsible for customer service. Below the service organization, you can define individual service units to represent the various service and support teams and departments within the service organization. → Tip If the services provided are sold to your customers, you will also need to define a sales organization. Relevance Select this function if the org unit represents a fixed entity for service reporting that can be referenced by all service documents.
Company	In the context of organizational management, a Company represents an org unit that is financially and legally independent, that is not tied to a geographical location, and could be registered under business law. Each org structure must contain at least one org unit that is defined as a company. Saving the org unit as a company automatically creates an associated business partner which is used in the context of seller party or output form determinations. You set the default currency for the organization within the company org unit.

6. Save your entries.
7. To select the [Company](#) function for the org unit, open the unit details and make your selection in the header section.

6.1.3 Add Employees and Managers to Org Units

As an administrator, you can identify an employee as a manager by defining that employee as the manager of an org unit. The manager of an org unit enjoys unique access privileges and can access the data of their subordinates.

i Note

Employees and managers need the Primary Indicator. An employee can be marked as main by using the primary indicator for only one org unit. An employee who is marked as main in an org unit may only be added as a secondary employee in another org unit.

Prerequisites:

- The employees you want to assign to the org unit have been created in the system.
- Alternatively, you can assign employees to org units when you create the employee data in the system.

Procedure

1. From the [Org Structure](#) screen, select the relevant org unit from the list.
2. On the [Employees](#) tab, click [Add \(+\)](#) to assign employees on the [Employees](#) view or a manager on the [Managers](#) view to the org unit.
3. Select the employee or manager you want to assign to the org unit and select the icon (✓).
4. [Save](#) your entries.

6.2 Employees

Learn how to create employees in your system.

i Note

All accounts, individual customers, employees, contacts, are considered as business partners in the system. Therefore, all functions that are common to business partners are applicable to them.

Additionally, detailed information on External IDs is displayed under the [Mapping for Integration](#) section, and you can view all changes made to employees under the [Change History](#) (🕒) section.

6.2.1 Create Employees

You can create employee records manually or upload employee data via migration tool in the implementation project activity.

Procedure

1. Navigate to your [User Menu](#) and select ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Employees](#) ►.
2. In the worklist of the employees view, choose [Create](#) (+).
3. Enter as much personal, organizational, and contact information as you require.
4. Choose [Save](#).

6.2.1.1 Edit Employees

Edit an employee record.

Procedure

1. In the worklist of the employees view, click the employee record.
2. In the quick view, click open details.
3. If you want to delimit the validity of an employee record, enter the appropriate date in the [Valid To](#) field.
From this date onward, this employee record cannot be used for new business processes.
4. System automatically saves the changes you made.

6.2.2 Manage Attachments

As an administrator, you can upload, view, delete, and download attachments.

6.2.2.1 Upload Attachments From Your Computer

Here is a step-by-step guide on how to import attachments from a front-end system, such as a computer, into the system.

Procedure

1. Navigate to *User Menu* and select ► *Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. In the quick view, click *Open in Detailed View* (🔗).
3. From the *Attachment* (📎) tab, select *Drop or Browse Files*.
4. Select the file from your computer you want to upload and click *Open*.
5. From the dropdown, select the *Attachment Type*.
6. Click *Upload*.

i Note

You can upload multiple files at the same time.

To download attachments, select an attachments and click *Download* (⬇).

Select an attachment and click *Delete* (🗑) to delete an attachment.

6.2.2.2 Add Attachments From SAP System

Find out how to add attachments from your SAP system.

Procedure

1. Navigate to *User Menu* and select ► *Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. In the quick view, click *Open in Detailed View* (🔗).
3. From the *Attachment* (📎) tab, select the *Add* button.
4. Choose *Add From Library*.

i Note

To add a link, select [Add Link](#) (🔗) and provide the link in the link field. Click [Save](#). You also can provide a title for this link.

5. Choose the documents you want to add and click [Add](#).

6.2.3 Working Hours

An employee's scheduled working hours.

6.2.3.1 Add Working Day Calendar

The working days of an organizational unit within a particular company or region.

Context

Find out how you can assign the working days of an organization to an employee.

Procedure

1. Navigate to [User Menu](#) and select ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Employees](#) ►.
2. Select an employee and in the quick view, click [Open in Detailed View](#) (🔗).
3. Select the [Working Hours](#) (📅) tab.
4. Under the [Working Day Calendar](#) section, fill the [Time Zone](#) and [Working Day Calendar](#) fields from the dropdowns.
5. System automatically saves this information.

6.2.3.2 Add Working Hours

The amount of time an employee is scheduled to spend working.

Procedure

1. Navigate to *User Menu* and select ► *Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. Select an employee and in the quick view, click *Open in Detailed View* (🔗).
3. Select the *Working Hours* (🕒) tab.
4. In the *Working Hours* section, select *Create* (+).
5. Select the *Days* and set the *Time* (🕒).
6. Click the icon ✓.
7. Employees' working hours are saved.

6.2.4 Create Sales Responsibility

Employees are always created with an assignment to an organizational unit. Therefore, you must create the necessary sales data required to use employees for various business needs within the org. structure of your company. You can also use territories, distribution channels and divisions for your employees.

Procedure

1. Navigate to *User Menu* and select ► *Settings* ► *All Settings* ► *Company* ► *Employees* ►.
2. Select an employee and in the quick view, click *Open in Detailed View* (🔗).
3. Select the *Sales Responsibility* tab.
4. Click *Create* (+).
5. Click *Edit* (✎) and using *Value Help* (🔍) add the *Sales Organization ID* field.
6. For *Distribution Channel* and *Division*, click *Edit* (✎) and choose from the dropdown.
7. To set default sales data, enable the toggle *Default Sales Responsibility*.

i Note

There can only be one default sales data.

8. The system saves the information automatically.

6.3 Distribution Channel

A channel through which saleable materials reach customers. Distribution channels include wholesale, retail, and direct sales. You can assign a distribution channel to one or more sales organizations.

6.3.1 Create Distribution Channels

Learn how to create a distribution channel.

Procedure

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Distribution Channel](#) ►.
2. Click [Create](#) (+).
3. Enter all the desired fields and [Save](#) the created distribution channel.

6.4 Division

A sales organization can sell materials from different divisions. Divisions can be used to form sales areas to refine the organizational sales structure.

Use divisions to organize materials or services, for example to form product lines.

6.4.1 Create Divisions

Learn how to create a division.

Procedure

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Company](#) ► [Divisions](#) ►.
2. Click [Create](#) (+).
3. Enter all the desired fields and [Save](#) the created division.

i Note

Once divisions are created for your company, they can be included in account sales data and sales documents, such as sales orders.

7 General

Learn how you can configure the broad settings in your solution.

[Language and Region \[page 45\]](#)

Configure language and region settings for all your users.

[Change User Interface Texts for Applications \[page 46\]](#)

As an administrator, you can translate or change field labels and UI texts for applications.

[Company Settings \[page 47\]](#)

Define your company settings that are applicable for all users, or a selected business role.

[Branding \[page 47\]](#)

Adapt the look and feel of your SAP solution to suit your company's need.

[Adaptation \[page 24\]](#)

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

[Extensibility \[page 54\]](#)

Extensibility allows you and your partners to extend the software platform without having to modify the original codebase. So, you can add to the base functionality, thereby offering new capabilities and outputs.

[Extensibility Using SAP Build Apps \[page 58\]](#)

SAP Build Apps enables you to build apps for all form factors including mobile, desktop, browser, TV, and others without writing code.

7.1 Language and Region

Configure language and region settings for all your users.

On the [Language and Region](#) page, you can configure the tabs:

- Currency: Set the switch to [Enabled](#), against the currencies that you want to enable for your users.
- Language: Set the switch to [Enabled](#) against the system languages and communication languages that you want to enable for your users.
- Time Zone: Set the switch to [Enabled](#), against the time zones that you want to enable for your users.
- Country/Region: Set the switch to [Enabled](#), against the countries/regions that you want to enable for your users. Expand the (+) icon, next to [Country/Region](#), to view more details.
- Fiscal Period: Define fiscal periods.
 1. Select [Fiscal Year](#), [Fiscal Quarter](#), or [Fiscal Period](#) option from the dropdown.
 2. Click the add icon (+) to add a new fiscal year, fiscal quarter, or fiscal period information.
 3. Enter all the information in the respective columns, and click the [Save](#) (💾) icon to save your fiscal period data.

i Note

All values entered for *Fiscal Year*, *Fiscal Quarter*, or *Fiscal Period* fields can be edited inline.

- Exchange Rate: Define the currency exchange rates on the *Exchange Rate* tab.
 1. Select a currency from the *Select Unit Currency* dropdown.
 2. Click the *Add* (+) icon to add a new currency to the list.
 3. Choose a currency from the *Quoted Currency* dropdown, define the exchange rate in the *Exchange Rate* column, and also edit *Quoted Date* to select the date on which the currency value was defined.
 4. Click the *Save* (💾) icon to save the entry.

7.2 Change User Interface Texts for Applications

As an administrator, you can translate or change field labels and UI texts for applications.

Context

You can use this feature to override the standard SAP texts and customize field label terminology to fit your business needs.

Procedure

1. Navigate to ► *Settings* ► *All Settings* ► *General* ► *Language Adaptation* ►. The system opens the **Language Adaptation** tool in a new window.
2. Select an application, source language, and target language for maintaining custom translations. Keep the source language and target language same if you just want to change the field labels or texts.
3. Click *Next*.
4. Enter the desired texts in the *Target Text* column.
5. Click *Save*.

i Note

If you override the standard SAP texts, you cannot revert the changes.

7.3 Company Settings

Define your company settings that are applicable for all users, or a selected business role.

7.3.1 Auto Sign Out

For security reasons, users are automatically logged off from the system if they've been inactive in the system for a certain period. If you leave this option empty, inactive users will be logged off from the system after 1 hour. You can set the auto log off time for all users in your company.

As an administrator, you can use this feature by following these steps:

1. Navigate to **User Menu** > **Settings** > **All Settings**.
2. Select **Company Settings** under **General**. You can also start typing the name in the search field to get the result.
3. Under **Auto Sign Out** tab, you can set the logged off time for the following:
 - Company
 - Business Users
4. To set the logged off time for all the inactive users in the company, select the drop-down under **Company** and choose any of the time periods.
5. To set the logged off time for the specific inactive business users, click the plus (+) icon under **Business Role** to create a new row.
6. Search and select the business role, choose the time, and click the ✓ icon to create. Similarly, you can add many business roles as required.
7. If you want to delete any of the rows, click the  icon.
8. Click **Save**.

7.4 Branding

Adapt the look and feel of your SAP solution to suit your company's need.

As an administrator, you can set the company logo, background screen, and login screen. The changes will be visible to all users the next time they log on to the system.

To use this feature, navigate to **User Menu** > **Settings** > **All Settings**. Select **Branding** under **General** or search for branding in the search field and select from the result.

7.4.1 Set Company Logo

You can set your company logo on your login page and shell bar for desktop and mobile.

Procedure

1. Go to the [Logo](#) tab, and under [Desktop](#), upload your logo on the login page by following one of the actions:
 - Drag and drop your image file to the field.
 - Click [Browse files](#) and select an image file.

i Note

The supported image file format is SVG. The size of the image can't exceed 1 MB.

2. If you want to set your logo on the shell bar, select [Shell Bar](#) on the right side of the screen and upload the image file.

Similarly, upload a logo on your login page and shell bar under [Mobile](#).

3. Click [Save](#) when you finish.

7.4.2 Set Background Screen

You can set your background screen when there are no tabs open.

Procedure

1. Go to [Tab Configuration](#) tab, and select one of the following options:
 - [Default](#): Shows a default screen when all the tabs are closed.
 - [Brand Logo](#): Shows your brand logo when all the tabs are closed.
 - [Plain Page](#): Shows a simple text when all the tabs are closed.
2. Click [Save](#).

7.4.3 Set Login Image

You can set your login image.

Procedure

1. Go to [Login Image](#) tab, choose one of the following options based on the requirements:
 - [Default](#): Provides five different images rotating for the login page.
 - [Custom](#): Add your own images in the login page.

i Note

The supported image file formats are PNG and JPG. The size of the image can't exceed 1 MB.

2. Click [Save](#).

7.5 Adaptation

As an administrator, you can change the look and feel of the solution for all users by changing layout settings, adding mashups and fields, as well as defining extension fields.

This document describes all functions that you can access as an administrator using the [Start Adaptation](#) button that you can find under the user profile dropdown list.

7.5.1 Customize Your Solution

As an administrator, you can customize the UI look and feel for all users.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#).

The system displays an orange line in the header to indicate that you are in adaptation mode.

2. Make your changes. Note that any change you make applies to only the main layout.
3. When your changes are complete, click [End Session](#) in the header area.

7.5.2 Manage Fields

Add, hide, and reorder fields.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange outline.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
 - a. To change the sequence of fields, drag the **Move** icon (☰) next to a field and move the fields around.
 - b. To hide fields, click the **Hide Field** icon (✕).
 - c. To add fields, select the **Add from list** icon (≡+) to bring up the fields available in the entity. You can select one or multiple fields at a time.
4. Click [Apply](#).

7.5.3 Add Extension Fields

Add extension fields for your objects.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
4. To add fields, select the **Add from list** icon (≡+) to bring up the fields available in the entity.
5. Find the [Extensions](#) node in the list and drill down inside the structure to view the list of available extension fields.
6. Select the **Add Field** icon (+) to add the desired field.

Note: You can add only part of structure fields, for example, you can add only the amount value and not the currency.
7. Click [Apply](#).

7.5.4 Add and Manage Sections

Add, hide, show, and reorder sections.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
2. Hover over the desired section to view the editing icons.
 - a. To move the section up or down, click the **Sort** icons ().
 - b. To hide the section from the screen, click the **Hide Section** icon ().
 - c. To add sections, follow these steps:
 1. Click the **Add Section** icon () on the orange/yellow outline.
 2. Select [Create Section](#). The system creates an empty section above the selected spot and a [Create Section](#) pop-up box opens up.
 3. Enter the section name.
 4. Optional: Select the [Show Section Header](#) checkbox to make the section name visible as header.
 5. Click [Apply](#).
 - d. To manage multiple sections together, follow these steps:
 1. Click the orange icon ().
 2. Select [Manage Sections](#). The system displays the list of visible sections in the [All Sections](#) pop-up box.
 3. You can perform the following actions:
 - To change the sequence of sections, drag the **Move Section** icon () next to sections and move them around.
 - To hide sections from the screen, click ().
 - To make hidden sections visible, select the **Add from list** icon () to bring up the [Add Section](#) pop-up box. Search and select the desired sections and click [Apply](#) to make those sections visible again in the UI.
 - To delete custom sections from the screen, select the **Delete Section** icon ().
4. Click [Apply](#) to make changes to all these explicit actions at one go.

Note

You can't delete standard sections; you can only hide them from the screen.

7.5.5 Add Mashups to Custom Sections

After creating a custom section, you can add mashups to that section.

Procedure

1. Open a custom section.
2. Click the **Drill Down** icon (▸) next to [Add Mashup](#) to drill down to the list of available mashups.
3. Search for the desired mashup and click the **Drill Down** icon (▸) to view the mashup properties.
4. If necessary, you can adjust the appearance of the mashup by editing the following options in the [Edit Properties](#) pop-up box.

Note: The options available to you'll vary depending on the mashup.

- Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
 - Define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of [Height\(%\)](#) to 100, the newly added mashup occupies the full height of the screen.
 - Configure the parameters to define which screens can be used in the mashup.
5. Click [Apply](#).

7.5.6 Create Custom Buttons

As an administrator, you can create custom buttons to trigger end user action such as release sales orders, print invoices, and so on. You can create them at designated sections on the screen, for example, near an action group.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange outline.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit** icon (✎) to open the [Section Items](#) pop-up box.
4. Select the **Add** icon (+) to open the [Add Buttons](#) pop-up box.

i Note

You can create a custom button under a standard button group or create a stand-alone button. To create a custom button under a standard button group, drill down inside the group.

5. Add a button name.

6. Add an event name. We recommend that you add a name that explains what the button is used for. For example, Create a Sales Order.

User actions, such as click of a button, or clicking a link triggers certain events. These events are associated to a button.

7. Optional: Add parameters.

Parameters are data that is passed when an event is triggered. You can use these data points as input for mashups.

For example, when you capture the current address and send it to third-party system during a button click, the system will pick up the address that you mention in the parameter section during a button creation.

8. Click [Apply](#).

7.5.7 Create and Manage Custom Tabs

As an administrator, you can create custom tabs to embed mashups.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange border.
2. Hover over the tab header section to view the editing icons.
3. Select the **Edit** icon () to open the [Edit Tabs](#) pop-up box.
4. Select the **Add** icon () to create a custom tab.
5. Add a name for your tab and press the **Return** key.
6. Click [Apply](#).

A custom tab is now created with the same title.

7. Optional: To hide the tab and launch it based on a button click, select the drill down icon () next to the tab and select the [Show on Event Trigger](#) checkbox. The system now hides the tab by default and makes it visible only after you click the button you associate with an event. Ensure that the desired parameters of the event are associated to the mashup present in custom tab.
8. Optional: To hide a custom tab, select the **Hide Tab**().
9. Optional: To delete a custom tab, select the **Delete Tab** icon ().
10. Optional: To change the sequence of tabs, drag the **Move** icon () next to the tabs and move them around.

Click [Apply](#) after performing all the optional steps for the changes to reflect.

7.5.7.1 Add Mashups to Custom Tabs

After creating a custom tab, you can add mashups to that tab.

Procedure

1. Open a custom tab.
2. Click the **Drill Down** icon (>) next to [Add Mashup](#) to drill down to the list of available mashups.
3. Search for the desired mashup and click the **Drill Down** icon (>) to view the mashup properties.
4. Optional: If necessary, you can adjust the appearance of the mashup by editing the following options in the [Edit Properties](#) pop-up box.

i Note

The options available to you'll vary depending on the mashup.

- Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
 - Define the height of HTML Mashups to be displayed in the available screen. For example, if you set the value of [Height\(%\)](#) to 100, the newly added mashup occupies the full height of the screen.
 - Configure the parameters to define which screens can be used in the mashup.
5. Click [Apply](#).

7.6 Extensibility

Extensibility allows you and your partners to extend the software platform without having to modify the original codebase. So, you can add to the base functionality, thereby offering new capabilities and outputs.

Extensions range from customizing through to the development of add-ons or composites. Extensions can either be made by customers, partners, or SAP. Extensibility allows customers and partners to support specific use cases and requirements that are not covered by standard software.

7.6.1 Create Extension Fields

Extension fields are additional fields that you, as an administrator, can add to a cloud solution from SAP.

Context

Extension fields provide the means to store additional data for specific use cases beyond what is natively present in the solution. You can create these fields for a screen that has been enabled for extension fields.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select [Settings](#) > [All Settings](#) > [General](#) and click [Extensibility Administration](#).

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Click the [Extendable](#) checkbox to view the entities that can be extended.
3. Drill down to the entity you want to add an extension field to, and select it. The system opens the corresponding worklist.
4. In the [Custom Fields](#) tab, click the add icon (+) to open the [Create Fields](#) quick create screen.
5. Enter the [Label](#). The system autopopulates the name.
6. Select the [Data Type](#) and the corresponding [Data Format](#).
7. Click [Save](#).

The extension field is now created and is listed under the [Custom Fields](#) tab.

i Note

- You can't delete extension fields currently.
- You can create up to 50 extension fields.

7.6.2 Create Validations

Validations help you, as an administrator, to implement a custom logic where you can prevent a save or issue a warning message to users during save based on certain conditions.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select [Settings](#) > [All Settings](#) > [General](#) and click [Extensibility Administration](#).

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and drill down to select an entity.
3. Navigate to the [Validations](#) tab and click the **Create Validation** icon (+). The system launches the validation editor tab.

The editor has three main blocks:

- Header: Specify the validation name and description here.
 - Condition: Set the desired conditions using a flow-based visual condition editor.
 - Action: Specify an error or warning message based on the conditions defined in the condition block.
4. In the header block, specify the validation name and description.

5. In the *Condition Block* set the desired conditions using a flow-based visual condition editor.
 - a. Select the standard or extension field from the dropdown.
 - b. Select the comparison operator.
 - c. Select the target comparison. You can compare either by a fixed value or with another field.
 - d. If you choose *Value*, enter a corresponding target value. If you choose *Field*, select a target field.

Add multiple conditions by clicking the **Add Condition** icon (+). To remove a condition, click the **Remove Condition** (X). To clear all conditions in a block, click the **Delete Condition Block** icon (🗑)

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

6. Optional: You can also create an OR condition by clicking the + *Or* button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

i Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

7. In the third block, under *Action* choose to show either an error or a warning message.
8. Add the message text.
9. Click *Save and Activate*.

i Note

You can create a maximum of five validations per entity.

In the *Determinations* tab, the system lists all validations for a specified entity. Click the **Edit** (✎) or **Delete** (🗑) icons to edit or delete the corresponding validation.

7.6.3 Create Determinations

Use determinations to calculate specific field value based on certain conditions. You can either default or propose the value or mandate a certain value using determinations.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list select **Settings** > **All Settings** > **General** and click *Extensibility Administration*.

The system opens a new screen and displays a list of all business entities in the left panel. These entities are grouped by services.

2. Select a service and click the *Drill Down* icon (⊞) to open and select an entity.
3. Navigate to the *Determinations* tab and click the **Create Determination** icon (+). The system launches the determination editor tab.

The editor has three main blocks:

- Header: Specify the determination name, event, and description here.
 - Condition: Set the desired conditions using a flow-based visual condition editor.
 - Assignment: Assign a value to specific fields. You can use each row to assign a single field value. For example, `total_amount = item_count * item_price`.
4. In the header block, perform the following actions:
 - a. Enter a name.
 - b. Choose one of the following events:
 - Pre Hook: You can use it to default or suggest standard and custom field values. The user can change the field values.
 - Post Hook: You can use it to mandate custom field values. The user cannot overwrite the field values.
 - c. Add a description.
 5. In the *Condition Block* set the desired conditions using a flow-based visual condition editor.
 - a. Select the standard or extension field from the dropdown.
 - b. Select the comparison operator.
 - c. Select the target comparison. You can compare either by a fixed value or with another field.
 - d. If you choose *Value*, enter a corresponding target value. If you choose *Field*, select a target field.

Add multiple conditions by clicking the **Add Condition** icon (+). To remove a condition, click the **Remove**

Condition (✕). To clear all conditions in a block, click the **Delete Condition Block** icon (🗑)

Note that when you create multiple conditions, the condition group is considered fulfilled only if all the conditions within a group are met.

6. Optional: You can also create an OR condition by clicking the + Or button. The system displays additional condition blocks. Follow the same steps as earlier to fill one or more OR condition blocks.

i Note

When you create multiple OR conditions, the condition group is considered fulfilled if any one of the block conditions are met.

Under the *Summary* header, the system shows you a summary of all the conditions that you've entered.

7. In the *Assignment Block* perform the following actions:
 - Use a row to assign or calculate value to a specific field.
 - Write the expressions and calculations in free-style format.
 - Mention the target field first, followed by "=" and then specify the expression. For example, `{status} = ('Open')`
 - Enclose all entity fields in curly braces: For example, `{status}`, `{ID}`, and so on.
 - Ensure that you add the extension/custom fields under the extensions structure. For example, `{extensions/customfield}`
 - Get the technical name of the fields using the dropdown in the *Condition Block*.
 - Use expressions consisting of operators and certain pre-defined functions specified in the *Assignment Block*.
 - Ensure that you check the feedback provided by the system in case of invalid expressions. The error messages are configured to give exact feedback so that you can correct the expression.

- Create additional assignment rows by clicking the **Add Assignment** icon (+). To remove a row, click the **Remove Assignment** icon (✕). Click *Clear All* to remove all assignment rows.
- Click + *Else* to create one more set of condition blocks that are triggered if the initial sets of conditions are not satisfied. Note that you can create an Else block only when you have at least one condition in the initial IF block. Follow the same steps to create Else conditions and assignments inside the Else block.

8. Click *Save and Activate*.

i Note

You can create a maximum of five determinations per entity.

In the *Determinations* tab, the system lists all determinations for a specified entity. Click the **Edit** (✎) or **Delete** (🗑) icons to edit or delete the corresponding determination.

7.7 Extensibility Using SAP Build Apps

SAP Build Apps enables you to build apps for all form factors including mobile, desktop, browser, TV, and others without writing code.

You can extend your system using SAP Build Apps by designing, developing, and embedding these extensions on your system. You can access Build Apps by clicking the Switch Application icon (🗑) in the Shell Bar.

i Note

If you've 100 or more users, SAP Build Apps base package is included as a default with SAP Service Cloud Version 2 . If you've fewer than 100 users, you need additional licensing for SAP Build Apps to build extensions. You must contact your account executive to purchase licenses.

i Note

If you've 100 or more users and SAP Build Apps isn't configured on your tenant, raise an incident on the CA-LCA-ACP component with the following details:

- SAP Service Cloud Version 2 tenant URL
- SAML 2.0 Trust Configuration for the identity provider used by SAP Service Cloud Version 2
 - XML metadata, name, and origin key
 - Existing IdP User Groups to BTP Role collection mappings, if available
- SAP Service Cloud Version 2 Discovery API Destination configuration
 - base path (if not same as SAP Service Cloud Version 2 tenant URL)
 - authentication type
 - OAuth2SAMLBearerAssertion
 - OAuth Client App Name
 - OAuth Client ID
 - Additional parameters as needed by other authentication types

Once the ticket is assigned, grant access to the following to the assignee:

- Customer Global Account (<https://controlcenter.ondemand.com>) to create subaccount.
- SAP Service Cloud Version 2 basic authentication administrator credentials for establishing trust between BTP destinations and SAP Service Cloud Version 2.

Related Information

[Design and Develop Extensions on SAP Service and Sales Cloud Version 2 Using SAP Build Apps](#) 

8 Users and Control

Users and control allow you, as an administrator, to manage employees, users, business roles, and security policies in the system.

- **Employees:** A group of people who contribute to creating goods or services in a company, based on a work contract or a contract for services. Unlike externals, internal employees are bound by instructions and obliged to adhere to the company's policies and regulations.
- **Users:** Refer to employees, business users, and other users such as partner contacts. Users are assigned to business roles and technically, access controls in your system are implemented on the business roles.
- **Business Roles:** Represents a set of authorizations to access your solution within your organization and system landscape. In the system, a business role defines a hierarchy of access controls that you can assign to application users in your organization. You can use roles to assign access to a large number of users at a time. As roles change, you need to update only the access rights within a role and the rights apply for all assigned users.
- **Security Policies:** A set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

8.1 Users

Users are created for employees to access the system.

Each business user must have at least one business role assigned that governs the access of the user to the system.

Business User List View

You can view all the business users in your system in a list format.

The following header fields are available on this page:

- **User ID:** the user ID of your business user.
- **Full Name:** the full name of your business user.
- **Created On:** the date on which your business user was created.
- **Valid Until:** the date until which your business user is valid in the system.

8.1.1 Business User Details

View more details of a business user on this page.

The [Details](#) tab provides the following information of a business user.

- [Employee Data](#): Shows the basic employee information such as Employee Name, Employee ID, Company, Email ID, and so on.
- [User Details](#): Displays the following user information.
 - [User ID](#): A unique ID to identify a business user. Only an administrator has the privileges to edit it.
 - [Created On](#): The date on which the business user was created in the system.
 - [Valid Until](#): The date until which the user is active in the system. An administrator can edit this date.
 - [Password Locked](#): A toggle that indicates if the password is locked or not. The following are the features of the [Password Locked](#) toggle.
 - The toggle is auto-enabled after too many failed login attempts.
 - If the toggle is enabled, an employee can't log on to the system from the login screen but can only log in using a certificate.
 - An administrator can disable this toggle to unlock the password, however, an administrator can't lock the password.
- [Security Policy](#): Displays the security policy assigned to the user. An administrator can change the assigned security policy.
- [Reset Password](#): Provides an option to reset the current password of the user. As an administrator, you can reset your user's password by clicking the [Reset Password](#) button and then set a new password for your user, which then triggers a password reset email to your user.

The [Assigned Business Roles](#) tab lists all the business roles assigned to a particular business user. An administrator can search and add a business role. The administrator can also delete an assigned role.

8.1.2 Create Business Users and Assign a Business Role

Administrators can create business users and provide appropriate authorizations by assigning business roles.

Context

Business users in the system are linked to employees. Business users have business roles assigned, and the access controls in the system are implemented on business roles.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.

3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Users](#) ►.
4. Choose [Business Users](#) from the dropdown on the page.
5. Click the add icon (+) on the [Business Users](#) page.
6. Search and select an employee for whom you wish to create the business user.

i Note

- The employee linked to a user must have a unique Email ID.
- If business user already exists for the employee, a message stating *The business user already exists for this employee* is displayed.

7. Once you select an employee, the [User Data](#) section is auto-populated. By default, the user ID is created with the last name followed by the first name of the employee. If you wish to modify the user ID details, you can modify them.
8. Once you've verified all the details, click [Create](#). The business user is now created and linked to the employee. Refresh the business users' list to view your newly created business user.
9. Choose your business user, and then navigate to the [Assigned Business Roles](#) tab.
10. Add business roles to your user by using the [Search and Add](#) functionality on the screen. All the business roles that you assigned to your business user now appear in the list format on the [Assigned Business Roles](#) tab.

You've now created a business user and assigned the user to a business role.

8.2 Business Roles

A business role defines the specific areas of the system the users can access when they're assigned to that role. Administrators can implement authorizations to business users by assigning business roles.

A business role represents a set of authorizations to access your solution within your organization and system landscape. A business role defines a hierarchy of access controls that you can assign to business users in your organization. As roles change, you need to update only the access rights within a role and the rights will apply for all assigned users.

Business Roles List View

Lists all the business roles in a list view. As an administrator, you can edit the following fields on this page:

- Business Role ID
- Business Role Name
- Description
- Status: You can set the status to [In Preparation](#), [Active](#), or [Obsolete](#).
- Admin: You can activate this switch to indicate that the role is for an administrator.

8.2.1 Business Role Details

View more details of a business role on this page.

The [Details](#) tab provides the following information of a business role.

- [Business Services](#):
 - All business services that are assigned to a business role are displayed in the [Business Services](#) list. As an administrator, you can add or delete business services to the business role.
 - All business services have unrestricted read and write access by default.
 - Some business services allow creation of restrictions. Such services have [Read/Write](#) dropdown enabled. On setting any one of them to [Restricted](#), the [Restriction Rule](#) link is enabled.

i Note

If one role blocks write access and another role allows it, the more permissive role wins.

- An administrator can select from the available restrictions for a given service to create a restriction rule.
- You can delete invalid business services that get flagged with the following message "Some of the business services assigned to the business role is no longer valid. Please delete all the services with a warning sign to proceed further."
- [App List](#):
 - Each business service can have one or more apps associated with it. These apps define the left navigation for the users having this particular business role assigned.
 - On adding the business service to a role, the corresponding apps are added to the app list.
 - Administrators can enable or disable these apps using the toggle against them.
 - You can also search for apps from the search bar. You can also filter the app list by [All Apps](#), [Generic Apps](#), and [Admin Apps](#).
 - As an administrator, you can also enable all apps at once using the [Enable All Apps](#) switch and enable all admin apps at once using the [Enable All Admin Apps](#) switch.

The [Assigned Business Users](#) tab lists all the users that have this business role assigned. An administrator can search and add users to this role. The administrator can also delete users from this role.

8.2.2 Create Business Roles and Assign a Business User

Administrators can create business roles with appropriate authorizations and assign them to business users.

Context

Use business roles to assign access rights to multiple business users who carry out similar activities. You can also define access restrictions in a business role.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Business Roles](#) ►.
4. Click the add icon (+) on the [Business Roles](#) page, and enter the business role name and description. The [Business Role ID](#) is auto-generated. You can also set this role as an admin role by enabling the [Admin](#) toggle.
5. Click the [Add Business Services](#) button, select all the services that are required for this role from the list of services, and then click [Add](#). You can now [Save and Activate](#) your business role.
6. Once the business role is saved, the [App List](#) section is now visible, which displays all the available apps for the selected business services. Select the toggle button against the app to make it available for the business role.

i Note

The [App List](#) section defines the left navigation on the UI for users that have this business role assigned.

7. Set restrictions for a role using [Read Access](#) and [Write Access](#) columns in the [Business Services](#) section. By default, both these columns are set to unrestricted for a service, and the [Restriction Rule](#) is set to unassigned.
8. Change either or both [Write Access](#) and [Read Access](#) to [Restricted](#), and click [Unassigned](#) to create a restriction rule. Select options from the list for creating a restriction rule for the selected service.
9. Navigate to the [Assigned Users](#) tab. A list of users who have been this role assigned is displayed. If there are no business users assigned yet, the [Search and Add Users](#) button is visible.
10. Click the [Search and Add Users](#) button to add business users from the quick view. Search and select users from the quick view to add them to the role.

Results

You've now created a business role and assigned the business role to the selected list of business users.

8.3 Security Policies

A security policy is a set of rules that defines password complexity, such as including numerical digits and password validity, like requiring a password change after a certain period of time.

You can define multiple security policies because work areas or departments of a company may have different password security requirements.

As an administrator, you can manage the security level, if desired, by editing and enhancing the security policy, for example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

8.3.1 Security Policy Features

Lists and describes the features of a security policy.

The security policy features are classified as:

- System password-related features
 - Enable Password Login - if enabled, allows users to log in to the system with a password.
 - Set Default - if enabled is set as the default security policy for all the users.
- System password complexity-related features
 - Minimum Number of Characters - defines the minimum number of characters in a valid password.
 - Minimum Number of Lowercase Alphabets - defines the minimum number of lowercase alphabets in a valid password.
 - Minimum Number of Uppercase Alphabets - defines the minimum number of uppercase alphabets in a valid password.
 - Minimum Number of Digits - defines the number of numerals in a password.
 - Minimum Number of Special Characters - defines the minimum number of special characters in a valid password.
- System password validity-related features
 - Allowed Number of Incorrect Password Attempts - defines the number of successive login failures before a password is locked. A message is displayed on the login screen once the password is locked.
 - Password History - defines the number of passwords to store in the memory to compare with the new productive password set by the user.
 - Maximum Password Validity (in Days) - defines the maximum number of days a password is valid for.
 - Minimum Waiting Time Before Changing Password (in Days) - defines the minimum number of days a user must wait before changing the existing password to a new password.
 - Initial Password Validity (in Days) - defines the validity of the initial password (system-generated password when a user is created in the system).
 - Unused Productive Password Validity (in Days) - defines the validity of the productive password after changing from the initial password.

8.3.2 Create Security Policies

You can define multiple security policies as different work areas or departments in a company can have different password security requirements.

Context

As an administrator, you can increase the security level by editing and enhancing a security policy. For example, by changing the complexity and validity for all passwords, in accordance with your company's security requirements.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to [All Settings](#) > [Users and Control](#) > [Security Policies](#) .
4. Click the add icon (+) on the [Security Policies](#) page.
5. Enter a [Policy ID](#), [Policy Name](#), and [Description](#) for the security policy. You can further choose to set this policy as the default using the [Set Default](#) switch.
6. Enhance security complexity for your policy by setting inputs for: [Minimum Number of Characters](#), [Minimum Number of Lowercase Alphabets](#), [Minimum Number of Uppercase Alphabets](#), [Minimum Number of Digits](#), [Minimum Number of Special Characters](#), [Allowed Number of Incorrect Password Attempts](#), [Password History](#), [Maximum Password Validity \(in Days\)](#), [Minimum Password Change Waiting Time \(in Days\)](#), [Initial Password Validity \(in Days\)](#), [Unused Productive Password Validity \(in Days\)](#), and then click [Save](#).
7. Refresh the [Security Policies](#) page to view your newly created security policy. All the security policies that you've created appear in the list format on the [All Policies](#) page.

You've now created a security policy.

8.3.3 Edit Security Policy

Administrators can edit security policies.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to [All Settings](#) > [Users and Control](#) > [Security Policies](#) .
4. Click an existing policy ID to view the details. The [Security Policy](#) quick view opens.
5. Hover the mouse over the fields that you want to edit. Click on pencil icon next to the field to edit the field.

Results

Once edited, the changes are saved automatically.

8.4 Identity Provider Configuration

Identity provider refers to an entity that manages identity information and provides authentication services to trusted service providers.

Every user type must authenticate to the system for regular browser-based front-end access, as well as for electronic data exchange, such as business-to-business communication. The system doesn't support anonymous access. We recommend using Single Sign-On (SSO) for basic security. To protect accounts further, configure the identity provider (IdP) of the SSO solution to provide enhanced security.

8.4.1 Configure Identity Provider

Administrators can configure identity provider to enable single-sign on for the system.

Enable Single Sign-On for the System

To enable Single Sign-On for the system, you've to connect the identity provider to the system. To do so, download the metadata of your system, upload the metadata XML file to your identity provider, and then download the configured metadata from the identity provider. Afterwards, choose the identity provider, and create the identity provider to start using the system.

Download the System Metadata

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [IdP Configuration](#) ►.
4. Click [Download Metadata](#). The XML file is downloaded to your local drive.

Configure System Metadata in Your Identity Provider (IdP)

1. Log on to the [SAP Identity Authentication Service \(IAS\)](#) as an administrator. You can also use an external identity provider.
2. On the IAS screen, navigate to ► [Applications & Resources](#) ► [Applications](#) ►, and click [Add](#) to create a new application for your system.
3. In the [Add Application](#) pop-up window, enter a name and click [Save](#). The system opens and displays the new application.
4. Under ► [Trust](#) ► [Single Sign-On](#) ►, configure the following settings:
 - Click [Type](#), and select SAML 2.0
 - Click [SAML 2.0 Configuration](#), and upload the SAC metadata XML from your local drive.
 - Click [Subject Name Identifier](#), and configure the attribute, which the application uses to identify the users.
Ensure that you use the same attribute that is maintained in your system. For example, if your system-specific application in the IdP is configured as **Login Name**, then use the same value in SAC-specific application as well.

5. Navigate to [Tenant Settings](#), and click [SAML 2.0 Configuration](#) to open a new screen.
6. Click [Download Metadata File](#) to download the IdP metadata file.

i Note

Use this procedure to configure IdP provided by SAP IAS. You can follow similar steps to configure external IdPs as well.

Upload IdP Metadata to Your System

1. Log in to your system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Users and Control](#) ► [Configure IdP](#) ►.
4. Click the add (+) icon under the [Trusted Identity Provider](#) section.
5. Under the [Upload IdP Metadata](#) quick view, enter an [Alias](#), and upload the configured metadata XML file (downloaded from your IdP) by either dropping or browsing to upload from the system.
6. Click [Save](#) to save your configuration.
7. You can further set your configuration to active by activating the [Set Active](#) toggle, and also set your configuration as default by activating the [Set Default](#) toggles.

You've now configured Single Sign-On to your system.

9 Accounts and Individual Customers

Account management capabilities gives you a holistic view of the customers, and allow you to capture, monitor, store, and track all critical business information about customers, prospects and partners.

You can use this information to focus on your most profitable customers, maintain customer satisfaction and loyalty, and enable consistent interactions with your customers across all channels.

i Note

All accounts, individual customers, employees, contacts, partners and competitors are considered as business partners in the system. Therefore, all functions that are common to business partners are applicable to them.

i Note

In the system, the *Customers* refer to corporate accounts, individual customer accounts and contacts. *Accounts* represent organizations with which business is done, and *Individual Customers* represent persons with whom business is done. Create business accounts to represent your customers in your system.

Additionally, detailed information on External IDs for accounts is displayed under the [Mapping for Integration](#) section, and you can view all changes made to accounts under the [Change History](#) (🕒) section.

9.1 Configure Accounts

As an administrator, you can configure accounts and individual customers in [Settings](#).

9.1.1 Configure Highlights

You can choose the highlights that are displayed on the account details page.

Prerequisites

You must define Fiscal Year. For more information, see the Related Information section.

Procedure

1. Navigate to ► [Settings](#) ► [All Settings](#) ► [Customer Insights](#) ► [Highlights](#) ►.
2. In the [Highlights](#) tab, use the switch to enable or disable the required highlights.
3. You can select the timeframe for displaying Appointments, Opportunities, and Expiring Quotes.
To use preset default values for all the highlight settings, you can select [Use Default Values](#).
4. You can select the number of top-ranked signals to display.
5. Save your preferences.

Related Information

[Language and Region \[page 45\]](#)

9.1.2 Configure Key Metrics

You can choose the key metrics that are displayed on the account details page.

Procedure

1. Navigate to ► [Settings](#) ► [All Settings](#) ► [Customer Insights](#) ► [Key Metrics](#) ►.
2. In the [Display Fields](#) tab, use the switch to enable or disable the respective metrics.
3. Save your preferences.

9.1.3 Configure Timeline Filters

You can choose the filters that are displayed in the [Timelines](#) section on the account details page.

Procedure

1. Navigate to ► [Settings](#) ► [All Settings](#) ► [Customer Insights](#) ► [Timeline Filters](#) ►.
2. In the [Interaction Filters](#) tab, use the switch to enable or disable the required filters.
3. In the [Object Filters](#) tab, use the switch to enable or disable the required filters.
4. Save your preferences.

9.1.4 Configure ABC Classifications

As an administrator, you can learn to configure the [ABC Classifications](#) for accounts and individual customers.

Procedure

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ►.
2. Search for [ABC Classifications](#).
3. In the new window that appears, click + ([Create](#)).
4. Edit the [Code](#) and [Description](#) fields to add a new classification.
5. The system saves the information automatically.

9.1.5 Configure Industries

As an admin, you can learn to configure [Industries](#) for accounts and individual customers.

Procedure

1. Navigate to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ►.
2. Search for [Industries](#).
3. In the new window that appears, click + ([Create](#)).
4. Edit the [Code](#) and [Description](#) fields to add a new industry.
5. The system saves the information automatically.

9.2 Create Accounts

Learn how to create an account so you can keep track of all the information about your customers.

Procedure

1. From the [Navigation Menu](#) (☰), search for [Accounts](#).
2. Click + ([Create](#)).

3. The system opens a quick view.

Enter the following details.

Name: Provide title for the account.

Role: Define default customer roles for a customer type. A customer type can be either a prospect or a customer. A prospect is someone who may become a customer.

ABC Classification: Categorize accounts based on their importance within your organization. The standard solution offers three levels of categorization, namely A, B, and C accounts.

Owner: Assign an owner to the account using the [Value Help](#) (□).

4. Enter the address details.

Optional: [House Number](#), [Street](#), [City](#), [Postal Code](#), and [Country/Region](#)

5. Enter the communication details.

Optional: [Language](#), [Website](#), [Phone](#), and [Email](#)

6. [Save](#) your details.

Note

In the detailed view of an account, the left-hand panel provides [Communication Details](#), [Address](#), and [General](#) account information. To add the account to favorites, click ★ ([Add to Favorites](#)). This can be done in the quick view as well. You can view the favorite accounts by going to ► [Navigation Menu](#) ► [Favorites](#) ►. Choose [Accounts](#) from the [Type](#) dropdown.

Related Information

[Configure ABC Classifications \[page 71\]](#)

9.3 Assign Status

Set an account status to **Active**, **Obsolete**, **In Preparation**, or **Blocked**.

Context

Active: Indicating that the account is valid and can be used.

Obsolete: An indication that the account has expired or is invalid.

In Preparation: A sign that the account is currently being worked on.

Blocked: A system status indicating that an account has been suspended.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Choose the account you want to set the status for.
3. The system opens a quick view.
4. Click  (*Open in Detailed View*).
5. In the left-hand panel, under the *About* () tab, click  (*Edit*) in the *Status* field to set the status for the account.
6. The system saves the changes automatically.

9.4 Assign Industry

Categorize accounts according to the primary business activity of the enterprise. These are standard defaulted values.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Click the account for which you wish to assign an industry.
3. The system opens a quick view.
4. Click  (*Open in Detailed View*).
5. In the left-hand panel, under the *About* () tab, edit () the *Industry* field to assign an industry for the account.
6. The system saves the changes automatically.

9.4.1 Configure Industries

As an admin, you can learn to configure *Industries* for accounts and individual customers.

Procedure

1. Navigate to  *User Menu*  *Settings*  *All Settings* .
2. Search for *Industries*.

3. In the new window that appears, click **+** (*Create*).
4. Edit the *Code* and *Description* fields to add a new industry.
5. The system saves the information automatically.

9.5 Create or Add Contacts

Find out how to create or add contacts for a person or an organization that is associated with your account.

Procedure

1. Go to *Navigation Menu* () and search for *Accounts*.
2. Select the account and *Open in Detailed View* ()
3. In the *Contacts* section, click **+** (*Create*).
4. Enter all the desired fields.

i Note

The *Account* field is automatically populated.

5. Click *Save* to create the contact.

i Note

You can also *Search and Add* a contact. To do this, click *Search and Add*, search for the contact, and select **+** (*Add*). The system adds the contact automatically. You can see the updated information in the contacts section.

Within the detailed view of the account, you can  (*Call*),  (*Email*), or  (*Edit*) the function field of a contact. The *Ellipsis* () on each contact allows you to *Delete* () or *Mark as Main*. When the contact is marked as main, it is prioritized, and the solution places it at the top of the list.

9.6 Add Accounts to Account Team

You can add multiple accounts as well as individual customers to your account team.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Select the account and *Open in Detailed View* () .
3. In the *Account Team* section, click *Search and Add*.
4. Search for the account and click **+** (*Add*).
5. The system adds the account automatically. You can see the updated information in the accounts section.

i Note

Within the detailed view of the account, you can  (*Call*),  (*Email*), or  (*Edit*) the function field of an account in *Account Team*. The *Ellipsis* (⋮) on each account allows you to *Delete* () or *Mark as Main*. When the account is marked as main, it is prioritized, and the solution places it at the top of the list.

9.7 Create Addresses

Learn how to add, edit, and delete an address.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Select the account for which you would like to add an address and *Open in Detailed View* () .
3. Click **+** (*Create*) in the *Address* section.
4. In the new screen, click **+** (*Add*) to add an address.
5. In the bottom section, enter the address and communication details.
6. The system saves the changes automatically. Navigate *Back* to the address section. You can see the updated information.

i Note

To edit any address, select *Full Screen* from the *Address* section and make the edits. Additionally, to delete an address, select the address and click  (*Delete*).

9.8 Maintain Attachments

You can upload, view, delete, and download attachments.

9.8.1 Upload Attachments From Your Computer

Here is a step-by-step guide on how to import attachments from a front-end system, such as a computer, into the system.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Select the account and *Open in Detailed View* () .
3. Choose the *Additional Data* tab.
4. From the *Attachments* section, select  (*Drop or Browse Files*) .
5. Select the file from your computer you want to upload and click *Open*.
6. From the dropdown, select the *Attachment Type*.
7. Click *Upload*.

i Note

You can upload multiple files at the same time.

Select an attachment and click  (*Download*) to download an attachment.

To delete an attachment, click  (*Delete*).

9.8.2 Add Attachments From SAP System

Learn to add attachments from your SAP system.

Procedure

1. From the *Navigation Menu* () , search for *Accounts*.
2. Select the account and *Open in Detailed View* () .
3. Choose the *Additional Data* tab.

4. From the [Attachments](#) section, select **+** ([Attachments](#)).
5. Select [Add From Library](#).

i Note

To add a link, select [🔗](#) ([Add Link](#)) and provide the link in the link field. Click [Save](#). You also can provide a title for this link.

6. Choose the documents you want to add and click [Add](#).

9.9 Create Sales Data

Accounts are often used within the purview of an org. structure. Therefore, create the necessary sales data required for accounts within the org. structure of your company. Use distribution channels and divisions for accounts. Assign one or more sales organizations, distribution channels, and divisions to an account.

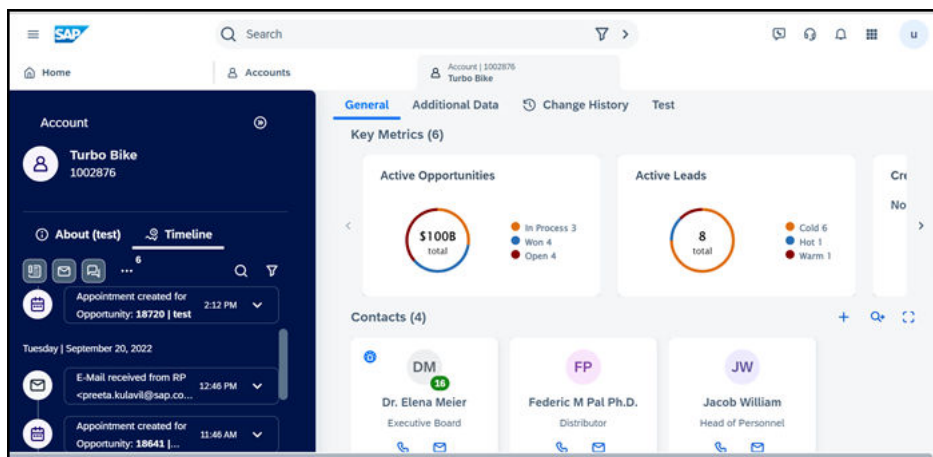
Procedure

1. From the [Navigation Menu](#) ([☰](#)), search for [Accounts](#).
2. Select the account and [Open in Detailed View](#) ([🔗](#)).
3. In the detailed view of an account, select the [Additional Data](#) tab.
4. In the [Sales Data](#) section, select **+** ([Create](#)).
5. A new window opens. Click **+** ([Create](#)).
6. [Search and Add Sales Organization](#), [Distribution Channel](#), [Division](#), [Sales Office](#), [Sales Group](#), and other desired fields.
7. The system saves the changes automatically. Navigate [Back](#) to the sales data section. You can see the updated information.

9.10 View Customer Insights

View customer insights inside the account details page.

The account details screen displays Customer Insights that help you, as a sales representative, to thrive in the digital-sales era by surfacing relevant insights and recommending actions to fully leverage growth opportunities and provide an enhanced customer experience.



You can see the following sections as a part of Customer Insights:

- **Key Metrics** The following key metrics are available:

- Active Leads
- Active Opportunities
- Credit Limit Usage
- Recent Invoices
- Purchase Orders
- Contracts
- Open Cases

Clicking on any of the metrics opens the a detailed view of the metric. You can choose the key metrics that you need on the account details page.

- **Highlights**

Summary of activities such as appointments, expiring quotes, leads, cases, top-ranked signals from external systems and so on. You can navigate to cases and appointments from highlights. You can choose the highlights and configure the timeframes to display the information in highlights.

- **Timeline**

Displays customer's interaction in the order of their occurrence for a selected time period. The date filter allows to filter timeline events for specific time periods such as:

- [Current Week](#)
- [Last 30 Days](#)
- [6 Months](#)
- [1 Year](#)
- [All Time](#)

You can also filter timeline based on interaction and entity types.

The following interactions are available:

- Phone Calls
- Emails
- Chats
- Tasks
- Appointments

The following entities are available:

- Cases

- Opportunities
- Leads
- Sales Quotes

You can choose filters that you want to display in the timeline.

Related Information

[Configure Key Metrics \[page 70\]](#)

[Configure Timeline Filters \[page 70\]](#)

[Configure Highlights \[page 69\]](#)

[Integration \[page 298\]](#)

10 Contacts

Use contacts to represent the relationship between contacts and accounts. Contacts can be maintained only for corporate accounts.

i Note

All accounts, customers, employees, contacts, partners, and competitors are considered as business partners in the system.

Contacts are persons that have a relationship with a corporate account and may be involved in business processes like activities, orders, opportunities, and so on. You can maintain the details for contacts by creating a contact or using the relation between a contact and an account. You can define a contact as sales area dependent and assign them to multiple accounts.

Additionally, use contacts to:

- Present a 360° view of the relationship that your organization has with a specific contact
- Set a status for a contact. For example, you can set a contact to [Active](#) or you can [Block](#) a contact.

10.1 Create Contacts

Find out how to create a contact.

Procedure

1. From the [Navigation Menu](#), search for [Contacts](#).
2. Click **+** ([Create](#)).
3. Enter all the desired fields.
4. Click [Save](#).

i Note

You can see the [Communication Details](#), [Address](#), and [General](#) contact information of a contact in its detailed view under the [About](#) (i) tab in the left panel.

10.2 Maintain Attachments

As an administrator, you can upload, view, delete, and download attachments.

10.2.1 Upload Attachments From Your Computer

Here is a guide on importing attachments from your front-end system.

Procedure

1. Go to ► [Navigation Menu](#) ► [Contacts](#) ► .
2. In the worklist of the contacts view, select the [Contact Name](#) and [Open in Detailed View](#) ().
3. Under the [General](#) tab, in the [Attachments](#) section, select  ([Drop or Browse Files](#)).
4. Select the file from your computer you want to upload and click [Open](#).
5. From the dropdown, select the [Attachment Type](#).
6. Click [Upload](#).

i Note

You can upload multiple files at the same time.

To download attachments, select the attachment and click  ([Download](#)).

Click  ([Delete](#)) to delete an attachment.

10.2.2 Add Attachments From SAP System

You can add attachments from your SAP system by following these steps.

Procedure

1. Go to ► [Navigation Menu](#) ► [Contacts](#) ► .
2. In the worklist of the contacts view, select the [Contact Name](#) and [Open in Detailed View](#) ().
3. Under the [General](#) tab, in the [Attachments](#) section, select  ([Add](#)).
4. Click  ([Add From Library](#)).

i Note

To add a link, select  ([Add Link](#)) and provide the link in the [Link](#) field. Click [Save](#). You also can provide a title for this link.

5. Choose the documents you want to add and click [Add](#).
6. You can view the attachments in the [Attachments](#) section.

11 Agent Desktop

Customer service agents can use the agent desktop to work on several business processes and communication items simultaneously.

The agent desktop is the central work environment for an agent in SAP Service Cloud Version 2. Agent desktop enables service agents to engage with customers efficiently across multiple communication channels. Includes supporting tools like automatic customer identification and search, detailed customer information, knowledge base integration, and case creation.

SAP Service Cloud Version 2 provides an integrated, cloud-based communication stack allowing multisession communication with customers in real time.

i Note

A third-party communication system provider is required for communication channels. Contracting a communication system provider, including all costs incurred by using the provider services, is your responsibility. These contracts and charges aren't part of SAP Service Cloud Version 2. The availability of the inbound and outbound call functionality depends on the services that your provider supports.

11.1 Configure Agent Desktop

Administrators can configure agent desktop in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

→ Tip

Expand a collapsible section heading to view the configuration steps.

11.1.1 Configure General Settings

Enable automatic activity creation and open agent desktop automatically when a service agent accepts an incoming interaction.

Context

You can configure the solution to automatically create activity entities for either incoming or outbound interactions, or for both. This screen is also where you enable agent desktop to open automatically upon accepting an incoming interaction.

Procedure

1. Go to your user profile and select **Settings** > **Agent Desktop** > **General Settings**.
2. Select the checkbox to enable one or more of these options:
 - **Create Activity from Interaction - Inbound** Create an activity record for each incoming interaction automatically.
 - **Create Activity from Interaction - Outbound** Create an activity record for each outgoing interaction automatically.
 - **Open Agent Desktop** Open the agent desktop when an agent accepts an incoming interaction.
3. Save your changes.

11.1.2 Configure Communication Provider

Configure communication system provider and widget size.

Prerequisites

A third-party communication system integrated with your solution. The communication system must be available as a widget or embedded frame inside the agent desktop interface. For more details, see the integration information in the related links that follow the steps.

Note

A third-party communication system provider is required for communication channels. Contracting a communication system provider, including all costs incurred by using the provider services, is your responsibility. These contracts and charges aren't part of SAP Service Cloud Version 2. The availability of the inbound and outbound call functionality depends on the services that your provider supports.

Procedure

1. Go to your user profile and select **Settings** > **Agent Desktop** > **Live Interaction Widget Settings**.
2. Under **Configure Widget and Provider**, enter the required provider details:
 - **Provider Name**
 - **Provider ID**
 - **Provider URL - URL for the communication system widget**
3. Add channels. Select the add icon (+) at the top of the table to add a new channel.

Set the parameters depending on the channel type:

Channel Parameters

Type	ID	Identify Customer By
Phone	(unique channel ID)	Email
Chat	(unique channel ID)	Phone
SMS	(unique channel ID)	Phone
(Social Media Service)	(unique channel ID)	Email

4. Activate channel. Set the [Active](#) switch for the channel entry to on.

Interactions on a channel are only possible if the channel is active.

You can configure multiple channels of each type for your solution. For example: different phone numbers or SMS short codes with your communication provider for each support team can be configured as separate channels with unique IDs.

5. Save your changes.

11.1.3 Configure Knowledge Base Mashup Integration

Use a mashup to integrate a knowledge base widget with your solution.

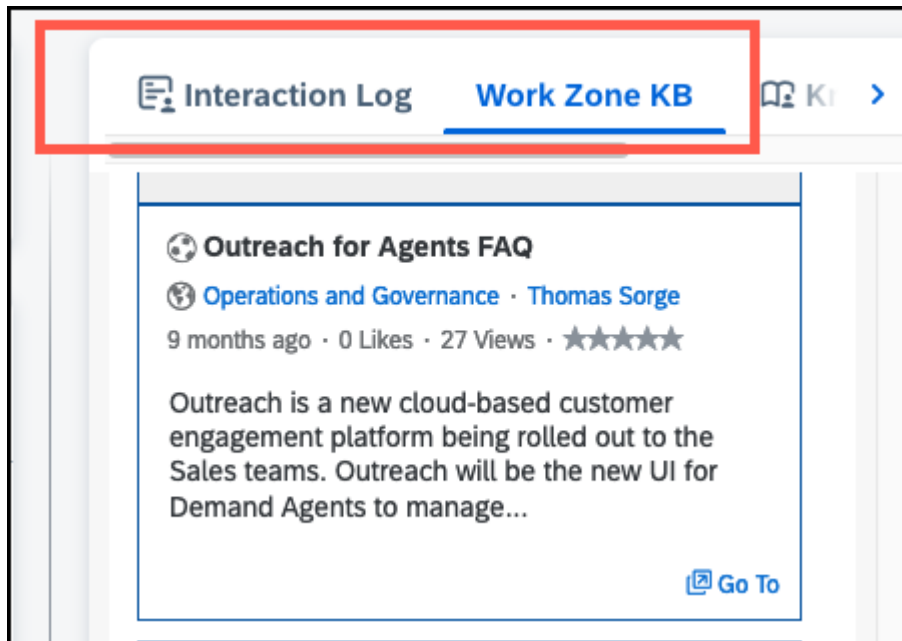
Context

At a high level, integrating a knowledge base as a mashup involves two processes: configure a mashup, then adapt the agent desktop screen to add the mashup.

Procedure

1. Request a widget URL from your knowledge base provider.
2. Create a mashup using the widget URL. See the related links section for detailed instructions on creating a mashup.
3. Adapt the agent desktop UI to add the knowledge base mashup to the screen. See the related links section for detailed instructions on adding a mashup as a tab.

We recommend that you add the mashup as a tab on the right-hand pane.



Related Information

[Mashups \[page 185\]](#)

[Create HTML Mashups \[page 186\]](#)

[Add Mashups as Tabs \[page 188\]](#)

11.1.4 Configure Knowledge Base Provider

Configure your knowledge base parameters.

Procedure

1. Go to your user profile and select **Settings** > **Knowledge Base** > **Provider Settings**.
2. Select **SAP MINDTOUCH** under **Knowledgebase Providers**.
3. Enter the required parameters:
 - **Site URL** - API URL
 - **User** - user name for account
 - **Key**
 - **Secret**
4. Set the **Active** switch to on and save the configuration.

11.1.5 Configure Inbound Email Channels to Create Cases

Email messages routed to your solution can automatically generate cases that are sent to service agents for processing. You, as an administrator can add the inbound email addresses to use for case creation.

Prerequisites

- Decide which email addresses to use for customer support. There is currently a limit of 20 channels per tenant.
- In your organization's mail server, configure email forwarding from each support address to the technical address of your SAP Service Cloud Version 2 tenant. Find the tenant technical address on the email channel settings page.

Context

When an email message is routed to the solution, the system creates a case for the corresponding email channel according to the settings you configure for the channel. In addition, the solution can then route cases to a team or agent using routing rules you specify.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Email](#) ► [Channel Settings](#) ►.
2. Add a channel for each inbound support address you wish to use. Select the create icon (+) at the top of the [All Email Channels](#) list to create a new channel.
3. Enter the support email address for this channel and enter a display name for the channel.

You can also enter an address to send a blind carbon copy (BCC) of outbound messages you send as replies to this channel. Some organizations use this secondary address as another way to track support responses.

4. Choose [Case](#) for [Object](#).

When you choose [Case](#), the [Type](#) and [Default Account](#) fields appear.

5. Choose the case type, and the default account for this channel.

All cases created from this channel use the case type you select here. The default account is assigned to the case if the solution is unable to determine a unique account or customer based on the **from** address on the email message.

6. Select [Save](#) to save the new email channel.

The technical email address for your tenant appears on this screen once you save the channel.

7. Select [Activate](#) to set the channel to [Active](#) status.

A two-step verification process is required before the channel is activated. A message is sent to the channel email address with a verification code. A second message is sent from the verification service with a link you use to activate the channel.

Email channels only create cases from incoming messages when the channel status is set to *Active* and the forwarding rules are in place on your organization's mail server.

Once a channel is in *Active* status, you can edit all attributes of the channel, **except** the channel email address. For example, you can edit the channel name, display name, object, case type, or default account.

Related Information

[Configure Case Routing to Employee \[page 133\]](#)

[Configure Case Routing to Team \[page 134\]](#)

11.1.6 Create Email Templates

Email templates streamline email messaging with preset text, graphics, signatures, and attachments. Create templates for service responses, signatures, branding, and campaigns by uploading an HTML-based layout file.

Prerequisites

Prepare an HTML template layout. You can include technical placeholders to fill in data from the object record when the template is applied in the context of an associated object. See the steps following for how to find a list of placeholders.

Context

You can create email templates based on HTML layouts you create in your preferred HTML editor. Upload the HTML layout file and include any standard attachments for the template. You can also use placeholders to automatically pull data from associated objects in the system. For example: you can automatically fill in the customer name and ticket ID for a service response.

Procedure

1. Go to your user profile and select  [Settings](#)  [Email](#)  [Templates](#) .
2. Add a new template. Select the create icon (+) at the top of the [Templates](#) list to create a new template.

3. Enter a name for your new template.

4. Select a *Template Type*:

- Response: use for service responses from your support teams.
- Signature: can include placeholders for employee information.
- Branding: create a custom brand-themed email template.
- Campaign: for newsletters and surveys.
- Approval Response Template: create a response template for approvals.
- Approval Template: create an approval request template.

The required fields for the template information vary depending on the type of template you select. Some types are only available for email channels. You can enter a description for response templates and campaign templates.

5. Select an *Object Type*.

- Account
- Contact
- Individual Customer
- Case

Object type selection is limited based on the template type. Not all objects are available for every template type.

6. Select a *Channel Type* if necessary.

The channel type defaults to *Email* for most template types.

7. Set up the formatting for your template.

Depending on the type of template and object you selected in the previous steps, some or all of the following tabs are available for working with template formatting:

- *Viewer*: Shows a preview of the template as it appears in the email editor. Placeholders are shown where relevant data is filled in once the template is applied.
- *Documents*: Upload or change the HTML file defining the template layout here. Use any HTML editor you prefer to create your template layout.
- *Attachments*: Upload any documents you want to include as standard attachments with email messages created from this template.
- *Placeholder*: Displays a list of available data fields associated with the object type you selected for your template. Consult this list when creating your HTML layout file to insert placeholders for object data. Placeholders are formatted as the technical field name surrounded by double curly braces. For example: `{{familyName}}`, `{{givenName}}`.

i Note

The Approval template has the `{{uiLink}}` placeholder for source document link. When approval email notification is received by the intended approvers, the approver can access the source document by clicking on the link in the email with manual login and then navigate to the source document approval step.

8. Save your changes when finished.

11.2 Customer Search

Automatically identifies the customer or contact for the incoming interaction. If no matching customer is found, or if there are multiple possible matches you can confirm the correct customer manually.

11.3 Communication Channels

Agent desktop enables customer service communication across multiple contact channels such as: phone, e-mail, and chat. The service process begins with incoming complaints/queries/requests from these sources.

Communication channel integration is enabled by third-party communication system providers.

11.3.1 Configure Communication Provider

Configure communication system provider and widget size.

Prerequisites

A third-party communication system integrated with your solution. The communication system must be available as a widget or embedded frame inside the agent desktop interface. For more details, see the integration information in the related links that follow the steps.

i Note

A third-party communication system provider is required for communication channels. Contracting a communication system provider, including all costs incurred by using the provider services, is your responsibility. These contracts and charges aren't part of SAP Service Cloud Version 2. The availability of the inbound and outbound call functionality depends on the services that your provider supports.

Procedure

1. Go to your user profile and select **Settings** > **Agent Desktop** > **Live Interaction Widget Settings**.
2. Under **Configure Widget and Provider**, enter the required provider details:
 - **Provider Name**
 - **Provider ID**
 - **Provider URL - URL for the communication system widget**

3. Add channels. Select the add icon (+) at the top of the table to add a new channel.

Set the parameters depending on the channel type:

Channel Parameters

Type	ID	Identify Customer By
Phone	(unique channel ID)	Email
Chat	(unique channel ID)	Phone
SMS	(unique channel ID)	Phone
(Social Media Service)	(unique channel ID)	Email

4. Activate channel. Set the [Active](#) switch for the channel entry to on.

Interactions on a channel are only possible if the channel is active.

You can configure multiple channels of each type for your solution. For example: different phone numbers or SMS short codes with your communication provider for each support team can be configured as separate channels with unique IDs.

5. Save your changes.

11.3.2 Phone

Agent desktop supports voice calls with widget-based computer telephony integration (CTI) with third-party communication system providers.

Embed a third-party telephony widget into your solution, providing a unified call handling experience. Agents can receive incoming phone calls, initiate outbound calls and record voice calls. Links to voice call recordings appear in the timeline. Call playback takes place in the communication system widget.

i Note

Telephony functions in agent desktop are dependent upon communication system feature availability.

11.3.3 Chat

Agent desktop integrates third-party chat and messaging solutions to support service processes.

View chat and SMS interactions and transcripts in the timeline.

11.3.4 SMS

Connect with customers using short message service (SMS) for real-time chat interactions.

The SMS channel in agent desktop supports the following functions:

- Inbound SMS chat
- Outbound chat initiated by service agent
- Chat transcription (plain text)

11.3.5 Email

Email messages routed to your solution can automatically generate cases that can be routed to service agents or teams for processing.

You have the option to reply to an email message associated with a case, or compose a new email message.

11.3.5.1 Configure Inbound Email Channels to Create Cases

Email messages routed to your solution can automatically generate cases that are sent to service agents for processing. You, as an administrator can add the inbound email addresses to use for case creation.

Prerequisites

- Decide which email addresses to use for customer support. There is currently a limit of 20 channels per tenant.
- In your organization's mail server, configure email forwarding from each support address to the technical address of your SAP Service Cloud Version 2 tenant. Find the tenant technical address on the email channel settings page.

Context

When an email message is routed to the solution, the system creates a case for the corresponding email channel according to the settings you configure for the channel. In addition, the solution can then route cases to a team or agent using routing rules you specify.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Email](#) ► [Channel Settings](#) ►.
2. Add a channel for each inbound support address you wish to use. Select the create icon (+) at the top of the [All Email Channels](#) list to create a new channel.
3. Enter the support email address for this channel and enter a display name for the channel.

You can also enter an address to send a blind carbon copy (BCC) of outbound messages you send as replies to this channel. Some organizations use this secondary address as another way to track support responses.

4. Choose [Case](#) for [Object](#).

When you choose [Case](#), the [Type](#) and [Default Account](#) fields appear.

5. Choose the case type, and the default account for this channel.

All cases created from this channel use the case type you select here. The default account is assigned to the case if the solution is unable to determine a unique account or customer based on the **from** address on the email message.

6. Select [Save](#) to save the new email channel.

The technical email address for your tenant appears on this screen once you save the channel.

7. Select [Activate](#) to set the channel to [Active](#) status.

A two-step verification process is required before the channel is activated. A message is sent to the channel email address with a verification code. A second message is sent from the verification service with a link you use to activate the channel.

Email channels only create cases from incoming messages when the channel status is set to [Active](#) and the forwarding rules are in place on your organization's mail server.

Once a channel is in [Active](#) status, you can edit all attributes of the channel, **except** the channel email address. For example, you can edit the channel name, display name, object, case type, or default account.

Related Information

[Configure Case Routing to Employee \[page 133\]](#)

[Configure Case Routing to Team \[page 134\]](#)

11.3.5.2 Create Email Templates

Email templates streamline email messaging with preset text, graphics, signatures, and attachments. Create templates for service responses, signatures, branding, and campaigns by uploading an HTML-based layout file.

Prerequisites

Prepare an HTML template layout. You can include technical placeholders to fill in data from the object record when the template is applied in the context of an associated object. See the steps following for how to find a list of placeholders.

Context

You can create email templates based on HTML layouts you create in your preferred HTML editor. Upload the HTML layout file and include any standard attachments for the template. You can also use placeholders to automatically pull data from associated objects in the system. For example: you can automatically fill in the customer name and ticket ID for a service response.

Procedure

1. Go to your user profile and select [Settings](#) [Email](#) [Templates](#).
2. Add a new template. Select the create icon (+) at the top of the [Templates](#) list to create a new template.
3. Enter a name for your new template.
4. Select a [Template Type](#):

- Response: use for service responses from your support teams.
- Signature: can include placeholders for employee information.
- Branding: create a custom brand-themed email template.
- Campaign: for newsletters and surveys.
- Approval Response Template: create a response template for approvals.
- Approval Template: create an approval request template.

The required fields for the template information vary depending on the type of template you select. Some types are only available for email channels. You can enter a description for response templates and campaign templates.

5. Select an [Object Type](#).
- Account
 - Contact
 - Individual Customer

- Case

Object type selection is limited based on the template type. Not all objects are available for every template type.

6. Select a *Channel Type* if necessary.

The channel type defaults to *Email* for most template types.

7. Set up the formatting for your template.

Depending on the type of template and object you selected in the previous steps, some or all of the following tabs are available for working with template formatting:

- *Viewer*: Shows a preview of the template as it appears in the email editor. Placeholders are shown where relevant data is filled in once the template is applied.
- *Documents*: Upload or change the HTML file defining the template layout here. Use any HTML editor you prefer to create your template layout.
- *Attachments*: Upload any documents you want to include as standard attachments with email messages created from this template.
- *Placeholder*: Displays a list of available data fields associated with the object type you selected for your template. Consult this list when creating your HTML layout file to insert placeholders for object data. Placeholders are formatted as the technical field name surrounded by double curly braces. For example: `{{familyName}}`, `{{givenName}}`.

Note

The Approval template has the `{{uiLink}}` placeholder for source document link. When approval email notification is received by the intended approvers, the approver can access the source document by clicking on the link in the email with manual login and then navigate to the source document approval step.

8. Save your changes when finished.

11.3.5.3 Monitor Inbound Email Messages

Use email monitoring to verify that inbound messages are received successfully.

Context

Identify technical errors with inbound email channels and resolve them to prevent problems with follow-on processes or with your business partner.

Procedure

1. Go to your user profile and select **Settings** > *Email* > *Monitor*.

2. Review the latest messages and check for any error messages.

Scroll the list and choose the page buttons to view additional messages.

11.4 Customer Hub

For real-time interactions, customer hub provides a place to access and enter customer information to help you resolve issues quickly.

Upon accepting an incoming call or chat request, customer hub opens automatically. If the interaction is from a known customer or contact, customer hub loads the relevant customer or contact information. The solution captures details of the interaction in the timeline.

- View customer details at the left of the screen.
- Scan interactions and notes in the [Timeline](#) tab.
- Access entity information, such as registered products, cases, and more in the [Objects](#) tab.
- Resolve issues quickly with the knowledge base pane at the right side of the screen.
- Create a new message or case quickly from the [What would you like to do?](#) area.

11.4.1 Timeline

The timeline shows phone call, e-mail, chat interactions, cases, and other touch points with the customer.

Select an item in the timeline to view more detailed information.

11.4.2 Objects

The [Objects](#) tab in customer hub shows entities in your solution related to the current customer such as cases, registered products, and so on.

Open the [Objects](#) tab to review related entities for the customer. Open an entity from the list to view details in a new tab.

11.5 Knowledge Base

Integrate an external knowledge base to enable agents to search for solutions and attach articles to responses without having to leave the agent desktop.

The knowledge base appears as a side panel on the agent desktop.

Search the knowledge base for relevant articles and preview the content in the side panel. Open the article in a new tab in the agent desktop (🔗), or view the article in the original knowledge base site in a new browser tab (🔗 [Open in help center](#)).

Use the [Copy](#) button (📋) to copy the article URL and paste into the interaction to share the article in your response back to your customer.

The email button (✉) creates a new email message with the article URL in the body of the message.

11.5.1 Configure Knowledge Base Mashup Integration

Use a mashup to integrate a knowledge base widget with your solution.

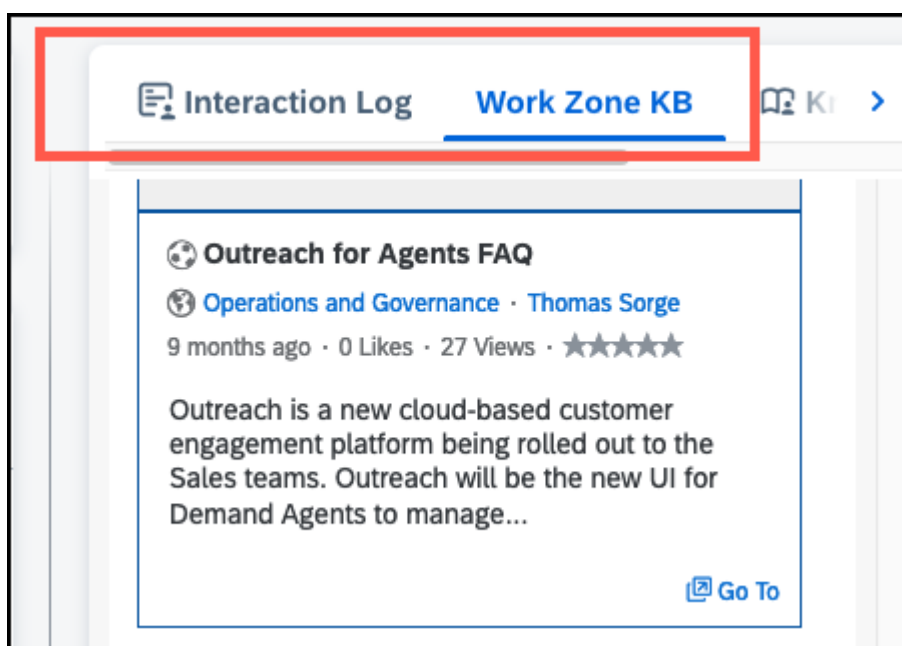
Context

At a high level, integrating a knowledge base as a mashup involves two processes: configure a mashup, then adapt the agent desktop screen to add the mashup.

Procedure

1. Request a widget URL from your knowledge base provider.
2. Create a mashup using the widget URL. See the related links section for detailed instructions on creating a mashup.
3. Adapt the agent desktop UI to add the knowledge base mashup to the screen. See the related links section for detailed instructions on adding a mashup as a tab.

We recommend that you add the mashup as a tab on the right-hand pane.



Related Information

[Mashups \[page 185\]](#)

[Create HTML Mashups \[page 186\]](#)

[Add Mashups as Tabs \[page 188\]](#)

11.5.2 Configure Knowledge Base Provider

Configure your knowledge base parameters.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Knowledge Base](#) ► [Provider Settings](#) ►.
2. Select [SAP MINDTOUCH](#) under [Knowledgebase Providers](#).
3. Enter the required parameters:
 - [Site URL](#) - API URL
 - [User](#) - user name for account
 - [Key](#)
 - [Secret](#)
4. Set the [Active](#) switch to on and save the configuration.

11.6 Session Information

View session details, change customer, check linked items, and make notes for this interaction.

The Session information panel on the right of the Agent Desktop screen shows the current session details passed from your communication system. It also includes the following capabilities:

- [Wrong Customer?](#) In case the initial customer identification was wrong customer, this option opens the customer search screen to find and select another customer.
- [Interaction](#): shows the interaction type.
- Enter the [Subject](#) of the interaction, and add or edit the phone number.
- The [Direction](#) of the call appears ([Inbound](#) or [Outbound](#)).
- The [Start Date & Time](#) and [End Date & Time](#) appear here.
- [Notes](#): enter your notes for this interaction. Notes are available to remind you or subsequent agents of details communicated to the customer in this interaction.
- Enter a [Category](#) and sub-category for the interaction.
- [Linked Items](#): View and add cases and registered products related to this interaction.

11.7 Widget Integration

Communication channels integrate to SAP Service Cloud Version 2 with a UI widget.

Agent desktop currently supports customer interactions through the following channels:

- Phone
- Chat
- SMS

→ Remember

- You need to contract with a third-party communication provider for communication channels.
- Your communication provider supplies you with a responsive widget that can be embedded into the agent desktop UI.
- The widget provides a front-end user interface for handling channels like phone, chat, and SMS.
- Routing interactions to agents is handled by the channel provider.

Integration between the vendor and service cloud is done on the client-side. That is, the widget communicates with SAP Service Cloud Version 2 through browser events using `window.postMessage()`.

To notify SAP Service Cloud Version 2 of an incoming message, the widget passes a set of parameters in the form of payload and then pushes the information using `postMessage`.

The following is sample code showing the payload construct for incoming phone call.

Sample Code

```
/**
 * construct the payload in XML format
 * @param sDestination
 * @param sOriginator
 * @returns payload in xml format
 * @private
 */
c4c.cti.integration.prototype._formXMLPayload = function(parameters){
var sPayload = "<?xml version='1.0' encoding='utf-8' ?> <payload> ";
for(var key in parameters){
var tag = "<" + key + ">" + parameters[key] + "</" + key + ">";
sPayload = sPayload + tag;
}
sPayload = sPayload + "</payload>";
return sPayload;
};
/**
 * Send information to Service Cloud
 * @param parameters
 */
c4c.cti.integration.prototype.sendIncomingCalltoC4C = function (parameters) {
var payload = this._formXMLPayload(parameters);
this._doCall(payload);
};
/**
 * post to parent window
 * @param sPayload
 * @private
 */
c4c.cti.integration.prototype._doCall = function (sPayload) {
window.parent.postMessage(sPayload, "*");
};
```

```
};
```

i Note

Format the payload as JSON or XML.

[Phone \[page 100\]](#)

Integration events and payloads for the phone channel.

[Chat \[page 103\]](#)

Integration events and payloads for the chat channel.

[SMS \[page 104\]](#)

Integration events and payloads for the SMS channel.

Related Information

[Configure Communication Provider \[page 84\]](#)

11.7.1 Phone

Integration events and payloads for the phone channel.

Notify

Your communication provider uses the **notify** event to alert the agent of an incoming phone call. Typically, the notify event is used when a call is assigned to a specific agent by a routing algorithm.

The notify event doesn't record a phone call activity in SAP Service Cloud Version 2.

A sample payload follows:

Sample Code

```
payload :-  
{  
  "Type": "CALL",  
  "EventType": "INBOUND",  
  "Action": "NOTIFY",  
  "ANI": "+358403573592",  
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",  
}
```

ExternalReferenceID specifies the interaction record in your third-party channel provider system. The activity document created in SAP Service Cloud Version 2 links to this external reference ID.

Accept

Your communication provider uses the **accept** event to indicate that an agent has accepted a call. Typically, the accept event is used when a call is connected between an agent and a caller.

SAP Service Cloud Version 2 creates a phone call activity along with start date and time upon receipt of the **accept** event.

A sample payload follows:

Sample Code

```
payload :-  
{  
  "Type": "CALL",  
  "EventType": "INBOUND",  
  "Action": "ACCEPT",  
  "ANI": "+358403573592",  
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",  
}
```

End

Your communication provider uses the **end** event to indicate that the agent has ended the call. Typically, the end event is used when a call is disconnected between an agent and a caller.

SAP Service Cloud Version 2 updates the phone call activity with the end date and time upon receipt of the **end** event.

A sample payload follows:

Sample Code

```
payload :-  
{  
  "Type": "CALL",  
  "EventType": "UPDATEACTIVITY",  
  "Action": "END",  
  "ANI": "+358403573592",  
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",  
}
```

Recording ID

The recording ID is used to link the recorded phone call with the activity document created in SAP Service Cloud Version 2. The recording ID ties the interaction record to the recording and enabled users to trigger playback directly from SAP Service Cloud Version 2.

Note

Phone recording and playback functions are provided by the widget.

Sample Code

```
payload :- {
  "Type": "CALL",
  "EventType": "UPDATEACTIVITY",
  "Action": "END",
  "ANI": "+358403573592",
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",
  "RecordingId": "REC_67F9EA535AE0EA1180E000505687F2F"
}
```

RecordingId is a unique identifier generated in the vendor system to reference the recorded phone call. To store the recording link, pass the RecordingId along with the phone call **end** event.

Playback

To play back a recorded phone call, SAP Service Cloud Version 2 launches the widget and sends a payload with the recording identifiers. Your communication channel provider interprets the payload and starts playing the specified recording in the widget.

Sample playback payload from SAP Service Cloud Version 2 to widget:

Sample Code

```
{
  type: "recordingPlayback"
  InteractionId: "67F9EA535AE0EA1180E000505687F2FD" -> external reference
  RecordingId: "REC_67F9EA535AE0EA1180E000505687F2F" -> Recording ID received
  by CCTR
}
```

Outbound Call

SAP Service Cloud Version 2 offers click-to-call functionality for phone numbers. For example, on the customer detail screen, agents can select a phone number to trigger a phone call to the customer. To initiate the outbound phone call, the solution sends a payload to the widget as follows:

Sample Code

```
{
  Direction: "OUT",
  PhoneNumber: "+1 6503054000",
  ActivityUUID: "00163EA718FC1EEABDD5608C895BDC71"
}
```

The widget uses the phone number parameter to trigger an outbound call upon receipt of the payload.

ActivityUUID is the identifier for the phone call activity created in SAP Service Cloud Version 2 that can be used by your communication provider to link the interaction record in their system.

11.7.2 Chat

Integration events and payloads for the chat channel.

Notify

Your communication provider uses the **notify** event to alert the agent of an incoming chat. Typically, the notify event is used when a chat is assigned to a specific agent by a routing algorithm.

The notify event doesn't record a chat activity in SAP Service Cloud Version 2.

A sample payload follows:

Sample Code

```
payload :- {  
  "Type": "CHAT",  
  "EventType": "INBOUND",  
  "Action": "NOTIFY",  
  "Email": "Mike.Ross@test.com",  
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",  
}
```

ExternalReferenceID specifies the interaction record in your third-party channel provider system. The activity document created in SAP Service Cloud Version 2 links to this external reference ID.

Accept

Your communication provider uses the **accept** event to indicate that an agent has accepted a chat request. Typically, the accept event is used when an agent begins chatting with a customer.

SAP Service Cloud Version 2 creates an inbound chat activity along with start date and time upon receipt of the **accept** event.

A sample payload follows:

Sample Code

```
payload :- {  
  "Type": "CHAT",  
  "EventType": "INBOUND",  
  "Action": "ACCEPT",  
  "Email": "Mike.Ross@test.com",  
}
```

```
"ExternalReferenceID":"ED4E4730D1C711EAAA5EBC019352B05E",
}
```

End (Update Transcript)

Your communication system uses the **end** event to upload a transcript with a chat activity. When an agent concludes a chat interaction with a customer, the external communication system uploads the transcript to the chat activity and records the ending time and date.

SAP Service Cloud Version 2 supports only plain text for transcripts. Your external communication system must convert transcripts to plain text before uploading.

A sample payload follows:

Sample Code

```
payload :-
{
  "Type": "CHAT",
  "EventType": "UPDATEACTIVITY",
  "Action": "END",
  "Email": "Mike.Ross@test.com",
  "ExternalReferenceID": "ED4E4730D1C711EAAA5EBC019352B05E",
  "Transcript": "support@sap.com : Hello Mike\n Mike.Ross@test.com: Hello\n Mike.Ross@test.com : I need to book an appointment for fixing my washing machine\n support@sap.com : Sure, will tomorrow 12.00 PM work for you? \n Mike.Ross@test.com: yes thanks bye\n support@sap.com : bye"
}
```

Transcript – Contains the plain text chat transcript.

11.7.3 SMS

Integration events and payloads for the SMS channel.

Notify

Your communication provider uses the **notify** event to alert the agent of an incoming SMS message. Typically, the notify event is used when a chat is assigned to a specific agent by a routing algorithm.

The notify event doesn't record an SMS chat activity in SAP Service Cloud Version 2.

A sample payload (formatted JSON data) follows:

Sample Code

```
payload :- {
  "Type": "MESSAGE",
```



```

"EventType": "INBOUND",
"Action": "NOTIFY",
"ANI": "+30040300597",
"Text": "This is SMS text",
"ExternalReferenceID": "77FB2990D24411EAAA5EBC019352B05E"
}

```

ExternalReferenceID specifies the interaction record in your third-party channel provider system. The activity document created in SAP Service Cloud Version 2 links to this external reference ID.

Accept

Your third-party communication system uses the **accept** event to indicate to SAP Service Cloud Version 2 that an SMS message has been accepted by an agent for further processing.

The accept event creates a new Message/SMS activity in SAP Service Cloud Version 2, and records the time and date.

Sample Code

```

payload :- {
  "Type": "MESSAGE",
  "EventType": "INBOUND",
  "Action": "ACCEPT",
  "ANI": "+30040300597",
  "Text": "This is SMS text",
  "ExternalReferenceID": "77FB2990D24411EAAA5EBC019352B05E",
}

```

End (Update Transcript)

Your communication system uses the **end** event to upload a transcript with a messaging activity. When an agent concludes a chat-like interaction with a customer using SMS, the external communication system can upload the entire transcript to one messaging activity.

SAP Service Cloud Version 2 supports only plain text for transcripts. Your external communication system must convert transcripts to plain text before uploading.

A sample payload (formatted JSON data) follows:

Sample Code

```

payload :- {
  "Type": "MESSAGE",
  "EventType": "UpdateActivity",
  "Action": "END",
  "ExternalReferenceID": "77FB2990D24411EAAA5EBC019352B05E",
  "Transcript": "+30040300597 : Hi\\nhelp@ssupport.com : Hi\\nhelp@ssupport.com :  
how you doing?\\n+30040300597 : Please reschedule appt to  
tomorrow.\\nhelp@ssupport.com :Sure! Will look into that."
}

```

Transcript – Contains the plain text SMS transcript

11.8 Integrate External Systems into Agent Desktop with Mashups

Use the screen adaptation function of Agent Desktop and the mashup capabilities built into SAP Service Cloud Version 2 to integrate external system functions into Agent Desktop. The example that follows integrates SAP S/4HANA or SAP S/4HANA Cloud Service Management to create service orders. As part of the standard integration, only service order is supported.

Overview

Incorporate a back-office application, such as SAP S/4HANA Cloud Service Management or SAP S/4HANA, into the front-office interface of SAP Service Cloud Version 2 Agent Desktop using a mashup. This example integration enables your organization to create and display service orders in SAP S/4HANA or SAP S/4HANA Cloud from within the Agent Desktop screen using the standard SAP S/4HANA or SAP S/4HANA Cloud Service Management UI screens.

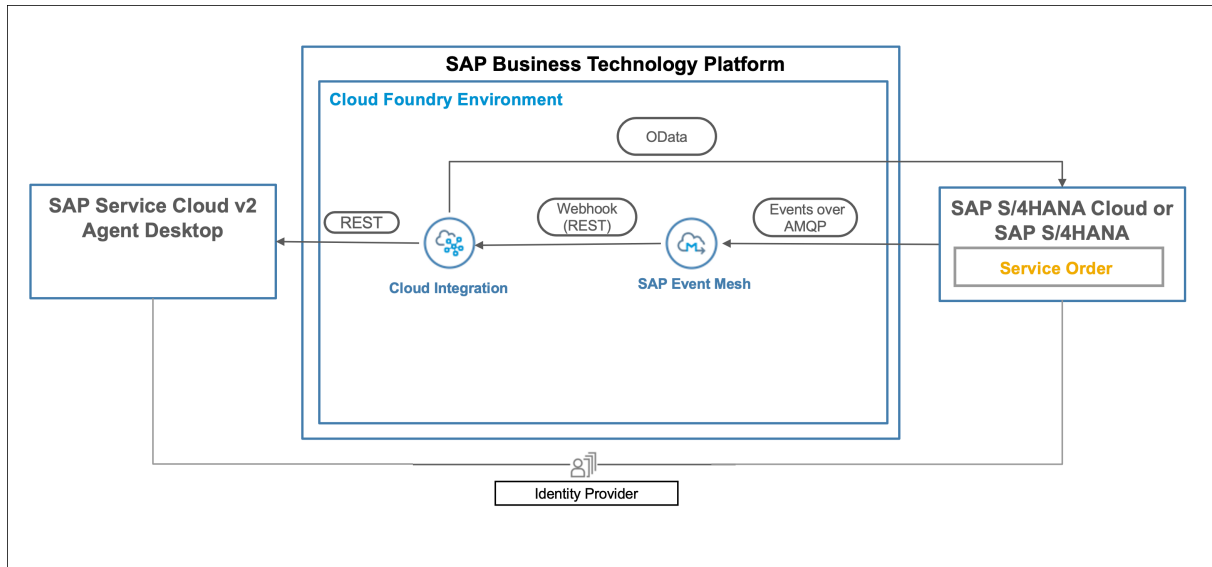
Prerequisites

You require the following software components for this example integration:

- SAP Service Cloud Version 2
- SAP Agent Desktop
- SAP S/4HANA or SAP S/4HANA Cloud Service Management
- SAP Event Mesh
- SAP Cloud Integration

Architecture and Design

The following diagram illustrates this integration architecture:



Related Information

[SAP Event Mesh documentation](#)

[SAP Cloud Integration documentation](#)

11.8.1 Configure SAP Event Mesh

Configure SAP Event Mesh for integration between SAP S/4HANA or SAP S/4HANA Cloud and SAP Service Cloud Version 2, Agent Desktop.

Context

Log on as an administrator to configure this integration.

Procedure

1. Go to the Cloud Cockpit of your BTP sub-account.

2. Navigate to the [Instance and Subscriptions](#) option.
3. Under [Instances](#), create a new [Service Instance](#) of the [Event Mesh](#) service, with a **Default** plan.

Provide the JSON parameters for the instance following this example:

Sample Code

```
{
  "options": {
    "management": true,
    "messagingrest": true,
    "messaging": true
  },
  "rules": {
    "topicRules": {
      "publishFilter": [
        "${namespace}/*"
      ],
      "subscribeFilter": [
        "${namespace}/*"
      ]
    },
    "queueRules": {
      "publishFilter": [
        "${namespace}/*"
      ],
      "subscribeFilter": [
        "${namespace}/*"
      ]
    }
  },
  "version": "1.1.0",
  "emname": "<Unique Instance name>",
  "namespace": "<Unique namespace value>"
}
```

Make a note of the [Namespace](#) for this service instance. You need it later for configuring the [Enterprise Event Enablement](#) in SAP S/4HANA Cloud.

4. Create a [Service Key](#) for this newly created event mesh instance.

Make a note of the **Client ID** and **Secret** of this key under the **amqp10ws** protocol. You need it later for configuring the [Communication System](#) in SAP S/4HANA Cloud.

5. Launch the [Event Mesh Administrator](#) user interface to create a subscription to the **Event Mesh** application.
6. Choose the service instance you created previously.
Find the service instance listed under [Message Clients](#).
7. Go to the [Queues](#) tab, and create a new queue. Prefix the queue name with the namespace you created and noted previously.

Once the queue is created, you can create subscriptions to the SAP S/4HANA Cloud outbound topics corresponding to service order events in the SAP S/4HANA Cloud solution.

8. Add subscriptions to your queue: select the queue you created in the previous step, and go to [Actions](#) [Create Subscriptions](#).

For events you want to relay to the agent desktop timeline, create subscriptions using the following pattern:

<Message Client Namespace>**ce**<S4 Hana outbound topic name>

For example: `s4toc4/v2/srvo/ce/sap/s4/beh/serviceorder/v1/ServiceOrder/Completed/v1`

Other events supported currently are:

- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Created/v1`
- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Released/v1`

9. Create a webhook in the event mesh instance. Go to the instance [WebHooks](#) tab. Create a new webhook and point it to the integration flow endpoint by providing the required authentication.

Related Information

[SAP Event Mesh documentation](#)

11.8.2 Configure SAP S/4HANA Cloud

Configure SAP S/4HANA Cloud for integration with SAP Service Cloud Version 2, Agent Desktop.

Context

Procedure

1. Log on to your SAP S/4HANA Cloud solution and go to [Communication Management](#).
2. Create a new communication arrangement for the communication scenario: SAP_COM_0092 (Enterprise Eventing Integration).

i Note

You also need to create two more communication arrangements once you finish this one.

3. Enter the value you noted for the event mesh instance namespace under [Additional Properties](#).
4. Enter a unique value for the property: [Channel](#).

Make a note of the channel property value. You need it later to configure the channel binding.

5. Create a [Communication System](#) with the host name set to your event mesh instance.

For example: **enterprise-messaging-messaging-gateway.cfapps.sap.hana.ondemand.com:443**

6. Enter the authentication and token endpoint in the **OAuth 2.0** settings. Use the **Service Key** token endpoint value that you created previously.
7. Enter the OAuth 2.0 client ID and client secret corresponding to the service key `amqp` protocol under [Outbound communication](#).

8. Enter your mesh instance with the `amqp` protocol under *Outbound services*.

For example: **`https://enterprise-messaging-messaging-gateway.cfapps.sap.hana.ondemand.com:443/protocols/amqp10ws`**

9. Activate the communication arrangement.

If you can't activate the communication arrangement, check and correct your entries made in the previous steps. Continue this procedure when the communication arrangement status is active.

10. Go to *Enterprise Event Enablement* *Configure Channel Binding* and search for the communication arrangement, using the value you entered for the **Channel** property.

Verify that the communication arrangement status is **Active**. If the status is anything other than **Active**, try reactivating the communication arrangement.

11. Go to the channel details and create *Outbound Topics* corresponding to the service order events that need to be relayed to the Agent Desktop timeline.

The topics configured in this channel must be the same as the subscriptions you created in the event mesh instance.

Events supported currently are:

- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Created/v1`
- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Released/v1`
- `/sap/s4/beh/serviceorder/v1/ServiceOrder/Completed/v1`

12. Go to **Administration** **Security** **Maintain Protection Allowlists**. In the *Clickjacking Protection* tab, click on the **+** icon and add the *Trusted Host Name, Schema, and Port* for your SAP Service Cloud tenant.

This step allows trusted access for SAP Service Cloud and Agent Desktop to your S/4 HANA Cloud tenant.

13. Create a new communication arrangement for the communication scenario: `SAP_COM_0350` (Service Order OData Integration).

This communication arrangement invokes the **Service Order OData API** from the integration flow. The log in credentials for the inbound user configured in this communication arrangement must be maintained in the integration flow.

14. Create a Communication Arrangement for the Communication Scenario: `SAP_COM_0008` (Business Partner, Customer, and Supplier Integration).

This communication arrangement invokes the **Business Partner OData API** from the integration flow.

11.8.3 Configure SAP S/4HANA

Configure SAP S/4HANA for integration with SAP Service Cloud Version 2, Agent Desktop.

Using the enterprise event enablement feature of SAP S/4HANA, events can be exchanged between SAP S/4HANA and event mesh. In order to enable SAP S/4HANA systems for enterprise messaging you need to create channels and maintain outbound event topic bindings for the channel. For more information, see [Creating, Configuring and Managing Channels](#)

11.8.4 Configure SAP Cloud Integration

Configure and deploy the SAP Cloud Integration flow (iFlow) between SAP Service Cloud Version 2, Agent Desktop, and SAP S/4HANA or SAP S/4HANA Cloud.

11.8.4.1 Configure iFlow Parameters for SAP S/4HANA Cloud to SAP Service Cloud Service Order Integration

Prerequisites

As a prerequisite, go to ► [Operations View](#) ► [Security Material](#) ► [Add](#) ► [User Credentials](#) ► and deploy the following:

- User credentials to connect to SAP S/4HANA Cloud system
- User credentials to connect to SAP Service Cloud Version 2.

Configure the iFlow Parameters

Configure the iFlow parameters for the SAP S/4HANA Cloud to SAP Service Cloud integration.

- Configure Sender

Parameter	Description
Sender	Select S4_HANA_Cloud
Adapter Type	HTTPS
Address	<Webhook configured in event mesh>
Authorization	User Role
User Role	ESBMessaging.send

- Configure Receiver
 - S4_HANA_Cloud

Parameter	Description
Adapter Type	HTTP
S4 host URL	<SAP S/4HANA Cloud host URL>
Proxy Type	Internet

Parameter	Description
Credential Name	<User credential created for connecting to SAP S/4HANA Cloud>
User Role	ESBMessaging.send
Timeout (in ms)	60000

- Service_Cloud_V2

Parameter	Description
Adapter Type	HTTP
C4C host URL	<SAP Service Cloud V2 host URL>
Credential Name	<User credential created for connecting to SAP Service Cloud V2>
Timeout (in ms)	60000

- More configurations

Parameter	Description
Type	All Parameters
Agent Desktop Add-on	• False: For standalone scenario • True: For side by side scenario
Receiver Communication	Default communication system of Service Cloud version 2
Sender Communication	Communication system of Service Cloud version 2 which is configured for the communication configuration Integrate Service Entities with SAP S/4HANA Cloud

11.8.4.2 Configure iFlow Parameters for SAP S/4HANA to SAP Service Cloud Service Order Integration

Prerequisites

As a prerequisite, go to [Operations View](#) [Security Material](#) [Add](#) [User Credentials](#) and deploy the following:

- User credentials to connect to SAP S/4HANA system
- User credentials to connect to SAP Service Cloud Version 2.

Configure the iFlow Parameters

Configure the iFlow parameters for the SAP S/4HANA to SAP Service Cloud integration.

- Configure Sender

Parameter	Description
Sender	Select S4
Adapter Type	HTTPS
Address	<Webhook configured in event mesh>
Authorization	User Role
User Role	ESBMessaging.send

- Configure Receiver

- S4_HANA

Parameter	Description
Adapter Type	HTTP
S4 host URL	<SAP S/4HANA host URL>
Proxy Type	On-premise
Credential Name	<User credential created for connecting to SAP S/4HANA>
User Role	ESBMessaging.send
Timeout (in ms)	60000

- Service_Cloud_V2

Parameter	Description
Adapter Type	HTTP
C4C host URL	<SAP Service Cloud V2 host URL>
Credential Name	<User credential created for connecting to SAP Service Cloud V2>
Timeout (in ms)	60000

- More configurations

Parameter	Description
Type	All Parameters
Agent Desktop Add-on	• False: For standalone scenario • True: For side by side scenario
Receiver Communication	Default communication system of Service Cloud version 2

Parameter	Description
Sender Communication	Communication system of Service Cloud version 2 which is configured for the communication configuration Integrate Service Entities with SAP S/4HANA Cloud

11.8.5 Configure SAP Service Cloud

Configure SAP Service Cloud, Service Agent Desktop for integration with SAP S/4HANA Cloud.

Prerequisites

Master data replication of **Business Partner (BP)** in SAP S/4HANA or SAP S/4HANA Cloud to **Customer** in SAP Service Cloud.

For this integration, every customer record in SAP Service Cloud requires a corresponding remote business partner in S/4HANA Cloud Service Management.

Context

! Restriction

Individual customers aren't supported currently.

Procedure

1. Review Section 4 of the guide: **Setting Up Opportunity-to-Order with SAP Cloud for Customer**.
Find a link to this guide in the Related Links section.
2. Follow the procedure in section 4 to set up master data integration for business partners (BP).

Related Information

[Setting Up Opportunity-to-Order with SAP Cloud for Customer](#) 

11.8.6 Configure Service Agent Desktop

Configure SAP Service Cloud, Agent Desktop for integration with SAP S/4HANA or SAP S/4HANA Cloud Service Management.

Context

An inbound communication configuration linking the Event Bridge service to the Inbound entity is delivered with the connector-service. Create the following artifacts for this inbound communication configuration:

- Communication System
- Communication Configuration

Procedure

1. Go to [User](#) > [Settings](#) > [Integration](#) > [Communication Configuration](#).
2. Choose [Integrate Service Entities with SAP S/4HANA Cloud](#).
3. Select [Copy](#) icon.
4. Create a new communication configuration for the communication system you created.
5. Activate the communication configuration you created.

You don't need an outbound configuration for this integration.

11.8.7 Configure Mashup: Create Scenario

Configure a create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA or SAP S/4HANA Cloud.

Prerequisites

- The S/4HANA or S/4HANA Cloud URLs that are included in the mashup need to be added to a list of allowed URLs in SAP Service Cloud Version 2. You can do this by navigating to your user profile on the top-right corner of your screen and click [Settings](#) > [All Settings](#) > [General](#) > [Content Security Policy Setting](#) and scroll to the section [Frame Source](#) and add the S/4HANA Cloud hostname URLs.

❖ Example

If the Service Order Create URL is `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create` then, add `https://my123456.s4hana.ondemand.com/` in the [Frame Source](#).

Similarly, add the login URL of S/4HANA Cloud system in the Frame Source.

- Assign the customer service manager role `SAP_BR_CUSTOMER_SERVICE_MGR` to the user in S/4HANA.
- Assign the business catalog `SAP_CRM_BC_SERV_PROCCORDER_PC` to the user in S/4HANA Cloud Service Management.

For further details on assigning business catalogs to S/4HANA Cloud users, see the related links section following the steps.

Context

Procedure

1. Go to **User** > **Settings** > **Foundation** > **Mashup Authoring**.
2. Create a new mashup by clicking the **+ Create** button.
3. Under the **General Information** section, enter a mashup name and description. Use a meaningful title. For example: Create Service Order.
4. Under the **Input Parameters** section, enter the service order create URL of your SAP S/4HANA or SAP S/4HANA Cloud tenant in the **URL** field.
 - Example URL for SAP S/4HANA: `https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-create?sap-processtype=SRV0&sap-soldtoparty=171000030`
 - Example URL for SAP S/4HANA Cloud: `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create?sap-processtype=SV01&sap-soldtoparty=17100004`
5. Click **Extract Parameters**.
6. Under the **Parameter, Constant and API Key** section, ensure that the **sap-processtype** is set to SV01 and **sap-soldtoparty** is blank. You configure the parameter binding of the sap-soldtoparty by using the extensibility feature in Agent Desktop.

Post extraction of the parameters from the URL, the section should appear as:

Input Parameters

URL *

https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-create?sap-processtype=SRV0&sap-soldtoparty=171000030

Parameter, Constant and API Key

sap-processtype	• SV01	• ☐ Mandatory	• No System Parameter a...
sap-soldtoparty	• No Constant added	• ☐ Mandatory	• No System Parameter a...

7. [Save](#) and activate your mashup.

- Ensure to create a new mashup to use with agent desktop integration.
- Set the screen binding parameters in Agent Desktop via Extensibility and not in Mashup authoring.

Related Information

[Add Business Catalogs to Business Users in SAP S/4HANA Cloud](#)

11.8.8 Configure Mashup: Display Scenario

Configure a display scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Prerequisites

- Assign the customer service manager role `SAP_BR_CUSTOMER_SERVICE_MGR` to the user in S/4HANA.
- Assign the business catalog `SAP_S4CRM_BC_SRVC_DSP_PC` to the user in S/4HANA Cloud Service Management.

For further details on assigning business catalogs to S/4HANA Cloud users, see the related links section following the steps.

Procedure

1. Go to **User > Settings > Foundation > Mashup Authoring**.
2. Create a new mashup by clicking the **+ Create** button.
3. Under the **General Information** section, enter a mashup name and description. Use a meaningful title. For example: Display Service Order.
4. Under the **Input Parameters** section, enter the service order display URL of your SAP S/4HANA or SAP S/4HANA Cloud tenant in the **URL** field.
 - Example URL for SAP S/4HANA: `https://my12345.org/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html?sap-client=210&sap-ushell-config=headerless#ServiceOrder-display?ServiceOrder=ID`
 - Example URL for SAP S/4HANA Cloud: `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-display&/C_SrvOrdDocListRptAndObjPg' {ID1}' /?FCLLayout=MidColumnFullScreen`

5. Click [Extract Parameters](#).
6. Under the [Parameter, Constant and API Key](#) section, ensure that the [FCLLayout](#) is set to `MidColumnFullScreen` and `{ID}` is blank. You configure the parameter binding of the `{ID}` by using the extensibility feature in Agent Desktop.

Post extraction of the parameters from the URL, the section should appear as:

Input Parameters

URL *

https://.../ui?sap-ushell-config=headerless#ServiceOrder-display&/C_SrvOrdDocListRptAndObjPg('{ID1}')/

Parameter, Constant and API Key

FCLLayout	• MidColumnFullScreen	• ☐ Mandatory	• No System Parameter a...
{ID1}	• No Constant added	• ☐ Mandatory	• No System Parameter a...

7. [Save](#) and activate your mashup.
 - Ensure to create a new mashup to use with agent desktop integration.
 - Set the screen binding parameters in Agent Desktop via Extensibility and not in Mashup authoring.

Related Information

[Add Business Catalogs to Business Users in SAP S/4HANA Cloud](#)

11.8.9 Adapt Agent Desktop for Mashup

11.8.9.1 Service Order

11.8.9.1.1 Adapt Agent Desktop for Mashup: Create Service Order

Adapt the agent desktop screen to include the create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Context

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to [User](#) [Start Adaptation](#) .
4. Select the [What would you like to do?](#) section and select the **Edit Section** icon () to open the [Section Items](#) pop-up box.
5. Select [Create](#) and click the **Create Button** icon ().
6. In the [Add Buttons](#) dialog box, enter the [Button Text](#) as **Service Order** and enter an [Event Name](#).
7. In the Parameters section, ensure the parameter binding of sap-soldtoparty is set to [Fields/External Business Partner - Default ID](#) and click [Apply](#).

This links the service order to a customer account.
8. Select the tabs section and click the icon () to open the [Edit Tabs](#) pop-up box.
9. Click the **Create Tab** icon () and provide a new tab name, for example: Create Service Order and press Enter.
10. Click the icon () to launch Edit Tabs pop-up dialog again. Select the event you provided for service order create and click [Apply](#).
11. Click the icon () under the Create Service Order screen area.
12. Click [Add Mashup](#) and select the mashup that you created for create service order.
13. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that sap-soldtoparty is set to [accountInfo/externalReferenceBusinessPartnerId](#) and click [Apply](#).
14. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

11.8.9.1.2 Adapt Agent Desktop for Mashup: View Service Order

Adapt the agent desktop screen to include the View Service Order Details mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Context

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to ► [User](#) ► [Start Adaptation](#) ►
4. Select the tabs section and click the icon (✎) to open the [Edit Tabs](#) pop-up box.
5. Click the **Create Tab** icon (+) and provide a new tab name, for example: Display Service Order and press Enter.
6. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service order display and click [Apply](#).
7. Click the icon (✎) under the Display Service Order screen area.
8. Click [Add Mashup](#) and select the mashup that you created for display service order.
9. In the Edit Properties window, set the value of [Height\(%\)](#) to 100 and ensure that {ID} is set to [s4ServiceOrderInfo/serviceOrderId](#) and click [Apply](#).
10. Click [End Session](#) to exit the adaptation mode and to view your adaptations on the screen.

11.8.9.1.3 Create an SAP S/4HANA Service Order from Agent Desktop Mashup

Create an SAP S/4HANA Service Order from Agent Desktop mashup by connecting to SAP Service Cloud Version 2.

Context

Now that the mashup is configured, and the Agent Desktop screen is adapted to show the mashup, service agents can create a service order directly in the back-office SAP S/4HANA Cloud Service Management application from Agent Desktop screen:

Procedure

1. Log in to your SAP Service Cloud Version 2 solution.
2. Choose ► [Service](#) ► [Agent Desktop](#) ►.
3. [Search](#) for an account and click [Confirm](#).
4. Under the [Customer Hub](#) tab, choose ► [Create](#) ► [Service Order](#) ►.

The [Service Order](#) tab loads the SAP S/4HANA Cloud system's [Create Service Order](#) screen using the mashup.

5. Enter a description and fill-in other details in the service order and click [Save](#).
The [Timeline](#) tab opens with a new [Service Order](#) created event on the timeline.
6. To see a quick view of service order, click on the [Service Order ID](#) in the timeline event.

To see a detailed view of the service order in new tab in the [Customer Hub](#), click the  icon.

i Note

Only users with a role that has access to the service sap.crm.service.s4hcServiceOrderService can view the Service Order quick view. You can assign business roles under ► [User](#) ► [Settings](#) ► ► [Users and Control](#) ►

11.8.9.2 Service Quotation

11.8.9.2.1 Create Service Quotation

Adapt the agent desktop screen to include the create scenario mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Procedure

1. Go to [Agent Desktop](#).
2. Under the [Search](#) tab, [Confirm](#) an account and navigate to the [Customer Hub](#) tab.
3. Go to ► [User](#) ► [Start Adaptation](#) ►
4. Select the [What would you like to do?](#) section and select the **Edit Section** icon () to open the [Section Items](#) pop-up box.
5. Select [Create](#) and click the **Create Button** icon () .
6. In the [Add Buttons](#) dialog box, enter the [Button Text](#) as **Service Quotation** and enter an [Event Name](#).

7. In the Parameters section, ensure the parameter binding of `sap-soldtoparty` is set to *Fields/External Business Partner - Default ID* and click *Apply*.

This links the service order to a customer account.

8. Select the tabs section and click the icon (✎) to open the *Edit Tabs* pop-up box.
9. Click the **Create Tab** icon (+) and provide a new tab name, for example: Create Service Quotation and press Enter.
10. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service quotation create and click *Apply*.
11. Click the icon (✎) under the Create Service Quotation screen area.
12. Click *Add Mashup* and select the mashup that you created for create service quotation.
13. In the Edit Properties window, set the value of *Height(%)* to 100 and ensure that `sap-soldtoparty` is set to *accountInfo/externalReferenceBusinessPartnerId* and click *Apply*.
14. Click *End Session* to exit the adaptation mode and to view your adaptations on the screen.

11.8.9.2.2 View Service Quotation

Adapt the agent desktop screen to include the View Service Quotation Details mashup in SAP Service Cloud Version 2 for integration with SAP S/4HANA Cloud.

Context

Procedure

1. Go to *Agent Desktop*.
2. Under the *Search* tab, *Confirm* an account and navigate to the *Customer Hub* tab.
3. Go to ► *User* ► *Start Adaptation* ►
4. Select the tabs section and click the icon (✎) to open the *Edit Tabs* pop-up box.
5. Click the **Create Tab** icon (+) and provide a new tab name, for example: Display Service Quotation and press Enter.
6. Click the icon (➤) to launch Edit Tabs pop-up dialog again. Select the event you provided for service quotation display and click *Apply*.
7. Click the icon (✎) under the Display Service Quotation screen area.
8. Click *Add Mashup* and select the mashup that you created for display service quotation.
9. In the Edit Properties window, set the value of *Height(%)* to 100 and ensure that `{ ID }` is set to *s4ServiceOrderInfo/serviceOrderId* and click *Apply*.
10. Click *End Session* to exit the adaptation mode and to view your adaptations on the screen.

11.8.9.2.3 Create an SAP S/4HANA Service Quotation from Agent Desktop Mashup

Create an SAP S/4HANA Service Quotation from Agent Desktop mashup by connecting to SAP Service Cloud Version 2.

Context

Now that the mashup is configured, and the Agent Desktop screen is adapted to show the mashup, service agents can create a service quotation directly in the back-office SAP S/4HANA Cloud Service Management application from Agent Desktop screen:

Procedure

1. Log in to your SAP Service Cloud Version 2 solution.
2. Choose **► Service ► Agent Desktop ►**.
3. **Search** for an account and click **Confirm**.
4. Under the **Customer Hub** tab, choose **► Create ► Service Quotation ►**.

The **Service Quotation** tab loads the SAP S/4HANA Cloud system's **Create Service Quotation** screen using the mashup.

5. Enter a description and fill-in other details in the service quotation and click **Save**.

The **Timeline** tab opens with a new **Service Quotation** created event on the timeline.

6. To see a quick view of service quotation, click on the **Service Quotation ID** in the timeline event.

To see a detailed view of the service quotation in new tab in the **Customer Hub**, click the  icon.

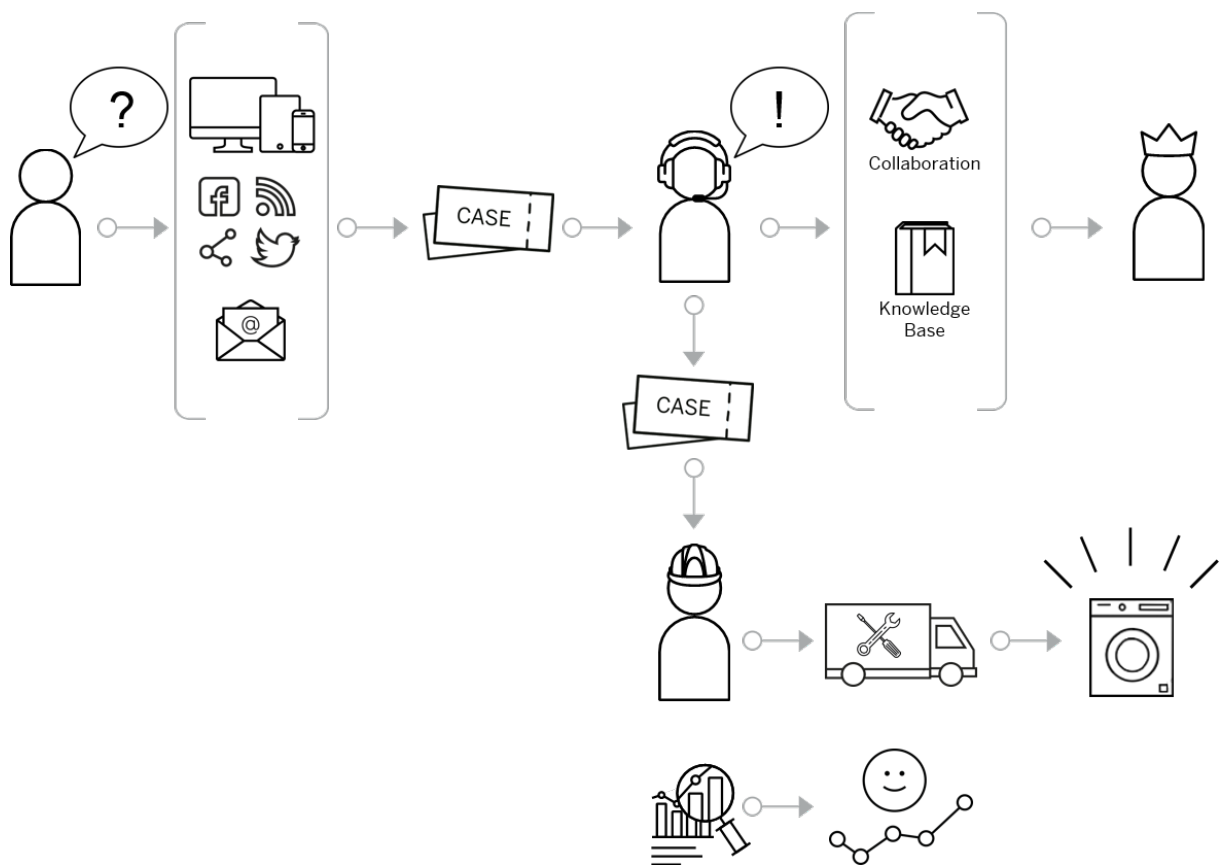
i Note

Only users with a role that has access to the service `sap.crm.service.s4hcServiceOrderService` can view the Service Quotation quick view. You can assign business roles under **► User ► Settings ► ► Users and Control ►**

12 Cases

Learn how the system creates cases, how to work with the case list, and how to respond to cases to resolve customer issues.

Cases are a record of a request for some type of service or support from an account, customer, or employee. This record allows you to track interactions with the service requester. Cases also record details like how much time has passed since the case was created, what actions were taken to resolve the issue, priority, associated products, or warranties, and much more. Cases are sometimes referred to as service requests.



- You, as an agent, create a case from an incoming support request. This request could arrive from any one of a number of channels, such as social media, a phone call, or an email. Cases can also be created automatically from service contracts.
- The system can do some automatic processing of the request to classify and route the case.
- As an agent, you can add crucial information such as customer contact information from a phone call, or specify how quickly the issue must be resolved.
- Use the knowledge base, collaborate with colleagues in real time, or work with other systems to access the information required to resolve the problem.
- Some cases can require a visit by a service technician. The solution can integrate with external systems to schedule an appointment and arrange for repair parts as needed.

- Respond to the requestor. Communicate the solution to the customer through the same channel used to make the initial request, or any other preferred contact point.
- Once the issue is resolved, you can close the case.
- Use case analytics and reports to help you fine-tune your service strategy and make your organization more efficient.

12.1 Configure Cases

Administrators can configure cases in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

→ Tip

Expand a collapsible section heading to view the configuration steps.

12.1.1 Configure Status Dictionary Entries

You, as an administrator, can use your own case statuses in your processes and schemas. Create the [Status Dictionary Entries](#) and then assign them a lifecycle and assignment status.

Context

The lifecycle statuses are used for service level agreement (SLA) calculations. Assignment statuses are used to identify the assignee of the case (the processor and requestor) to help case duration determination.

The following table lists the predelivered status values that you can select in the case and the next status you can reach from there. When you create your statuses, keep these possibilities in mind.

Current Status	Reachable Statuses	Non-Reachable Statuses	Description
Open	• In Process • Customer Action • Completed	• Closed	This status isn't reachable from any other value.
In Process	• Customer Action • Completed	• Open • Closed	If the status changes from In Process to Closed the system completes and closes the case.

Current Status	Reachable Statuses	Non-Reachable Statuses	Description
<i>Customer Action</i>	<i>In Process</i> <i>Completed</i>	<i>Open</i> <i>Closed</i>	
<i>Completed</i>	<i>In Process</i> <i>Customer Action</i> <i>Closed</i>	<i>Open</i>	
<i>Closed</i>	None	<i>Open</i> <i>In Process</i> <i>Customer Action</i> <i>Completed</i>	If a case is closed, it isn't possible to open it again. This status can only be used for cases still in the system. New cases aren't allowed to use <i>Closed</i>.

i Note

Once a case is set to **closed**, you can't change the case status.

Procedure

- Go to your user profile and select [Settings](#) [Case Management](#) [Status Dictionary Entries](#).
- Select the add icon (+) at the top of the table to add a new status code.
- Enter a code and a description for your new status.

Custom status codes have to start with **Z** or **Y**.

- Select the *Lifecycle Status* and *Assignment Status* for your new code.

Allowed Combinations

Lifecycle Status	Assignment Status
Open	Processor Action
In Process	Processor Action
In Process	Requestor Action
Completed	Requestor Action
Closed	Not Assigned

- Select the confirm icon (✓) to save changes.

You can delete a status code by selecting the delete icon (🗑) in the *Action* column.

i Note

You can change the description of the predelivered statuses, but you can't delete them.

Related Information

[Configure Status Schemas \[page 127\]](#)

12.1.2 Configure Status Schemas

You, as an administrator, can edit the standard status schemas, and create new schemas for cases.

Context

Change the visibility, and sequence of the standard statuses. Use status schemas to realize different processes in your company. For example: Create a status schema for use with your internal employee support, and a different status schema valid for interactions with partners.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Case Management](#) ► [Status Schemas](#) ►.
2. Select the add icon (+) at the top of the table to add a new status schema.
3. Enter an ID for your new schema in the [Status Schema](#) field. The ID consists of two characters and starts with either **Z** or **Y**.
4. Enter a description for your new schema.
5. Select the add icon (+) at the top of the table to [Add Status to Schema](#).

This step adds a row to the table where you can select from the status dictionary entries you've previously defined. Select the option if this is the [Initial Status](#) for the schema, and if you want the [Status Visible](#) for the case.

- Only the [Open Life Cycle Status](#) can be selected as initial status.
 - You can select the [Closed Life Cycle Status](#) but you can't set it to visible.
6. Save your changes.

You can [Hide](#) a status, or [Set as Visible](#) from the [Actions](#) menu (⋮).

i Note

Only statuses not currently in use can be deleted.

Related Information

[Configure Status Dictionary Entries \[page 125\]](#)

12.1.3 Configure Priority Color

Assign a new color to any of the four priority codes.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Case Management](#) ► [Priority Settings](#) ►.
2. Select the edit (✎) icon to change the color assigned to a priority code.
3. Select a new color from the menu.

Your changes are saved automatically.

12.1.4 Configure Service Levels

You, as an administrator, can create new service levels in your solution.

Context

Set up service level based on target milestones with response times, and specifying working hours and timezone.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Case Management](#) ► [Service Levels](#) ►.
2. Enter a name for the new service level.
3. Add target milestones.
4. Select a milestone type.
 - [Due Date for Completion](#)
 - [Due Date for Initial Review](#)
 - [Due Date for Resolution](#)
 - [Due Date for Response](#)
5. Add details for a milestone. Select an entry in the [Milestones](#) table and then add rows to the Details of Milestone table as needed. Select [Type of Service](#), [Priority](#), and [Net Labor Time](#) for each row.

You can maintain multiple service types and priorities for a milestone. Add a new row for each priority.

→ Tip

If you enter the duration in hours, the system only considers operating hours. If you enter the duration in days, the system uses the [Working Configuration](#), but not the time ranges. You enter this information under [Operating Hours](#) in the next step.

❖ Example

Suppose that your working days are from Monday to Friday, from 8:00 to 18:00 and the reporting date is Wednesday 06 April 2022, 15:00.

- If you enter a duration of 2 days the system calculates a due date on Friday, 08 April 2022, 15:00.
- If you enter a duration of 48 hours the system calculates a due date on Wednesday, 13 April 2022, 13:00.

6. Set the [Working Day Calendar](#), [Time Zone](#), and [Working Configuration](#) daily time ranges.

[Working Configuration](#) allows you to set the hours when your service staff is available. If you don't enter operating hours, the solution assumes that service is available 24x7.

→ Tip

If you want to intentionally set up 24x7 availability, create a calendar with all 7 days as working days and 12 AM to 12 AM as working hours.

You can use customer working hours for milestone calculations by maintaining a working day calendar, time zone, and working configuration consistent with the customer location.

7. Save the new service level. The service level now appears in the [Service Levels](#) list.
8. Activate the new service level.

Related Information

[Configure Service Level Agreements \[page 129\]](#)

12.1.5 Configure Service Level Agreements

You, as an administrator, can set up rules that assign service levels to cases automatically.

Context

When you create or edit a case, the system uses rules to evaluate which service level to apply. When a rule is found that matches the case, the system applies the appropriate service level. The service level pushes details about reaction times and due date calculations to the case.

Procedure

1. Go to your user profile and select [Settings](#) [Case Management](#) [Service Level Agreements](#).
2. Set up the rules table with [Adapt Columns](#).

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon ().

6. Select the confirm icon () to save changes to the row when finished.
7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

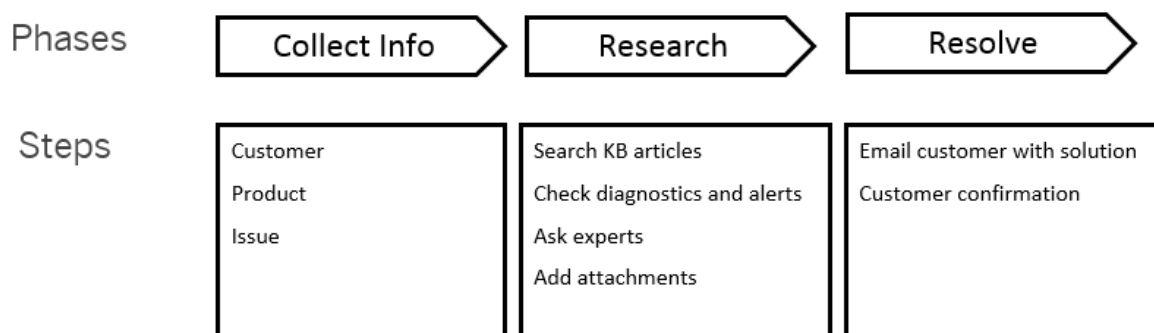
[Configure Service Levels \[page 128\]](#)

12.1.6 Configure Case Types

Customize case types to match the workflow used by your organization.

Context

Create case types to model any type of service workflow based on your business processes. You can add phases, steps, approvals, and automated flows to your case type. Phases and steps guide your agents visually through processing the case.



Standard Case Statuses

Status Code	Status
01	Open
02	In Process
04	Customer Action
05	Completed
06	Closed

Procedure

1. Go to your user profile and select [Settings](#) > [Case Management](#) > [Case Types](#).

→ Tip

Select the pin icon (📌) to add [Case Types](#) to the [Dashboard](#) for quick access.

2. Select a case type in status [Active](#) that you want to use as a starting point to customize and create your own case type. You can use the [Sample Case Type](#) provided by SAP to get started.
3. Select [Duplicate Case Type From Version](#) from the [Action](#) menu (⋮) for an active version of the case type template.
4. Enter a case type code starting with the letter **Z**. The type code can be up to four characters in length, with no spaces or special characters.

5. Enter a description for your new case type and select [Create](#).

The [Case Template](#) screen opens where you can edit and add phases and steps. **Phases** are represented by a screen within the case detail view, containing one or more specific steps. **Steps** are discrete actions that can be:

- **To do items:** Tasks or actions that can include explanatory text. If a step is mandatory, the agent must complete the task or action before the case can proceed to the next step. The agent must confirm manually once they complete the step.
 - **Automated flows (autoflows):** Triggers an autoflow action. For example: update a field in an object, create a time line event, or send an email message. You can use autoflow steps in addition to autoflows triggered at the case level.
 - **Approvals:** Use approval steps to require that the agent request approval before continuing to process the case. Agents can include notes before submitting the approval request. You can use approval steps in addition to approvals triggered at the case level.
6. Add, edit, reorder, or remove phases and steps to set up the case template to reflect your desired business process.

Drag the move icon () to reorder phases and steps.

7. (Optional) Select [View/edit number range table](#) to select an existing number range for the case type, or create a new number range.
8. (Optional) Select a service catalog for the case type.
9. Save your changes.
 - [Activate](#) sets your case type to active status.
 - [Close](#) saves your case type as a draft. You can activate it later.

Related Information

[Autoflow \[page 193\]](#)

[Approval \[page 190\]](#)

12.1.7 Configure Involved Party Settings

You, as an administrator can configure [Involved Parties](#) for cases.

Context

You can determine automatically all involved parties for business transactions and their related documents using party roles and determination rules. These rules allow you to streamline account team assignments, and ensure that business partners are correctly assigned to business objects in a way that matches your company processes. You can decide which party roles to use in your organization, however some party roles are mandatory by default, and can't be deactivated.

Procedure

1. Go to your user profile and select [Case Management](#) [Involved Party Settings](#).
2. Add or edit party roles and under [Party Role Assignments](#). Select the add icon (+) at the top of the table to add a new party role assignment.

You can also select the edit (✎) icon to edit the existing party role assignments.

3. Select a [Party Role](#) from the menu.
4. Select the appropriate toggles to set the role to: active, mandatory, or unique.
5. To add or edit determination rules, select the **review determination steps** icon (⌕) in the [Actions](#) column.
6. Enter a step number. For example: 100, 150, 200.
7. Select a determination method from the menu.
8. Select a source party role and whether to bring over one or all parties from the source.
9. Save your changes.

12.1.8 Configure Case Routing to Employee

You, as an administrator, can set up rules that route cases to employees automatically.

Procedure

1. Go to your user profile and select [Settings](#) [Case Management](#) [Case Routing to Employee](#).
2. Set up the rules table with [Adapt Columns](#).

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon (⬆).

6. Select the confirm icon (✓) to save changes to the row when finished.
7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

[Configure Case Routing to Team \[page 134\]](#)

12.1.9 Configure Case Routing to Team

You, as an administrator, can set up rules that route cases to service teams automatically.

Procedure

1. Go to your user profile and select [Settings](#) > [Case Management](#) > [Case Routing to Team](#).
2. Set up the rules table with [Adapt Columns](#).

The first time you set up business rules you must configure the rules table. You can have up to 15 columns in the table, including the results column.

→ Tip

Use only the columns required for assigning objects. Extra columns can impact performance. You can add additional columns and rules at any time.

3. Select the add icon (+) at the top of the table to add a new rule. A new row with an asterisk in each field of the row is added to the table.
4. Click the asterisk symbol (*) in a cell to enter a value for that field. The rules table offers a variety of logical operators to define your values.

You can add values to more than one field in a row. The fields in one row evaluate with a logical **AND** relationship. The field values must **all** evaluate to true in order for the rule to evaluate to true.

5. Continue to add rows for each desired result.

The system compares rows using the logical **OR** operator. The system compares each rule, top to bottom, until **ONE** evaluates to **true**.

Rearrange rows by selecting the row and selecting the sort icon (↕).

6. Select the confirm icon (✓) to save changes to the row when finished.
7. Click [Activate](#).

Enable your changes by activating the new rules.

→ Tip

Once you've set up your columns and you're familiar with the operators and table properties, you can export the table to a Microsoft Excel file. If you have a large number of service level rules, you can edit the file off line and then import the rules file back into your solution.

Related Information

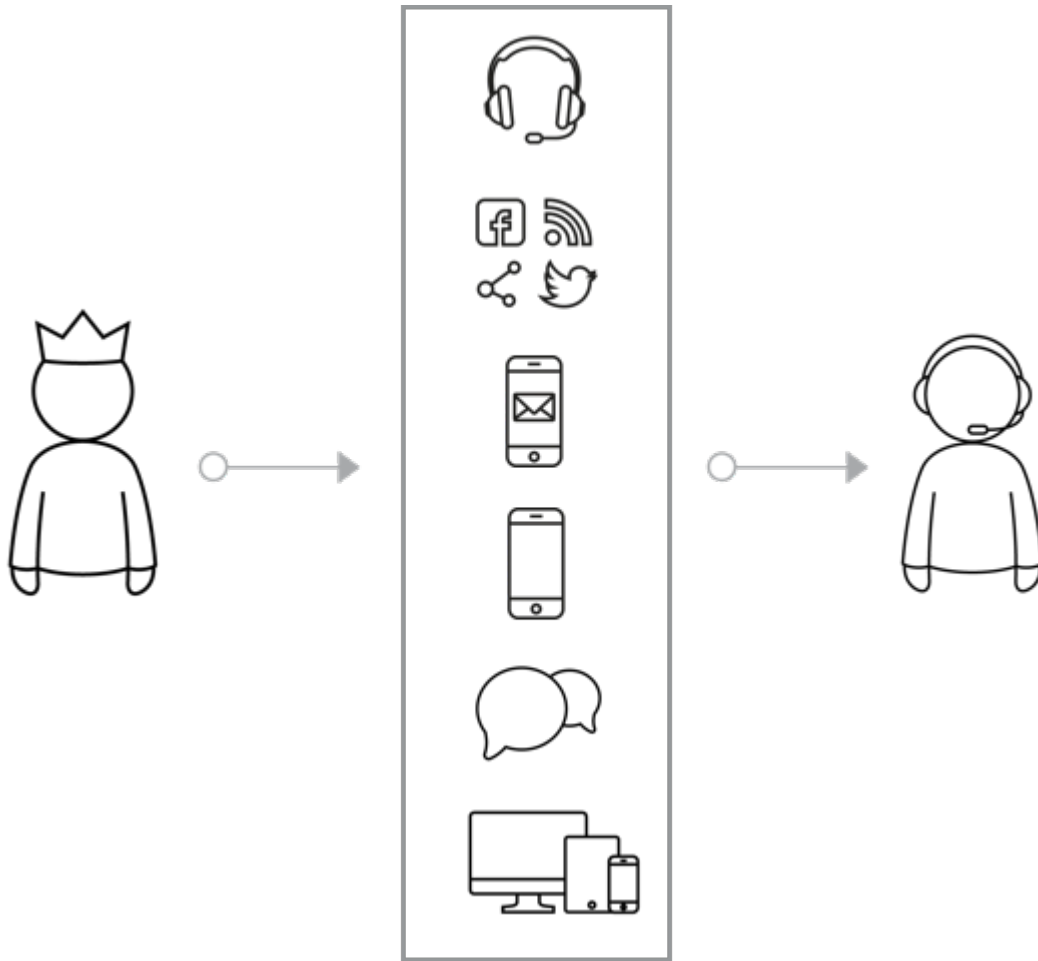
[Configure Case Routing to Employee \[page 133\]](#)

12.2 Create Cases

You can create a case manually from the [Case](#) worklist, or from the agent desktop based on incoming communication from various available channels.

- From [Agent Desktop](#), click the + [Create](#) button and select [Case](#).
- From the [Case](#) worklist, click the create icon (+) to create a new case.

The [Subject](#), [Type](#), and [Account](#) fields are mandatory. We recommend that you also select the source of the incoming communication that led to the case creation in the [Source](#) field. The menu displays communication channels based on what is configured and integrated in your system.



Your organization can choose to create cases automatically from incoming email messages to customer support addresses. See the following related link for more information.

Related Information

[Configure Inbound Email Channels to Create Cases \[page 87\]](#)

12.3 Manage Cases

Learn about options and parameters for case setup and management in the solution.

You have a wide variety of options for classifying, organizing, linking, and routing cases in the solution. Your administrator or service manager sets many of these options based on how your organization uses cases.

[Case Worklist \[page 137\]](#)

The [Case](#) worklist displays all cases in the solution.

[Use Case Types \[page 137\]](#)

Select a case *Type* to define the kind of case you're creating in the system.

[Work with Case Status \[page 139\]](#)

You can maintain dictionary entries for the user status, combine them into status schemas, and assign the status schemas to case types.

[Analyze Case Data with Qualtrics XM Discover \[page 142\]](#)

Access Qualtrics XM Discover from your SAP Service Cloud Version 2 solution to create and view widgets on a dashboard that display insights based on aggregated data from case information.

12.3.1 Case Worklist

The *Case* worklist displays all cases in the solution.

Choose *Case* from the app menu (☰) to view all cases that currently exist in the solution. You can search (🔍) for a case by keyword, such as ID, description, or account name. Filter the list to view a specific set of cases. You can filter by:

- Date (created, reported, or changed in the last week, month, or year)
- Priority
- Status
- Type

12.3.2 Use Case Types

Select a case *Type* to define the kind of case you're creating in the system.

SAP provides one standard case type: *Sample*. Copy the sample case type to define your own case type and associated processes.

Based on the case type, you can define the status schema; whether it's relevant for external pricing, contract use, work cases, or ATP (available to promise).

Example

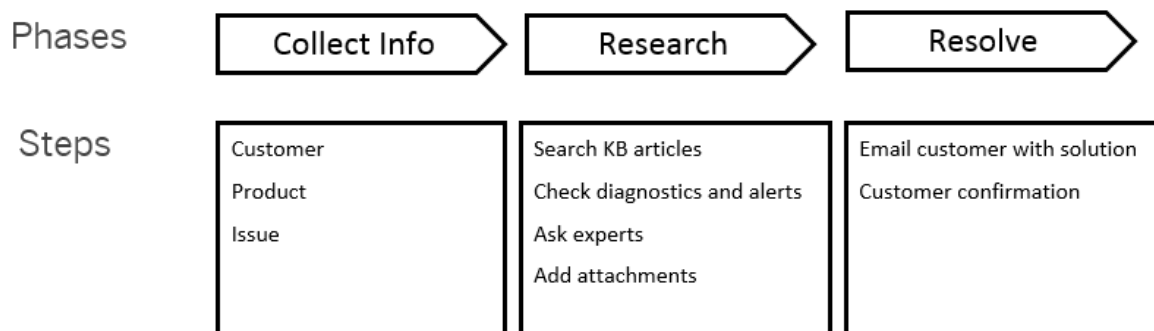
If you'd like to handle inquiries, feedback, or complaint processes differently in your organization, you would create three different case types.

12.3.2.1 Configure Case Types

Customize case types to match the workflow used by your organization.

Context

Create case types to model any type of service workflow based on your business processes. You can add phases, steps, approvals, and automated flows to your case type. Phases and steps guide your agents visually through processing the case.



Standard Case Statuses

Status Code	Status
01	Open
02	In Process
04	Customer Action
05	Completed
06	Closed

Procedure

1. Go to your user profile and select **Settings** > **Case Management** > **Case Types**.

→ Tip

Select the pin icon (📌) to add **Case Types** to the **Dashboard** for quick access.

2. Select a case type in status **Active** that you want to use as a starting point to customize and create your own case type. You can use the **Sample Case Type** provided by SAP to get started.
3. Select **Duplicate Case Type From Version** from the **Action** menu (⋮) for an active version of the case type template.
4. Enter a case type code starting with the letter **Z**. The type code can be up to four characters in length, with no spaces or special characters.

5. Enter a description for your new case type and select [Create](#).

The [Case Template](#) screen opens where you can edit and add phases and steps. **Phases** are represented by a screen within the case detail view, containing one or more specific steps. **Steps** are discrete actions that can be:

- **To do items:** Tasks or actions that can include explanatory text. If a step is mandatory, the agent must complete the task or action before the case can proceed to the next step. The agent must confirm manually once they complete the step.
 - **Automated flows (autoflows):** Triggers an autoflow action. For example: update a field in an object, create a time line event, or send an email message. You can use autoflow steps in addition to autoflows triggered at the case level.
 - **Approvals:** Use approval steps to require that the agent request approval before continuing to process the case. Agents can include notes before submitting the approval request. You can use approval steps in addition to approvals triggered at the case level.
6. Add, edit, reorder, or remove phases and steps to set up the case template to reflect your desired business process.

Drag the move icon () to reorder phases and steps.
 7. (Optional) Select [View/edit number range table](#) to select an existing number range for the case type, or create a new number range.
 8. (Optional) Select a service catalog for the case type.
 9. Save your changes.
 - [Activate](#) sets your case type to active status.
 - [Close](#) saves your case type as a draft. You can activate it later.

Related Information

[Autoflow \[page 193\]](#)

[Approval \[page 190\]](#)

12.3.3 Work with Case Status

You can maintain dictionary entries for the user status, combine them into status schemas, and assign the status schemas to case types.

In a case, only the entries that you've maintained via configuration display in the user [Status](#) menu.

12.3.3.1 Configure Status Dictionary Entries

You, as an administrator, can use your own case statuses in your processes and schemas. Create the [Status Dictionary Entries](#) and then assign them a lifecycle and assignment status.

Context

The lifecycle statuses are used for service level agreement (SLA) calculations. Assignment statuses are used to identify the assignee of the case (the processor and requestor) to help case duration determination.

The following table lists the predelivered status values that you can select in the case and the next status you can reach from there. When you create your statuses, keep these possibilities in mind.

Current Status	Reachable Statuses	Non-Reachable Statuses	Description
Open	• In Process • Customer Action • Completed	• Closed	This status isn't reachable from any other value.
In Process	• Customer Action • Completed	• Open • Closed	If the status changes from In Process to Closed the system completes and closes the case.
Customer Action	• In Process • Completed	• Open • Closed	
Completed	• In Process • Customer Action • Closed	• Open	
Closed	None	• Open • In Process • Customer Action • Completed	If a case is closed, it isn't possible to open it again. This status can only be used for cases still in the system. New cases aren't allowed to use Closed .

i Note
Once a case is set to **closed**, you can't change the case status.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Case Management](#) ► [Status Dictionary Entries](#) ►.
2. Select the add icon (+) at the top of the table to add a new status code.
3. Enter a code and a description for your new status.

Custom status codes have to start with **Z** or **Y**.

4. Select the [Lifecycle Status](#) and [Assignment Status](#) for your new code.

Allowed Combinations

Lifecycle Status	Assignment Status
Open	Processor Action
In Process	Processor Action
In Process	Requestor Action
Completed	Requestor Action
Closed	Not Assigned

5. Select the confirm icon (✓) to save changes.

You can delete a status code by selecting the delete icon (🗑) in the [Action](#) column.

i Note

You can change the description of the predelivered statuses, but you can't delete them.

Related Information

[Configure Status Schemas \[page 127\]](#)

12.3.3.2 Configure Status Schemas

You, as an administrator, can edit the standard status schemas, and create new schemas for cases.

Context

Change the visibility, and sequence of the standard statuses. Use status schemas to realize different processes in your company. For example: Create a status schema for use with your internal employee support, and a different status schema valid for interactions with partners.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [Case Management](#) ► [Status Schemas](#) ►.
2. Select the add icon (+) at the top of the table to add a new status schema.
3. Enter an ID for your new schema in the [Status Schema](#) field. The ID consists of two characters and starts with either **Z** or **Y**.
4. Enter a description for your new schema.
5. Select the add icon (+) at the top of the table to [Add Status to Schema](#).

This step adds a row to the table where you can select from the status dictionary entries you've previously defined. Select the option if this is the [Initial Status](#) for the schema, and if you want the [Status Visible](#) for the case.

- Only the [Open Life Cycle Status](#) can be selected as initial status.
 - You can select the [Closed Life Cycle Status](#) but you can't set it to visible.
6. Save your changes.

You can [Hide](#) a status, or [Set as Visible](#) from the [Actions](#) menu (⋮).

i Note

Only statuses not currently in use can be deleted.

Related Information

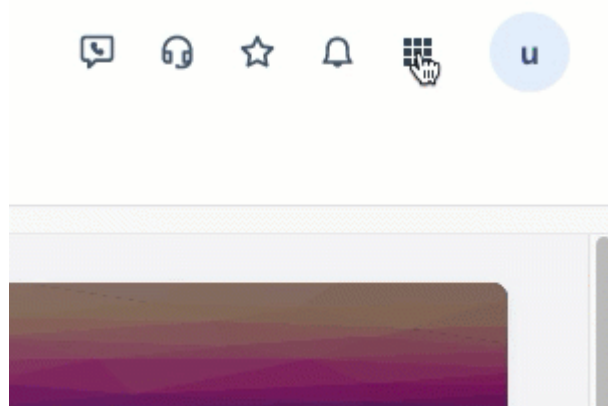
[Configure Status Dictionary Entries \[page 125\]](#)

12.3.4 Analyze Case Data with Qualtrics XM Discover

Access Qualtrics XM Discover from your SAP Service Cloud Version 2 solution to create and view widgets on a dashboard that display insights based on aggregated data from case information.

Qualtrics XM Discover provides insights into case data from your SAP Service Cloud Version 2 solution. Qualtrics XM Discover can be purchased separately for use with SAP Service Cloud Version 2.

Access Qualtrics XM Discover from the [Switch Application](#) menu.



The Qualtrics XM Discover logon screen opens. Enter your account and password to continue. Find data connector setup information in the related links section. Requires Qualtrics XM Discover account and password.

Related Information

[SAP Service Cloud connector configuration information \(requires Qualtrics XM Discover account\)](#) ➡

More information at: <https://www.qualtrics.com/discover/> ➡

13 Service Categories

Service categories are used to capture consistent information, to allow for reporting and benchmarking, and for determining service level assignments.

Creating and defining categories helps build valuable knowledge about the customer and also collects data that can be analyzed to understand business requirements and market demands.

Service categories are stored in catalogs to accomplish the following:

- Establish a multilevel category hierarchy.
- Link catalogs to different entities such as cases, and interactions such as email, chat, and phone calls.
- Manage the catalog status.

13.1 Create Service Catalogs

You, as an administrator, can create catalogs to organize service categories and sub categories.

Context

Create and organize service categories to sort cases and other entities into different service catalogs.

Procedure

1. Go to your user profile and select **Settings** > **Service Catalog** > **Catalog Definition**.
2. Select the Create icon (+) to create a new catalog, or select a catalog name to edit an existing catalog.
3. If you're creating a new catalog, enter a **Name**, **ID**, and **Description**, then save your changes.
4. Select or edit the entities with which to use this catalog. For example: case, email, chat, or phone call.
5. Add or edit a service category in the catalog. Select the **Add Parent Category** icon (+) to add a new category.
6. Enter or edit the **Name**, **ID**, and **Description**, for the category, then save your changes.

Once you enter an ID for the category, it can't be changed. However, you can edit the name and description.

7. To add subcategories to your top-level category, select the **Add Subcategory** icon (+) in the **Action** column. Enter a **Name**, **ID**, and **Description**, for the subcategory, then save your changes.

i Note

You can have up to three levels of subcategories under your top-level category, providing a total of four hierarchy levels for your category catalogs.

Add additional categories and subcategories to build your catalog.

→ Tip

You can export a spreadsheet template, create your catalog hierarchy structure, and then import your catalog structure. For more details on importing data, see the related information section.

8. Activate your new catalog. Select the [Activate](#) button in the catalog header.

! Restriction

Once you activate a catalog, you can deactivate it, but you cannot activate it again.

You can also activate and deactivate categories in the catalog with the [Activate](#) and [Deactivate](#) icons (⚙ and ⏻) in the [Action](#) column.

Results

Once you create and activate your service catalog, users in the system can assign categories to the entities you specified for the catalog.

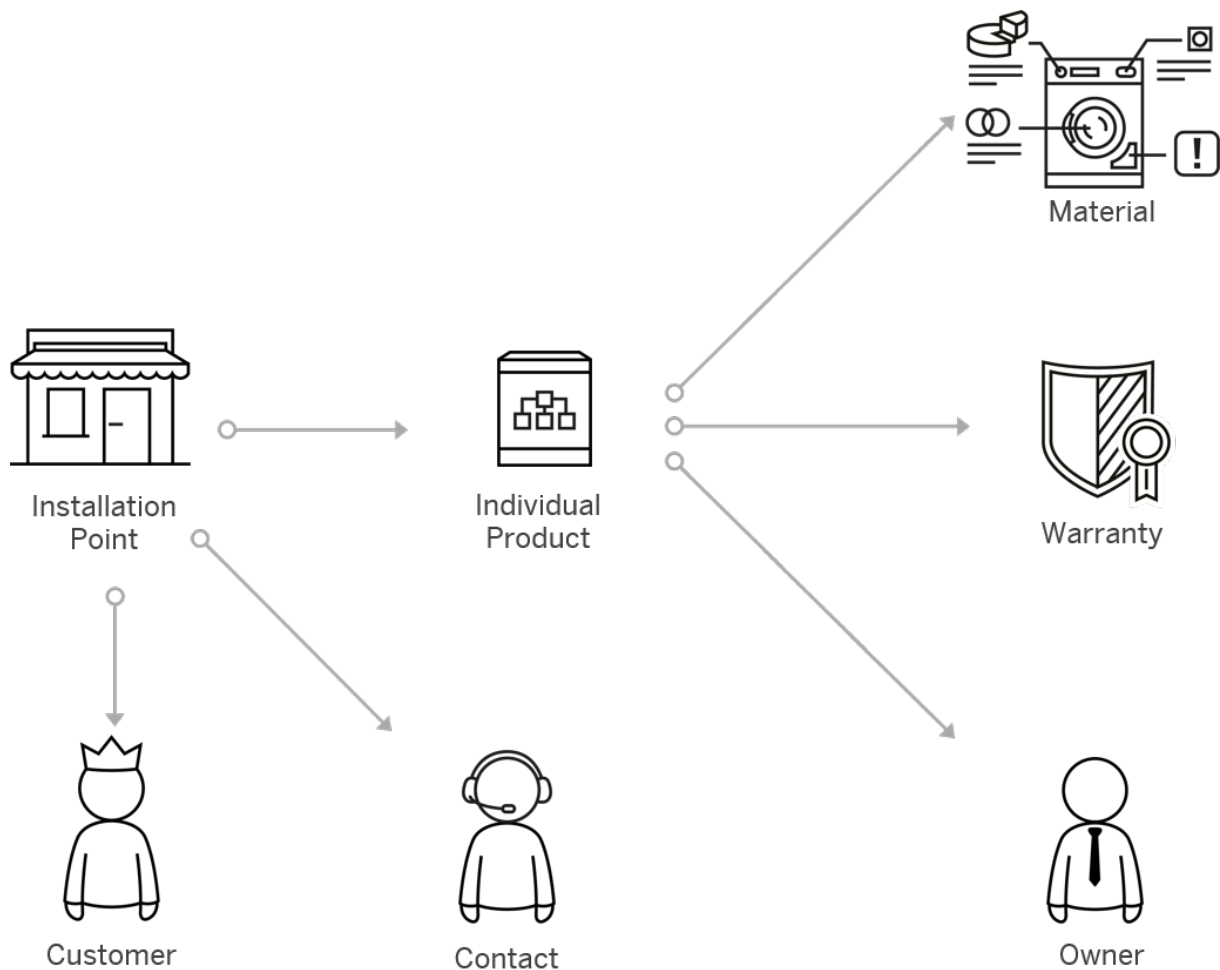
Related Information

[Data Import and Export \[page 214\]](#)

14 Registered Products

A registered product is an instance of a product that is associated with a specific customer and generally has a serial number.

When a ticket is created, the registered product information allows the service agent to identify the customer product and any associated warranties or service entitlements. Additional information, such as product location is used in the service process.



You can do the following with a registered product:

- Create serialized registered products with customer, location, and warranty information
- Create a ticket that refers to a registered product
- Create a hierarchy of registered product items

14.1 Configure Registered Products

Administrators can configure Registered Products in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

→ Tip

Expand a collapsible section heading to view the configuration steps.

14.2 Create Registered Products

Register a product to create a unique instance of it in the system.

Context

When you create a registered product, the relevant information is automatically copied from the customer. For example, the customer location and involved parties such as Ship-to-Party, Bill-to-Party, and service technician.

When you create a registered product from the agent workspace, the customer and product information in the ticket are automatically copied into the form.

To register a product, follow these steps:

Procedure

1. Navigate to [Registered Products](#), and click the [+ Create](#) button.
2. Enter data for Customer, Product, and Serial ID for identifying the registered product.
3. Save your entries.

14.3 Use Registered Products

Create a new registered product with information such as [Customer](#), [Serial ID](#), and [Location](#).

Using the [Registered Products](#) functionality, you can do the following:

- Add notes and attachments

- Maintain involved parties, user status, sales, and distribution data
- Track the history of changes
- Create tickets for a registered product

If there are changes made to the following fields on [Registered Products](#), the system administrative data (such as the Changed On field) is not updated:

- Registered Product Description
- Serial ID
- Reference Date
- Warranty-related fields

14.4 Search Registered Products

Search for registered products using various criteria.

14.4.1 Basic Search in Registered Products

On the [Registered Products](#) lists screen, the basic search supports searching with the following fields:

- Description
- Reference Product Description
- Serial ID
- Customer
- Contact
- Country
- Postal Code
- City
- Street

i Note

Fields such as Contract, Country, Postal Code, City, and Street in the Serial ID value help are personalization hidden and can be enabled as required.

Using [Search](#), with the type ahead feature, in an input field (with value help) displays search results based on the supported fields in the basic search of that value help.

14.4.2 Enterprise Search for Registered Products

Search for registered products within the solution.

You can search for a registered product using the following search criteria:

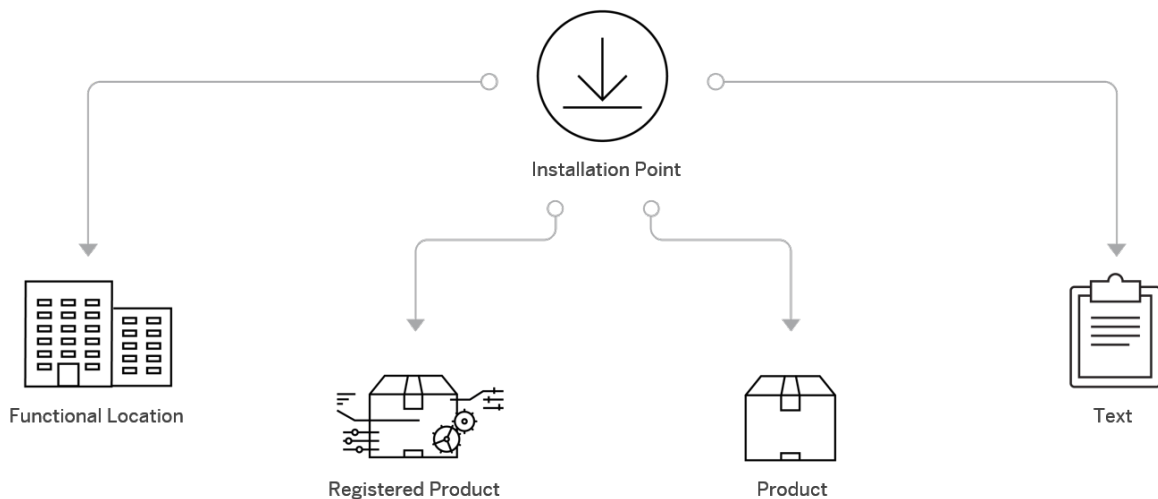
- Registered Product ID
- External ID
- Registered Product Category
- Sales Territory
- Service territory
- Country
- State
- Postal Code
- City

15 Installation Points

An installation point is the physical location of the object that requires servicing or maintenance based on the service or maintenance plan for that object. It also consists of product and text information for an installation point.

It consists of these categories:

- Functional Location
- Product
- Text
- Registered Product



15.1 Create Installation Points

From the [Installation Point](#) tab, you can create a new installation point only with the category of functional location.

In addition to the common features, this category of installation point supports warranty, sales and distribution data, and replication from SAP S/4 HANA. The customer information and installed base sections are prepopulated when you create an installation point from an installed base.

Installation points that are categorized as registered products are only displayed in the [Registered Product](#) tab. You can assign a registered product to an installation point. You can also assign an existing registered product to a current installation point.

To assign an installation point, with product or text category, to an installed base, create the installation point from the [Installed Base](#).

i Note

You can add a functional location to an installed base using a migration template.

15.2 Use Installation Points

Know more about using installation points.

15.2.1 Create Items for Installation Points

Items consist of objects that need servicing, maintenance, or repair. Items must be assigned to installation points in your SAP solution.

You can maintain the hierarchy structure of an installation point by creating a functional location and assigning items to it. You can maintain another functional location as the parent of an installation point using the [Belongs to Installation Point ID](#) field. You can also create a new functional location or registered product, or add an existing registered product as an item to the functional location.

The hierarchy view of the items is displayed in the [Items Hierarchy](#) tab of the [Installation Point](#) details view, but it's hidden by default. You need to set it visible through Adaptation.

i Note

If a functional location contains equipment, then that equipment also appears in the list on the [Items](#) tab.

15.2.2 Export Installation Points to Microsoft Excel File

Export all entries of installation points to a Microsoft Excel file.

In the [Installation Points](#) worklist, choose [Export to Microsoft Excel](#)® from the action menu to export all entries of the installation points based on the query you choose.

15.3 Search Installation Points

Search for installation points using various criteria.

On the [Installation Points](#) screen, the basic search supports searching with the following fields:

- ID

- Description (for all categories of installation Points)
- Serial ID
- External ID
- Address (Country, Region, City, Street, Postal Code, Building, House, Floor, Room)
- Customer
- Installed Base

16 Installed Base

Installed base is a hierarchical arrangement of items installed at your customer's location.

In the *Installed Base* work center, you can do the following:

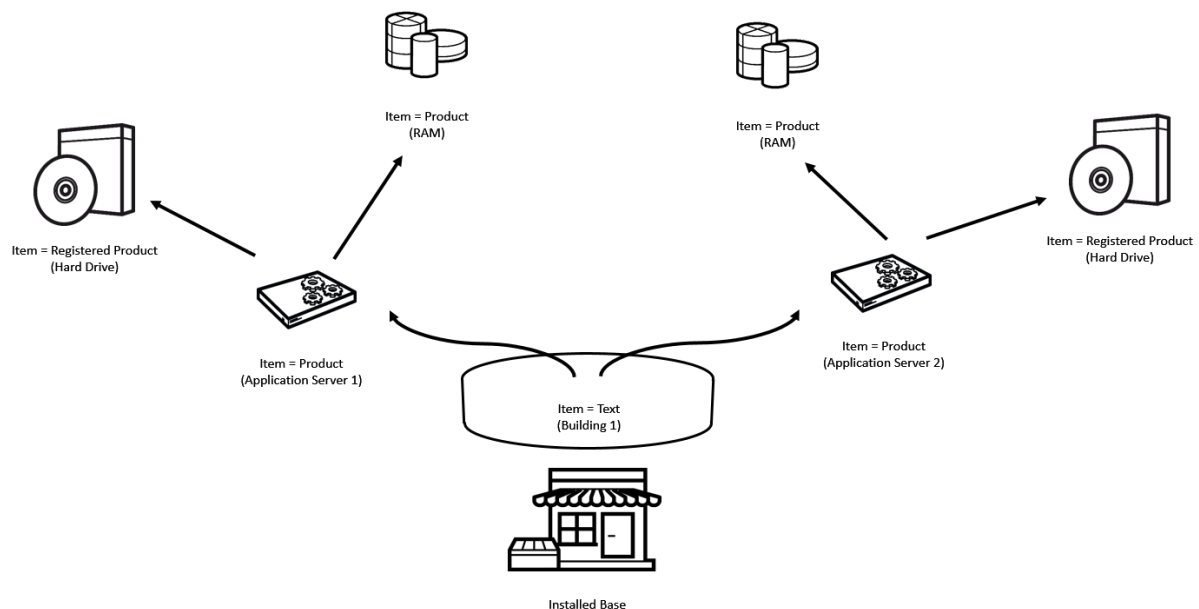
- View the list of maintained installed bases.
- Get information about customer, contact, location, and status, and access an installed base
- Create new installed bases

Certain information can be maintained at the installed base header level. For example, *Involved Parties*, *Attachments*, *Notes* and *Competitor Products*. The status of the installed base changes to *In Preparation* after it is saved.

Example

You can maintain an installed base for Company ABC where your servers are installed.

Here's an example of an Installed Base set up:



Here's what you can do with the Installed Base management feature:

- Create installed bases to track installed assets.
- Create a hierarchy of Installed Base items with any number of levels.
- Create a ticket with reference to an Installed Base or Installed Base item.
- Maintain competitor products for your Installed Base.
- Maintain involved parties for Installed Base or Installed Base items.

- Track changes, add attachments/notes, and create reports.
- Use the Installed Base and Installation Points (installed base items) migration templates for data load.

16.1 Configure Installed Bases

Administrators can configure Installed Bases in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

16.2 Use Installed Base

Create a new installed base with information such as [Name](#), [Customer](#), and [Owner](#).

Using the [Installed Base](#) functionality, you can:

- Maintain competitor products for your installed base
- Add notes and attachments
- Track history of changes
- Create tickets for installed base
- Associate installed base with contracts

17 Warranty

You can assign a warranty to a registered product or installation point, and determine its coverage in a ticket.

A warranty gets determined in the ticket header. Warranty determination is based on installation point, registered product, service date, and ticket incident category. You can assign warranty coverage in a ticket, which is determined on the ticket header, with inheritance to items. It influences external pricing (item) based on the accounting indicator in SAP S/4 HANA.

18 Utilities

Utilities is an enterprise that provides certain classes of services to the public including Electricity, Gas, and Water for a business partner.

The focus is inclusive of the following utilities industry segments that classifies as sectors within the Utilities:

- Electricity- Includes the generation, trading, transmission, distribution, retail, metering, and customer care segments of electricity providers.
- Gas- Includes the distribution, metering, retail, and customer care segments of end-use natural gas providers.
- Water - Includes the supply, treatment, distribution, metering and customer care segments of water and waste water providers.

The solution helps you to optimize call center operations using modern customer engagement tools for service processes. The utilities are fully integrated with IS-Utilities (IS-U) and SAP S/4 HANA, supports seamless transition between the different support channels.

18.1 Contract Account

In the contract account, you can store controlling data for long-term business relationships with a business partner.

This data controls processes in invoicing, contract accounts receivable and payable, and correspondence processing. You can define several contract accounts for each business partner.

All the contract accounts within the application is listed in a list view.

You can use the search option to select the list of relevant contract accounts.

i Note

The search option is available only for the following fields:

- Contract Account ID
- Contract Account External ID
- Contract Account Name
- Collective Contract Account External ID
- Collective Contract Account Name

18.1.1 Contract Account Workspace

View all details of contract account.

When you click an contract account, it opens in the detail view, where you can get a glance of all the important details.

18.1.1.1 Header Pane

The header pane gives a quick view of the contract account.

View more the information about the contract account such as:

- **Financials:** The financials section shows the following details of the contract account.
 - **Amount Due:** Sum of all unpaid invoices for which the due date has elapsed.
 - **Open and Payable:** Total amount the business partner owes to the utility company with respect to the opened contract account.
 - **Payment:** Latest posted payment amount and respective posting date.
 - **Scheduled Payment:** Next scheduled payment date and payment amount.
- **Customer:** The Customer section shows the following details of the contract account
 - Customer name and external ID
 - Credit worthiness of the customer
- **Collective Contract Account:** Appears only when the contract account opened has a collective contract account tied to it. This section display the collective contract account ID, Name, Type and Payment Method.
- **Security Deposit:** Shows the total amount requested/due from the customer and the amount of deposit paid by the customer.
- **Premises:** Shows all the premises associated with the contract account which have active contract.

18.1.1.2 Central

This tab display cards pertaining to the contract account.

The cards are as follows:

1. **Open Items:** The unpaid items for a billing document are grouped and displayed for each contract account. The open item card shows the following details:
 1. Time Slice: View the summation of total amount due against the range
 2. Amount Due: View the sum of all unpaid invoices for which the due date has elapsed
 3. Make Payment: Click this option to make a one time payment with the existing or new bank account. You can make payments for contract account as follows:
 1. Click create (+) option to make payment.
The payment screen opens.
 2. Under **Open Items**, the respective **Open & Payable**, **Amount Due** and **Other Amount** fields appears where you can select the amount you want to pay.

3. Under [Bank Information](#), select the bank details from the latest four bank IDs displayed.
You can also create new bank account by clicking add (+) icon.
4. After adding all the necessary details and click [Make Payment](#).
2. [Budget Billing](#): View the total amount and eligibility along with the next budget billing amount and due date.
Also you can create the plan using (+) icon.
How to create budget billing plan refer to [Create Budget Billing Plan \[page 160\]](#).
3. [Installment Plan](#): View the latest installment plan for the contract account with the following details:
 1. Installment plan type, next installment amount and due date of installment
 2. Total Plan Agreement
 3. Create plan using (+) icon.
How to create installment plan refer to [Create Installment Plan \[page 158\]](#).
4. [Promise to Pay](#): Promise to Pay is a commitment from customer for payment of an outstanding receivable within a specified time. Also you can create the promise to pay using (+) icon.
How to create promise to pay refer to [Create Promise to Pay \[page 159\]](#).
5. [Dunning](#): View the latest dunning notice and the dunning level provided to the customer and the simulated value for the next dunning activity for the contract account. You can also simulate the dunning activity by clicking the simulate button.
6. [Recent Invoices](#): View the latest four invoices for the contract account.
7. [Recent Cases](#): View the latest four cases created for the contract account.
8. [Active Contracts](#): View all the contracts for the selected contract account which are not moved out.

18.1.1.3 Payment Arrangements

This tab displays all the plans for the contract account along with the status.

You can create and edit each plans and make payments from here.

i Note

Payment Arrangements displays both active and deactivated plans.

18.1.1.3.1 Create Installment Plan

As an administrator, you can create installment for open items. The Utilities Contact Center solution supports create, change, and deactivation of an installment plan.

Procedure

1. From the [Navigation Menu](#) (≡), search for [Contract Account](#).
2. Select the contract for which you need to create a installment plan.

3. Click (+) icon from payment arrangement tab or from the Installment Plan card in the central tab.
4. Select [Open Items](#) as per your requirement.
5. Under [Plan Selection](#) specify the each parameter and click [Generate Proposal](#) option.
6. Review the installment plan summary under the [Confirm Installment Plan](#) screen and click [Save](#).

Edit or Deactivate Installment Plan

- To edit the existing installment plan, select the plan and click Edit (✎) icon.
- If you add or delete installments, then you must ensure that the total installment plan corresponds to the amount of the source receivable.
- If the installment plan is unpaid or partly paid, select Deactivate to cancel the interest and charge documents and deactivate the plan. Click [Deactivate](#) option to deactivate the plan.

18.1.1.3.2 Create Promise to Pay

Promise to Pay is a commitment from customer for payment of an outstanding receivable within a specified time.

Procedure

1. From the [Navigation Menu](#) (≡), search for [Contract Account](#).
2. Select the contract for which you need to create promise to pay.
3. Click (+) icon from payment arrangement tab or from promise to pay card in the central tab.
4. Select [Open Items](#) and enter the fields as per requirement.
5. Enter the desired fields for [Company Code](#), [Plan Selection](#), and [Plan Details](#) and click [Generate Proposal](#) option.
6. Review the promise to pay summary under the [Confirm Promise to Pay](#) screen and click [Save](#).

Edit Promise to Pay

To edit the existing promise to pay, select the plan and click Edit (✎) icon.

You can perform the following actions in edit screen:

- **Withdraw:** Select Withdraw to disable an open plan. You need to specify the **Reason for Withdrawal** and **Initiator of Withdrawal**.
- **Replace:** Allows you to withdraw an existing plan and create a new plan.
- **Status Change:** Allows you to modify status of an existing plan. Enter the **Levels of Fulfillment** and maintain comments. This option is only available for the deactivated promise to pay plans.
- **Reactivate:** Enables and activates a closed plan. The system prompts you to confirm if you need to reactivate a plan. This option is only available for deactivated promise to pay plans.

18.1.1.3.3 Create Budget Billing Plan

Utility company normally bills for its services at the end of a supply period. To remain solvent, therefore starts charging budget billing amounts throughout the current period based on the expected bill amount due. Specify the budget billing amounts and the dates that are due in the budget billing plan.

Procedure

1. From the *Navigation Menu* () , search for *Contract Account*.
2. Select the contract for which you need to create budget billing plan.
3. Click () icon from payment arrangement tab or from budget billing card in the central tab.
4. Select *Contract* as per requirement.
5. Enter the desired fields for budget billing period and click *Generate Proposal* option.
6. Review the budget billing plan information under the *Confirm Budget Billing Plan* screen.
7. Enter the data in the *Text for Budget Billing* field.
8. Optional: Change the due date and plan amount for each line item. You can also use the **Update Amount** action to update the changes for all the plan amount in the **Items** list.
9. Click *Save*.

Edit or Delete Budget Billing

To edit the existing budget billing plan, select the plan and click Edit () icon.

Modify the **Plan Amount** and **Text for Budget Billing** for a budget billing plan.

To delete the existing plan, select the budget billing plan and click *Delete* () .

i Note

You can only delete plans with no payments or further activities.

18.1.1.3.4 Create Payment Plan

The utility company and the customer together decide the amount to be paid with each bill. Instead of paying the actual bill amount, which varies according to season, the customer then has either a fixed payment plan or a payment plan in which the fluctuation of the amount is kept to a minimum. The utility company records the difference between the payment plan amount and the actual bill amount and charges it at a later date.

Procedure

1. From the *Navigation Menu* () , search for *Contract Account*.

2. Select the contract for which you need to create payment plan.
3. Click (+) icon from payment arrangement tab or from payment plan card in the central tab.
4. Select *Contract* as per requirement.
5. Enter the desired fields and click *Generate Proposal* option.
6. Review the payment plan details under the *Confirm Payment Plan* screen and click *Save*.

Edit or Delete Payment Plan

To edit the existing plan, select the payment plan and click Edit (✎) icon.

To delete the existing plan, select the payment plan and click *Delete* (🗑) icon.

i Note

You can only delete plans with no payments or further activities.

18.2 Premise

The premise is an enclosed unit which allows to perform a host of utilities functions such as creating or changing move-in.

All premises within the application is listed in a list/card view.

You can also view more details of a premise on detailed screen.

18.2.1 Premise Workspace

View all details of premise.

When you click a premise, it opens in the detail view, where you can get a glance of all the important details.

18.2.1.1 Header Pane

The header pane gives a quick view of the premise.

View the information of premises such as:

- **Header:** View details about the premise address and premise type.
- **Service Status:** View the service status based on the installation status
- **Customers:** View the customer linked to premise along with status
- **Owner:** View owner details linked to the premise
- **Details:** View more information about the premise

Service Status

Service status is displayed based on the installation status. The installation status is in turn derived from the device status.

The device status criteria are:

Active: If valid from date <= current date and valid to date > current date

Outdated: If valid to date < current date

Future dated: If valid with future date

No device: If no devices are present

Installation status criteria are:

Installation status will be **Off**, if at least one device is outdated though it has active devices

Installation status will be **On**, if there is no outdated devices and at least one active device

Installation status will be **future**, if there is no outdated and active devices

Service Status Combination

Case	Each Installation Status				Service Status
	Installation 1	Installation 2	Installation 3	Installation 4	
1	Off	Off	-	-	Off
2	Off	On	Future	-	Off: 1/3
3	Off	Off	Off	Future	Off: 3/4
4	On	On	Future	Future	On
5	Future	Future	Future	-	-
6	-	-	-	-	-

Customers Status

Customer Types

The types of customers are:

- **Active:** Customer who are present in premise
- **Outdated In:** Customer who were available earlier
- **Future:** Customer who occupy the premise at later point

The customer status are:

1. **Active:** Customer has moved in without move out date specified in system.
2. **Moving In:** Customer has a future dated move-in specified.
3. **Moving Out:** Customer has a future dated move-out specified.
4. **Inactive:** Customer as all ready moved out as of today.

18.2.1.2 Central

This tab displays cards such as Services, Services Orders, and Service Notifications.

Services

Gives an overview about the services associated with the premise.

The Installation ID, rate category, device ID of a meter is displayed. If there are more than one meter, then the count of the remaining devices are displayed.

Also shows the count of the active service order and the implausible reading and next scheduled meter reading date.

Service Orders

Display the latest four active service orders logged in for the premise along with the validity.

Service Notifications

Display the latest four active service notification logged in for the premise along with the validity .

i Note

For Service Order and Service Notification there is a standard BADI delivered considering the status as Document Received, Installation Pending, Cancelled, New, In process, and Completed. In order to use the custom defined user status, S4 BADI has to be implemented.

Recent Cases

Display the latest cases created for the premise along with status. This is not integrated with premise for now.

18.2.1.3 Services

This tab displays all the services associated with the premise.

The scaling of consumption depends on the S4 configuration. Based on the S/4 HANA configuration the consumptions are displayed in monthly, quarterly, or annually pattern.

- **Monthly:** The readings is plotted for every month in a year. If there are more than one reading in a month, then the latest reading is displayed. One year data from the time the last meter reading is taken into consideration.
- **Quarterly:** The readings are plotted for every quarter in a year. Three years data is considered for the same.
- **Annually:** The readings are plotted for every year. Three years data from the time of the last meter is considered for the same.

The header display the installation ID, point of delivery ID and rate category.

The services are displayed at the item level in 3 views such as:

- **Graphical View:** The power consumption (current consumption, expected consumption and last year consumption) of a time period for a customer is displayed in list view. Along with the consumption, the temperature of a day is also displayed.

i Note

Have a custom defined BADI, in order to view the temperature in the graph.

- **Table View:** Display the meter reading data in table view.
- **Card View:** Display the meter reading data in card view.

18.2.1.4 Cases

Display the latest cases created for the premise along with status. This is not integrated with premise for now.

18.2.1.5 Customers

In this tab, view all the customer along with their status associated with the premise.

You can also view the information about the move in, move out date, balance amount the customer owes, the due date from the last open item date till today, and the contract account, and so on.

18.2.1.6 Work Orders

In this tab, a call center agent can find an overview of service orders and service notifications logged in for the premise.

i Note

For Service Order and Service Notification there is a standard BADI delivered considering the status as Document Received, Installation Pending, Cancelled, New, In process, and Completed. In order to use the custom defined user status, S4 BADI has to be implemented.

18.2.2 Move-In Process

Move-in of a service means that a customer is signing up for a utilities service such as gas, electricity, water, and so on. A move-in created in SAP Cloud Solution creates a corresponding move-in in the S4HANA system.

Agent can view the details in Move-In tool bar as and when he fills in details under the different steps of guided move in. The tool bar shows the **Customer Name**, **Move-In Address**, **Move-In Date**, **Services Selected**, **Security Deposit Amount**, and **Mailing Address**.

18.2.2.1 Guided Move-In

For guided move-in, you can only use either Move-In or Move-In using MDT.

Context

Launch guided move-in in the Premise right panel using the Move-In button. Steps in a guided move-in are discussed as follows:

Procedure

1. Customer and Premise
 - a. Who: Enter the customer name by selecting the customer using the value help. Once the customer is selected the corresponding Total Amount Due, Open and Payable, Payment and Credit Worthiness cards will be displayed.
 - b. Where: The Premise address is defaulted under the Address field. You can view the further details under Premise Details, Customers, Service Status, and Work Order cards.
 - c. When: You can select the Move in date from the calendar when the customer wants to move in to the premise.
2. Service: The Service step shows the types of services (Electricity/Gas/Water) available for the customer to perform the move in.

18.3 Agent Desktop

Utilities service agents can use the agent desktop to work on several business processes and communication items simultaneously.

The agent can search for a contract account and premise by entering the respective Contract account ID and Premise ID in the search field under agent desktop.

Once the search result appears you can confirm the customer corresponding to contract account and premise that they want to investigate more.

After confirming the customer the [Customer Hub](#) screen appears and then navigate to the objects tab in the right panel to view the Utility specific objects such as :

- [Contract Account](#): Displays the contract account ID & name, account class, and amount due in the card.

Note

For the contract accounts where there is active contract then that contract's address is displayed in the card.

- [Premise](#): Displays the premise address and type where the customer has a move in and the services for which the move in has happened.

The agent can also make payments for contract account as follows:

- Click create (+) option to make payment.
The payment screen opens.
- Under [Contract Account](#), select the account for which you want to make the payment.
- Under [Open Items](#), the respective **Open & Payable**, **Amount Due** and **Other Amount** fields appear where you can select the amount you want to pay.
- Under [Bank Information](#), select the bank details from the latest four bank IDs displayed.
You can also create new bank account by clicking add (+) icon.
- After adding all the necessary details and click [Make Payment](#).

For more information on Agent Desktop refer to [Agent Desktop \[page 83\]](#) .

19 Intelligent Services

Use case data to create and train machine learning models to recognize similar cases, predict case categories, and sentiment.

Intelligent Services categorize your cases and improves accuracy. Intelligent Services predict the case category and recommends similar cases that can provide you with potential answers for the customer issue. The solution can scan the case text to identify sentiment, translate case text, and identify profane language. The model constantly captures feedback to be retrained for improved accuracy over time.

19.1 Case Categorization

Use machine learning to automate the case categorization process.

Save agents' time by automatically categorizing cases for your most commonly used case categories. Machine learning analyzes your past cases and category catalog to learn how cases are categorized in your organization. Once you train and activate the model and if the confidence thresholds are met, the system automatically populates the proposals into the case category.

Case categorization increases agent productivity, provides better prioritization of incoming cases and automatic classification based on model accuracy.

Case categorization is currently supported in the following languages:

- Chinese
- English
- French
- German
- Japanese
- Portuguese
- Russian
- Spanish

Prerequisites

- Data volume: Includes minimum 1000 categorized cases and an additional 300 instances of each category or subcategory values
- Data quality: Includes subject, description, and proper category

19.1.1 Case Intelligence Readiness Report

Use the readiness report to analyze your case data to determine if you can get useful predictions for case categorization.

Before you start creating and training the *Case Categorization* model, you must run the readiness report.

Find the readiness report under ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Categorization](#) ►. Select *Readiness Report* at the top right.

When the check is complete, the readiness report displays one of the following status:

- Ready to use machine learning
- Data improvements required
- Requirements not fulfilled

Each readiness factor shows you the minimum required value, the recommended value, and the actual value in your system. You can adjust the data in your system to meet the minimum and recommended thresholds and run the readiness check again. Each check factor should meet the recommended value to ensure reliable prediction results. If the check factor value falls between the minimum value and the recommended value, the report shows a caution. If the value is less than the minimum value, the report shows an error message. You can still proceed with the training, however the prediction result will be of lower accuracy.

Note

The data volume, or total number of objects (cases) shown is an **approximate** number of case records from the previous 12 months that could potentially be used to train the machine learning model.

Example

Let's look at a simplified example for case categorization readiness.

Suppose you have 10 categories in your service category catalog. The readiness report shows that only 33% (3) of these categories are assigned to the required number of cases to meet the qualification threshold. The percentage is below the recommended minimum threshold. However, 80% of your qualifying cases are assigned to these three categories. If you train a model based on this data you can potentially automate up to 80% of your incoming cases, despite using only 33% of your category catalog. In this case, machine learning could still yield significant agent productivity gains.

19.1.2 Configure Case Categorization

As an administrator, you must add, train, test, and activate the Case Categorization model.

Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Categorization](#) ►.
2. Under [Case Categorization](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Use up to four levels of categories to categorize cases.
5. Click [Save](#).

A new model is added with status **Created**.

6. Locate your model and click [Train](#) from the [Actions](#) (☰) menu to start training the model.

i Note

If you update your service category catalog you must map new category IDs to existing categories and retrain your model.

- a. Select the model for which you want to update the category mapping.
- b. Click [Download Mapping](#) to download the existing category mapping of the model.

i Note

The model does not have a mapping document, click [Prepare Mapping](#) to create one and then download it.

- c. Edit the category listing as required and upload the updated mapping document.
- d. Click [Save](#).
- e. Select [Train](#) to retrain the model.

The training takes time to complete depending on the volume of the historical data. The mode goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) ►.

The system stops the training if data extraction fails and updates the status as [Data Extraction Failed](#).

You can refresh and update the training status of the model by selecting [Get Status](#) from the [Actions](#) (☰) menu.

7. After the training is complete, click [Activate](#) from the [Actions](#) (☰) menu to activate the model.

You can deactivate a model by selecting [Deactivate](#).

i Note

You can enable automated recommendations using the [Enable](#) toggle button in the [Settings](#) tab of your model. Adjust the confidence level for automated prediction. The default value of the confidence level slider is 60 in the solution.

! Restriction

You cannot add new categories from the settings dialog. You can only deselect, or select the categories you included during model creation.

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Test](#).

The test console invokes the prediction model API using the sample case data you provide.

- The model considers the content of both the case [Subject](#) and [Description](#) fields. To test, combine text from both fields in the input box, or into one cell in the spreadsheet file. For testing purposes, a single entry has a limit of 1000 characters. In actual use, the model analyzes all text in the subject and description. Entries in the upload file don't have a maximum character limit.
- You can test the model quality from the test console by activating the model, but set the automation [Enabled](#) toggle to [No](#) in the model settings. With automation disabled, the activated model doesn't start categorizing tickets in your productive system, but can be tested in the console. Testing with automation disabled can be useful if you have data quality issues with your test tenant, but good quality data in your production system.

Case Categorization Test File Example

Case ID	Description
12341	Are boxes allowed as checked luggage?
12342	Pump is making a loud noise when activated.
12343	Coffee maker is making cold coffee!

19.1.3 View Case Categorization Prediction

Find category prediction results in cases.

Procedure

1. Go to [Cases](#).
2. Create a new case and enter the subject and description fields.
3. Select [Save](#).

Case categorization prediction works for all case creation channels such as manual, e-mail, social media, chat, messaging, and so on. The main usage for case category prediction is for cases generated from incoming e-mail messages. Cases are categorized during case creation. You will only see predictions for cases created after the model is set to active status.

19.2 Similar Case Recommendation

Find solutions from similar cases from the past that you can apply to the current case.

Note

To ensure accurate similar case recommendation with machine learning you require at least one year worth of cases with change history data. It is recommended to have at least 22,000 completed cases.

The machine learning algorithm scans the subject and description of categorized cases over the last 12 months and returns the top three most similar cases to the current case. The model learns the semantic relationship of words in the description and subject text and uses this context to find cases similar to the current case. This semantic, sentence-level representation of case text yields more accurate results than a simple keyword search. You can check similar cases for possible solutions to the current case.

Similar case recommendation is currently supported in the following languages:

- Dutch
- English
- Finnish
- French
- German
- Greek
- Italian
- Polish
- Slovak
- Spanish

19.2.1 Configure Similar Case Recommendation

As an administrator, you must add, train, and activate the Similar Case Recommendation model.

Procedure

1. Navigate to your user profile, and select  [Settings](#)  [All Settings](#)  [Machine Learning](#)  [Intelligent Service](#)  [Similar Case Recommendation](#) .

2. Under *Similar Case Recommendation* , click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click *Save and Train* to save your model and start the training.

The training takes time to complete depending on the volume of the historical data. The mode goes through the following statuses during the training: ► *Created* ► *Training in Preparation* ► *Data Extraction is Pending* ► *Data Extraction is in Progress* ► *Data Preprocessing is Pending* ► *Data Preprocessing is in Progress* ► *Training Triggered* ► *Training is Pending* ► *Training in Progress* ► *Training Completed or Training Failed* ► *Active or Inactive* ►.

The system stops the training if data extraction fails and updates the status as *Data Extraction Failed*.

You can refresh and update the training status of the model by selecting *Get Status* from the *Actions* (⋮) menu.

i Note

You can also save your model and train your model later.

Locate your model and click *Train* from the *Actions* (⋮) menu to start training the model.

5. After the training is complete, click *Activate* from the *Actions* (⋮) menu to activate the model.
You can deactivate a model by selecting *Deactivate*.

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the *Input* field and click *Test*.

19.2.2 View Similar Case Recommendation

View similar case recommendations in cases.

Procedure

1. Go to the *Case* worklist.
2. Create a new case and enter the subject and description fields.

Similar case recommendation happens at case creation. Hence, you'll only see predictions for cases created after the model is set to active status.

3. Select *Save*.

Find similar cases in the the side pane by clicking the *Similar Cases* (🔍) icon. A confidence score of 0–100 shows the level of accuracy of the match.

19.2.2.1 Similar Cases - Show More

View all similar cases to check for more solutions that apply to your current cases.

To view more similar cases, select [Show More](#) at the top right of the [Recommendations](#) side pane. The [Solution Center](#) opens and displays up to ten similar cases in the top table.

19.3 NLP Classification

Use Natural language processing (NLP) to identify sentiment from email.

The NLP Classification scenario includes the **Sentiment** output field that determines if the vocabulary used in the email from which a case is created, is positive or negative and the degree of positivity or negativity.

! Restriction

Sentiment analysis only identifies sentiment for cases created from email channels.

NLP sentiment classification returns the following possible values:

- Not Available
- Strong Positive
- Weak Positive
- Neutral
- Weak Negative
- Strong Negative

The sentiment value from an email from which a case is created, indicates the overall emotion.

The magnitude of sentiment indicates how much emotional content is present within the email, and is often proportional to the length of the text.

The natural language algorithm differentiates between positive and negative emotion in a case, but doesn't identify **specific** positive and negative emotions. For example, **angry** and **sad** are both negative emotions. However, the sentiment classification only indicates that the text is **negative**, not sad, or angry.

An email with a neutral value can indicate a low-emotion, or email text expressing mixed emotions; containing both high positive and high negative values that cancel out each other.

19.3.1 Configure NLP Classification

As an administrator, you must add, train, and activate the NLP Classification model.

Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [NLP Classification](#) ►.
2. Under [NLP Classification](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Turn on the [Sentiment](#) toggle button to enable sentiment detection.
5. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) ►.

You can refresh and update the training status of the model by selecting [Get Status](#) from the [Actions](#) (⋮) menu.

i Note

You can also save your model and train your model later.

Locate your model and click [Train](#) from the [Actions](#) (⋮) menu to start training the model.

6. After the training is complete, click [Activate](#) from the [Actions](#) (⋮) menu to activate the model.
You can deactivate a model by selecting [Deactivate](#).

i Note

You can adjust the threshold confidence level. The default value of the confidence level is 60.

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Classify](#). The test console invokes the prediction model API using the sample case data you provide.

19.4 Machine Translation

Use machine translation to translate emails.

Machine translation is available for the languages in the following table.

Machine Translation Supported Languages

Source Language	Code	Target Language	Code
Arabic	ar	English (United States)	en
Catalan	ca	English (United States)	en
Chinese	zh	English (United States)	en
Danish	da	English (United States)	en
Dutch	nl	English (United States)	en
English (US – applicable for below rows)	en	Arabic	ar
English	en	Bulgarian	bg
English	en	Catalan	ca
English	en	Chinese (Simplified)	zh
English	en	Chinese (Traditional)	zf
English	en	Croatian	hr
English	en	Czech	cs
English	en	Danish	da
English	en	Dutch	nl
English	en	Estonian	et
English	en	Finnish	fi
English	en	French (France)	fr
English	en	German	de
English	en	Greek	el
English	en	Hebrew	he
English	en	Hindi	hi
English	en	Hungarian	hu
English	en	Italian	it
English	en	Japanese	ja
English	en	Kazakh	kk
English	en	Korean	ko
English	en	Latvian	lv
English	en	Lithuanian	lt

Source Language	Code	Target Language	Code
English	en	Malay	ms
English	en	Norwegian	no
English	en	Polish	pl
English	en	Portuguese (Brazil)	pt
English	en	Romanian	ro
English	en	Russian	ru
English	en	Serbian (Latin)	sh
English	en	Slovak	sk
English	en	Slovenian	sl
English	en	Spanish (Spain)	es
English	en	Swedish	sv
English	en	Thai	th
English	en	Turkish	tr
English	en	Ukrainian	uk
English	en	Vietnamese	vi
Finnish	fi	English	en
French (France)	fr	English	en
German	de	Bulgarian	bg
German	de	Chinese (Simplified)	zh
German	de	Croatian	hr
German	de	Czech	cs
German	de	English	en
German	de	French (France)	fr
German	de	Hungarian	hu
German	de	Italian	it
German	de	Polish	pl
German	de	Romanian	ro
German	de	Russian	ru
German	de	Serbian (Latin)	sh
German	de	Slovakian	sk
German	de	Slovenian	sl
German	de	Spanish (Spain)	es
Hindi	hi	English	en
Italian	it	English	en

Source Language	Code	Target Language	Code
Japanese	ja	English	en
Korean	ko	English	en
Norwegian	no	English	en
Portuguese (Brazil)	pt	English	en
Romanian	ro	English	en
Russian	ru	English	en
Spanish (Spain)	es	English	en
Swedish	sv	English	en

19.4.1 Configure Machine Translation

As an administrator, you must add, train, test, and activate the Machine Translation model.

Procedure

1. Navigate to your user profile, and select [Settings](#) > [All Settings](#) > [Machine Learning](#) > [Intelligent Service](#) > [Machine Translation](#).
2. Under [Machine Translation](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: [Created](#) > [Training in Preparation](#) > [Data Extraction is Pending](#) > [Data Extraction is in Progress](#) > [Data Preprocessing is Pending](#) > [Data Preprocessing is in Progress](#) > [Training Triggered](#) > [Training is Pending](#) > [Training in Progress](#) > [Training Completed or Training Failed](#) > [Active or Inactive](#).

You can refresh and update the training status of the model by selecting [Get Status](#) from the [Actions](#) (⋮) menu.

Note

You can also save your model and train your model later.

Locate your model and click [Train](#) from the [Actions](#) (⋮) menu to start training the model.

5. After the training is complete, click [Activate](#) from the [Actions](#) (⋮) menu to activate the model.
You can deactivate a model by selecting [Deactivate](#).

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Translate](#). The test console invokes the prediction model API using the sample case data you provide.

19.4.2 View Machine Translated Text

View the translated description for the email in the user logon language.

Procedure

1. Navigate to the [Emails](#) worklist and select an email.
2. Click [Show Translation](#).

The machine-translated text appears in the description below the original text.

19.5 Case Summarization

Use Natural Language Processing (NLP) to summarize the text in the subject field and email interactions.

! Restriction

- Text summarization is tested only for English language texts.
- Text summarization is applicable only for email based interactions between a customer and a service agent.
- Text summarization is available only for email interactions generated from Outlook and Gmail.
- For existing cases, a summary is generated only when a new interaction is added (after model activation).
- There's a delay between sending or receiving interaction and summary updates. Text summarization takes place asynchronously from the email communication. The delay depends on the number of interactions in the queue and the number and size of the interactions.

19.5.1 Configure Case Summarization

As an administrator, you must add, train, test, and activate the Case Summarization model.

Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Summarization](#) ►.
2. Under [Case Summarization](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) ►.

[Get Status](#) from the You can refresh and update the training status of the model by selecting [Actions](#) (⋮) menu.

i Note

You can also save your model and train your model later.

You can refresh and update the training status of the model by selecting [Locate your model](#) and click [Train](#) from the [Actions](#) (⋮) menu to start training the model.

5. After the training is complete, click [Activate](#) from the [Actions](#) (⋮) menu to activate the model.
You can deactivate a model by selecting [Deactivate](#).

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a Case ID in the [Input](#) field and click [Summarize](#). The test console invokes the prediction model API using the sample case data you provide.

! Restriction

There's an upper limit on the number of interactions summarized for test purposes. Hence, you may not see all interactions in the summary.

19.5.2 View Case Summarization

View the case summary results.

Procedure

1. Log into your solution and go to [Cases](#).
2. Open a case in detailed view using the  icon.
3. Go to the [Summary](#) tab to view the summary of each interaction.

You can see an overall summary, a summary of agent-generated interactions, and a summary of customer interactions.

19.6 Case Topic Analyzer

Use the [Case Topic Analyzer](#) machine learning model to view trending topics from cases.

The [Case Topic Analyzer](#) machine learning model analyzes keywords from case descriptions and groups cases based on labels assigned to a model. A label is a topic which acts as a heading for a list of similar keywords.

Prerequisites

To ensure accurate results from the model, your system must contain historical data (cases) in order to train the model. Ideally around 3000 cases (minimum) with cases having meaningful text content in the case description (multiple sentences which conveys the issue).

i Note

When setting up a demo or test environment, you need to consider the following points:

- If most cases are talking about totally different distinct problems, topic detection will not happen properly.
- Cases must have meaningful text which conveys the issue correctly.

19.6.1 Configure Case Topic Analyzer

As an administrator, you must add, train, test, and activate the Case Topic Analyzer model.

Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Case Topic Analyzer](#) .
2. Under [Case Topic Analyzer](#), click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click [Save and Train](#) to start training the model.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) .

You can refresh and update the training status of the model by selecting the ↻ (refresh) icon.

5. After the training is completed, select your model to assign a label to the list of keywords.

i Note

A label is a topic which acts as a heading for a list of similar keywords.

You must set appropriate topic label which matches your business.

6. Select [Settings](#) to see a number of topics with keywords associated with each one.
7. Enter a topic label.
8. Click [Save](#).
9. Click the ⚙️ (activate) icon to activate the model.

i Note

- You can deactivate a model by selecting the ⏻ (deactivate) icon.
- You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field, and click [Test](#).
- Only English language is supported currently for the Case Topic Analyzer models.

19.6.2 View Topic Analyzer

View trending topics from case descriptions.

After you train and activate the case topic analyzer model, you can view trending topics in [Topic Analyzer](#). New cases created in the system are grouped according to the topics they belong to.

You can choose to view the trending topics in the form of cards or lists that display the topic name, the rank, number of cases, and percentage trend. The topics are ranked according to the number of cases for the topics for the calendar year. The percentage trends are calculated on monthly basis.

You can filter trending topics using a date filter with values such as *This week*, *This Month*, and *This Year*. By default, the date filter is set to filter trending topics for the current week. For a selected filter, topics will appear only if it has cases from that selected time period. However, the overall topic rank for the calendar year and percentage trend (month to month) does not change.

The trending topics display information for all topics in the following forms:

- **Bar charts** - displays the number of cases for topics
- **Graphs** - displays topic trends across months.
- **Keywords clusters** - displays list of trending keywords as word clouds and as lists.

When you select a trending topic, the list, graph and keywords display information specific to the selected topic only. You can view the list of associated cases and navigate to specific cases.

19.7 Profanity Check

Use profanity check to define a set of pejorative vocabulary.

Profanity check is currently available only in English.

19.7.1 Configure Profanity Check

As an administrator, you must add, train, test, and activate the Profanity Check model.

Procedure

1. Navigate to your user profile, and select ► [Settings](#) ► [All Settings](#) ► [Machine Learning](#) ► [Intelligent Service](#) ► [Profanity Check](#) .
2. Under [Profanity Check](#) , click the create icon (+) to add a model.
3. Enter a name for your model.
4. Click [Save and Train](#) to save your model and start the training.

The model goes through the following statuses during the training: ► [Created](#) ► [Training in Preparation](#) ► [Data Extraction is Pending](#) ► [Data Extraction is in Progress](#) ► [Data Preprocessing is Pending](#) ► [Data Preprocessing is in Progress](#) ► [Training Triggered](#) ► [Training is Pending](#) ► [Training in Progress](#) ► [Training Completed or Training Failed](#) ► [Active or Inactive](#) .

You can refresh and update the training status of the model by selecting [Get Status](#) from the [Actions](#) (⋮) menu.

i Note

You can also save your model and train your model later.

Locate your model and click [Train](#) from the [Actions](#) (☰) menu to start training the model.

5. After the training is complete, click [Activate](#) from the [Actions](#) (☰) menu to activate the model.
You can deactivate a model by selecting [Deactivate](#).

i Note

You can use the test console to test if the model is functioning correctly. Select your model, enter a description in the [Input](#) field and click [Check Profanity](#). The test console invokes the prediction model API using the sample case data you provide.

19.7.2 View Profanity Check

View the results of profanity check.

Procedure

1. Navigate to the [Email](#) worklist.
2. Reply to an email.
3. Draft a reply with profane language.

The system opens a pop up with the profane text.

19.8 Intelligent Services FAQ

Here are some frequently asked questions (FAQ) and corresponding answers about Machine Learning.

What are Intelligent Services?

Intelligent Services is a collection of machine learning models and scenarios embedded in SAP Service Cloud CNS. Intelligent Services is not a separate product. You don't require any additional licensing, deployment, or upgrades. You can manage everything through the SAP Service Cloud CNS administrator interface.

Is there any impact on the system performance as the data is cached?

There's no impact on day-to-day system performance as the data is cached and stored in the solution. All compute-intensive operations, such as training the model, happen outside of your SAP Service Cloud CNS tenant.

How many models are active at any time?

Only one model can be active in production. However, you can create multiple models.

Are there extension fields to the model?

Extension fields aren't available right now.

How many category levels does Case Categorization support?

Case Categorization supports upto four levels.

Related Information

[Service Ticket Intelligence](#)

20 Mashups

Learn how to configure, use, and access mashups.

Mashups are used to integrate data from SAP's cloud solution with data provided by an online Web service or application. Users can access the content provided by these Web services and applications, and use it in their daily work. Mashups can include Web searches, company or industry business information, or online map searches.

Prerequisites

As an administrator, you must configure the following settings:

- Enable the mashup service for business roles
 1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► and under [Users and Control](#), select [Business Roles](#).
 2. From the [Business Roles](#) window, click any of the [Business Role ID](#) to open the details.
 3. Under [Business Services](#), click the **+** icon to open the [Add Business Services](#) window.
 4. Select the check box for [mashupService](#) and click [Add](#).

i Note

You can create new HTML mashups based on a mashup category provided by SAP.

- Add the mashup URL in the Content Security Policy settings
 1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► and under [General](#), select [Content Security Policy](#).
 2. Scroll down to the section [Frame Source](#) and click the **+** icon.
 3. Enter the hostname URL and click the **✓** icon.

❖ Example

If you're creating a mashup with the URL `https://my123456.s4hana.ondemand.com/ui?sap-ushell-config=headerless#ServiceOrder-create` then, add `https://my123456.s4hana.ondemand.com/` in the [Frame Source](#).

How to Access Mashups

Once a mashup has been activated and made visible on screens, then the mashup can be displayed as an embedded screen.

Features

We support only HTML Mashup that embeds an HTML or JavaScript based Web page directly on a screen.

Configuring Mashups

You can configure mashups as follows:

- **Centrally:** Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► and under *Foundation*, select *Mashup Authoring*.
- **On a particular screen:** Use the adaptation function on any screen enabled for mashups.

You have the flexibility to configure mashups in the following ways:

- Activate mashups for use by all users or deactivate unwanted mashups.
- Change the way in which mashups are displayed, and test and publish the changes to all users.
- Create new mashups by copying an existing mashup and adapting the settings in the copied mashup.

20.1 Create HTML Mashups

Create HTML mashups to embed an HTML or JavaScript based Web page into a screen of your SAP cloud solution.

i Note

Some Web services may pass your business data to a third-party organization, for example, account data is passed to a search engine when performing a reverse lookup in an online address book. We recommend that you check whether the mashup conforms to your company's data privacy policies before activating the mashup.

Prerequisites

You must know the URL of the Web site that you want to embed.

Procedure

1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► and under *Foundation*, select *Mashup Authoring*.
2. From the new tab, click the plus icon (+) to create a mashup.
3. Under *General Information*, enter a name and a description for the mashup.

i Note

By default, the status of the mashup is *Active*. You can also change the status to *Inactive* if necessary.

4. Adjust the display height, as required. The default height is 500 pixels.
5. Under *Input Parameters*, enter the URL in the *URL* field and click *Extract Parameters*.
The system extracts the parameters from the URL and adds them to the table, where you can edit constants and bind parameters to the screen out-ports available for the selected port binding.

i Note

- For URLs that don't use queries you can manually add curly brackets around terms that should act as placeholders. For example, in the URL <https://mail.google.com/mail/#search/SAP>, you can replace the word **SAP** with a search term in curly brackets, for example, **{term}**. If you then enter <https://mail.google.com/mail/#search/{term}> in the *URL* field and click *Extract Parameters*, the word in brackets is extracted as a parameter.
- If you're using parameters, make sure that you don't add any sensitive information.

6. Once you finish, click *Show Preview* to check the details.
7. Save your changes.

20.2 Copy Existing Mashups

You can create a copy from the existing mashups.

1. Go to ► *User Menu* ► *Settings* ► *All Settings* ► and under *Foundation*, select *Mashup Authoring*.
2. From the new tab, select the mashup you want to copy by clicking on the mashup name.
3. From the *Mashup Details* screen, click *Duplicate*.
The system opens a copy of the mashup.
4. Enter a new name for the mashup and adapt the configuration settings, as required.
5. Save your changes.

20.3 Delete or Deactivate Mashups

Delete or deactivate mashups that you and other users have created.

1. Go to ► *User Menu* ► *Settings* ► *All Settings* ► and under *Foundation*, select *Mashup Authoring*.
2. Select a mashup from the list.
3. Click the *Delete* icon, or use the toggle option to *Deactivate*.
If you select deactivate, the mashup is no longer visible on screens. If you select delete, the mashup is removed from the *Mashup Authoring* view and is also deleted from all screens for which it had been made visible.

20.4 Add Mashups as Tabs

As an administrator, you can add a mashup as a new tab.

Using page layouts and visibility property, you can control who is allowed to use the mashup.

To add a mashup as a tab, follow these steps:

1. **Create a tab on a screen.**

1. Navigate to a screen where you want to add the mashup.
2. Go to ► [User Menu](#) ► [Start Adaptation](#) ►. The system opens in the adaptation mode.
3. Hover over the area where you see a plus icon (+).
The system highlights the area with a red border along with a tool tab
4. From the tool tab, select the pencil icon to edit. A new tab opens.
5. Click the plus icon (+) and enter a tab name.
6. Click [Apply](#).
The new tab is added to the header of the screen.

2. **Add a mashup to the tab.**

1. Open the new tab and hover over the empty space to enable the red highlight with the tool tab.
2. From the tool tab, select the pencil icon to edit. A new tab opens.
3. Select the arrow available next to [Add Mashup](#) to open a new window.
4. Click the arrow available next to the required mashup to display the properties.
5. If necessary, you can adjust the appearance of the mashup by selecting one of the following options under [Edit Properties](#).

i Note

The options available to you may vary depending on the mashup.

- [Show Header](#): Select the [Show Header](#) checkbox if you want to see the mashup title on the screen.
 - [Height \(%\)](#): You can define the height of HTML Mashups to be displayed in the available screen.
For example, if you set the value of Height (%) to 100, the newly added mashup occupies the full height of the screen.
6. Click [Apply](#).
 7. To save your settings, click [End Session](#) on the top of the screen.

21 Notifications

Notifications are a way to let you know that something new has happened so you don't miss anything that can be worth your attention.

i Note

Notifications is currently available only for **Approvals**.

Notifications display the category of actions, the time occurred, unsubscribe from the email, and more details. Clicking or tapping the icon , opens a dropdown list that has the recent notifications with the entity type information, actions, and option to view the entire detail about a particular notification. The dropdown displays maximum five items in the list.

Each category of notification has its own set of actions. Subscribe, Unsubscribe, Undo, Approve, Reject, Dismiss are few of the common actions. You can add notes when you approve or reject a notification. When you click, [View All](#) you're directed to the notification center where you can see the entire list of notifications in a separate tab.

In the notification center, you can select the kind of notifications you want to see from the variant dropdown or search for a particular notification. Notification category indicates the type of entity and is differentiated by the icon used in front of each notification item. Click or tap the [More](#) link to navigate to a particular entity in a separate tab.

22 Approval

As an administrator, you can create approvals for streamlining your business processes.

Approvals are typically routed to your reporting line manager, though there may be multiple levels of approval required for certain objects or conditions. By default, once the approval workflow is triggered, the assigned approvers receive in-app notifications.

Important Facts About Approvals

- One business transaction can have multiple active approval processes.
- The approval request notification appears in the notification area of the system for the involved employees and managers.
- If the approver finds an unsatisfactory transaction, the approver can add a comment and return it to the employee for revision. The employee then revises the transaction and submits it again for approval. Once the business transaction is correct, it can be further processed.
- You can only delete approval processes that are in the *Draft* or *Active* state.

22.1 Configure Approvals

As an administrator, you can configure approvals. You define the entity, event, and condition based on which a notification is sent to the approvers.

Procedure

1. Go to your user menu and select ► *Settings* ► *All Settings* ► *Business Flow* ► *Approval* ►.
2. Choose *+ New*. You can see a list of approvals that are available.
3. Enter a name in the *Process Name* field.
4. Select your language and enter subject and description in the *Task subject* and *Task Description* fields.
5. From the *Entities* dropdown, select an entity.

i Note

Approvals are available only for *Case*.

6. Select the *Event Type*.

i Note

You can add multiple approval steps using the [Add](#) button and configure conditions and approval for each step.

7. Optional: Define conditions to specify criteria that must be fulfilled to trigger approvals.

Conditions can be based on standard fields or extension fields.

You can define more than one conditions using logical operators [AND/OR](#). Use the [AND](#) operator to define conditions within the same group. To add more than one condition, use the [OR](#) operator. Add a new group to define the [OR](#) operator amongst the approval conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the approval process is triggered.

- a. Choose [+OR Condition](#). By default, the trigger is set to [Conditions are met](#). While using operands, the first condition can only be defined using the [AND](#) operand.
- b. Select the field, comparator and the value or field. When previous value and current value influence the action, choose [Previous value](#) to consider the value before the event occurred.
- c. Specify an appropriate operator, and a value.

i Note

You can also create conditions using custom fields. Set up custom fields before you create autoflow rules.

- d. If desired, specify conditions with the following options:
 - To add an [AND](#) expression, click the [Add](#) icon ([+](#)).
Conditions within a group are logical [AND](#) expressions. If all the conditions within a group are met, the condition group is considered fulfilled.
 - To add an [OR](#) expression, click [+ OR Condition](#).
If all conditions in at least one group are met, then the overall condition for the rule is considered fulfilled.
 - To delete a condition group, click the [Delete](#) icon ([🗑](#)).
 - To remove a condition, click the [Dismiss](#) icon ([×](#)).
8. Set up the approval.
 - a. To enable email notifications for approvals, select [Enable Email Notifications](#).

i Note

Your administrator must define email templates for [Approval Request](#) and [Approval Response](#) template types for [Case](#) object type. For more details on defining templates, see Related information.

1. Select the [Approval Request](#) and [Approval Response](#) templates.
- b. In the [Define Work Distribution](#) section, define the approvers. You can choose from the following options:
 - [Direct Approvers](#): You can select specific employees responsible as direct approvers, provided that they possess the corresponding access rights to the task type and business transaction. Such approvers also require read access to the relevant business transaction data and write access to the notes of the business transaction.

i Note

You can enable approvers to trigger actions directly from the details of the item to be approved. To do so, from the [Approval](#) tab of any related item, make the [Actions](#) column visible on the user interface.

- [Responsibles](#): You can select any responsible employees associated with the Approval. It can include [Processor of case](#), [Manager of processor for case](#), [Suggested recipients](#) and so on.

i Note

- To include multiple approvers, enable the [All Approvals Required](#) toggle button. However, sequential approval is not supported.
- To enable automatic approval of your business process (only when there are no approvers determined for a step), turn on the [Automatic Approval](#) switch.

9. Choose [Save](#) and [Activate](#).

You can select the approvals that you define here in the [Approval](#) step type of [Case](#) designer.

Related Information

[Create Email Templates \[page 88\]](#)

23 Autoflow

Learn how to automate your business processes using [Autoflow](#).

Defining an autoflow involves creation of rules that can trigger actions such as send event notifications to integrated solutions, in-app notifications to users, and trigger automatic emails for business objects.

23.1 Configure Autoflows

As an administrator, you can configure autoflows in [Settings](#).

You define the entity, event, and conditions based on which the autoflow triggers actions.

23.1.1 Configure Conditions

As an administrator, you define the conditions based on which the actions are triggered for an autoflow.

Context

Conditions are structured in groups and include criteria that must be fulfilled if a workflow action must be triggered. More than one condition can be defined using logical operators [AND/OR](#). The [AND](#) operator can be defined using condition within the same group. To add more than one condition, use the [OR](#) operator. Add a new group to define the OR operator amongst the workflow conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the workflow rule has been triggered.

Procedure

1. Go to your user profile and select [Settings](#) > [All Settings](#) > [Business Flow](#) > [Autoflow](#) .
2. Choose [+ New](#). You can see a list of autoflows that are available.
3. From the [Entities](#) dropdown, select an entity.
4. Select the [Event Type](#).

This determines when an autoflow rule must be evaluated and triggered. Timing is relative to an event that occurs to an entity. For example,

- On create - The rule is applied at every create of an entity instance.

- On update - The rule is applied at every update of an entity instance.

→ Recommendation

- Autoflow rules that are set to **on create** and **on update** are run synchronously while the transaction object is being created or saved in user session. Too many rules affect system performance. Therefore, it is recommended that you consolidate these rules. Based on the entity you have selected, there are various event types for managing autoflows.
- Autoflow rules affect all users using the entity within a process. Plan and define rules accordingly.

5. Optional: Define conditions to specify criteria that must be fulfilled to trigger an autoflow.

The following operators are available for defining conditions:

- Equal to
- Not Equal to
- Greater than
- Greater than or equal to
- Less than
- Less than or equal to
- Contains
- Does not contain
- Updated

Conditions can be based on standard fields or extension fields.

You can define more than one conditions using logical operators **AND/OR**. Use the **AND** operator to define conditions within the same group. To add more than one condition, use the **OR** operator. Add a new group to define the **OR** operator amongst the autoflow conditions such that if conditions in at least one group are met, then the overall condition is considered fulfilled and the autoflow process is triggered.

If u want to consider the previous value before the event occurred as a part of a condition, select the **Previous Value** checkbox. The value - **Before Entity Change**, can be used to define conditions based on field value changes. For example, an email is sent when the field - **Escalation Status** is changed from **Not Escalated** to **Escalated**.

- a. Choose **+OR Condition**. By default, the trigger is set to **Conditions are met**. While using operands, the first condition can only be defined using the **AND** operand.
- b. Select the field, comparator and the value or field. When previous value and current value influence the action, choose **Previous value** to consider the value before the event occurred.
- c. Specify an appropriate operator, and a value.

i Note

You can also create conditions using custom fields. Set up custom fields before you create autoflow rules.

- d. If desired, specify conditions with the following options:
 - To add an **AND** expression, click the **Add** icon (+) .
Conditions within a group are logical **AND** expressions. If all the conditions within a group are met, the condition group is considered fulfilled.
 - To add an **OR** expression, click **+ OR Condition**.

If all conditions in at least one group are met, then the overall condition for the rule is considered fulfilled.

- To delete a condition group, click the delete icon ().
- To remove a condition, click the dismiss icon ().

After you define conditions, you must define actions for the autoflow.

Related Information

[Configure Actions \[page 195\]](#)

23.1.2 Configure Actions

Configure actions for autoflows.

After you have defined conditions for an autoflow, you can configure actions. You can configure autoflows to trigger the following actions:

- [Send Emails \[page 195\]](#)
- [Send Event Notifications \[page 197\]](#)
- [Send Info Notifications \[page 198\]](#)

Related Information

[Configure Conditions \[page 193\]](#)

23.1.2.1 Send Emails

You configure an autoflow to send email messages when the conditions are met for business objects.

Context

After you configure conditions, define sending email as the action of the autoflow.

Procedure

1. To send an email, choose [Send Email](#) from the [Action](#) dropdown.
 - a. In the panel that appears, enter the following details:

Field Name	Description
Subject	Enter the subject of your email.
Template	Choose a template.
Sender Name and Email	Select the name and email ID of the sender. This depends on the email channels defined for the entity selected.
Role Based Recipients	Select the recipients of this email based on their role. These roles vary based on the entity you have selected. i Note If there are no roles associated with the entity, you cannot see any recipients.
Recipients	Select the recipients of this email irrespective of their roles. You can select from a list of accounts, individual customers and contacts that are maintained in the master data.

2. To add more actions, select the [Add](#) icon (+) .

The [Action](#) dropdown displays.

- Select [Send Emails](#) to send email.
- Select [Send Event Notifications](#) to send event notification to a third-party communication system.
- Select [Send Info Notifications](#) to send in-app notifications to selected users.

3. You can disable or delete actions from the [Action](#) menu:
 - a. To delete an action, you can select the [Delete](#) icon (🗑).
 - b. To disable an action, you can select the [Disable](#) icon.

4. Select [Save](#).

5. Select [Activate](#).

You can select the autoflows that you define here in the [Autoflow](#) step type of [Case](#) designer.

Related Information

[Configure Conditions \[page 193\]](#)

[Send Event Notifications \[page 197\]](#)

[Send Info Notifications \[page 198\]](#)

23.1.2.2 Send Event Notifications

You configure an autoflow to send event notifications to third-party systems, when the conditions are met for business objects.

Context

After you configure conditions, define sending event notification as the action of the autoflow:

Procedure

1. To send event notifications to a third-party communication system integrated with your solution, select [Send Event Notification](#) from the [Action](#) dropdown.
 - a. In the panel that appears, enter the following details:

Field Name	Description
Event Name	Enter a name.
Communication System	Choose a communication system. For more information, see Communication Systems [page 300] .
Subscribe Path	Enter the URL of the third-party system.

2. To add more actions, select the [Add](#) icon (+).

The [Action](#) dropdown displays.

- Select [Send Emails](#) to send email.
- Select [Send Event Notifications](#) to send event notification to a third-party communication system.
- Select [Send Info Notifications](#) to send in-app notifications to selected users.

3. You can disable or delete actions from the [Action](#) menu:

- a. To delete an action, select the [Delete](#) icon (🗑).
- b. To disable an action, select the [Disable](#) icon.

4. Select [Save](#).

5. Select [Activate](#).

You can select the autoflows that you define here in the [Autoflow](#) step type of [Case](#) designer.

Related Information

[Configure Conditions \[page 193\]](#)

[Send Info Notifications \[page 198\]](#)

[Send Emails \[page 195\]](#)

23.1.2.3 Send Info Notifications

You configure an autoflow to send in-app notifications to selected users, when the conditions are met for business objects.

Context

The info notifications appear in the Notifications area.

After you configure conditions, define sending info notification as the action of the autoflow:

Procedure

1. To send info notifications to users, select [Send Info Notification](#) from the [Action](#) dropdown.
 - a. In the panel that appears, enter the following details:

Field Name	Description
Language	Select a language from the dropdown.
Subject	Enter a subject.
Description	Enter a description for the notification.
Role based	Select roles from the dropdown, to send notifications to users with specific roles.
Employee	Select employees from the list, to send notification to specific users.

2. To add more actions, select the [Add](#) icon (+) .

The [Action](#) dropdown displays.

- Select [Send Emails](#) to send email.
- Select [Send Event Notifications](#) to send event notification to a third-party communication system.
- Select [Send Info Notifications](#) to send in-app notifications to selected users.

3. You can disable or delete actions from the [Action](#) menu:
 - a. To delete an action, select the [Delete](#) icon (🗑).
 - b. To disable an action, select the [Disable](#) icon.

4. Select [Save](#).
5. Select [Activate](#).

You can select the autoflows that you define here in the [Autoflow](#) step type of [Case](#) designer.

Related Information

[Configure Conditions \[page 193\]](#)

[Send Event Notifications \[page 197\]](#)

23.2 Monitor Autoflow Logs

As an administrator, you can monitor logs for autoflow rules.

When the system triggers an autoflow rule, an entry gets added in [Rule Logs](#). You can use the rule logs to troubleshoot autoflow errors. The [Rule Logs](#) screen displays logs for the last two weeks.

You can use the following filters to filter logs:

- [Date](#) - based on the time intervals
- [Service Name](#) - based on entities used while creating the rule
- [Rule Name](#) - name defined for the rule
- [Action Type](#) - action type defined for the rule
- [Action Status](#) - status of the action

To view the Autoflow logs, go to [User Menu](#) and select [Settings](#) [All Settings](#) [Business Flow](#) [Autoflow](#) [More](#). Select [More](#) and then select [Rule Logs](#).

The [Rule Logs](#) tab displays logs for custom rules and SAP rules. You can search a rule by entering [Source Document ID](#) in the search field. You can view and sort the logs using the following fields:

- Business Object - Business entity for which the rule triggered
- Source Document ID - Display ID of the business object
- Event Name - Event for which the rule gets triggered
- Execution Time - Time at which the rule was executed
- Execution Status - Overall status for the rule.
One of the following status displays:
 - [Not Executed](#) - for rules triggered, however, not executed due to condition evaluation being false
 - [Completed](#) - for rules triggered and executed successfully
 - [Not Completed](#) - for rules triggered and executed with errors

When you select a rule from the [Rule Name](#) column, the log messages display the status and more details for the actions that were triggered for the rule.

For every action of either same or different types, there will be an entry in [Log Messages](#). For example, if there were two emails and an event notification sent for an autoflow rule, there will be three entries in the log messages for the rule.

You can switch between the following modes to log rules and view their status:

- The [Default](#) mode logs only the rules for which the condition evaluation result is successful.
- The [Log All Rules](#) mode logs all rules irrespective of the conditions evaluation result is successful or not.

i Note

The [Log All Rules](#) mode switches back to the [Default](#) mode after 5 minutes.

24 Products

Products worklist helps you to maintain the various aspects of products in the application.

A product is the item offered for sale and it can be a service or a physical item. A product can be of real or virtual form. Each product is made at a cost and sold at a price and the price depends on the quality, market, the marketing, and the targeted segment. In the application, generally, products have a status, name, and ID.

As a service agent, you can use the SAP Service Cloud Version 2 to work with products as a criterion for categorizing, routing, or analyzing the text of customer tickets and perform appropriate action.

24.1 Configure Products

Administrators can configure products in [Settings](#).

Find [Settings](#) in the user profile menu at the top of the screen.

24.1.1 Configure Product Groups

As an administrator, you can categorize products into product groups based on the shared attributes like usage, features, production processes and so on. A product group can also be the market or customer segment in which the products are sold or the prices at which they're offered.

You can configure a product group by combining other products and classifying them as a single level or multilevel group.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Product](#) ► [Product Groups](#) ►.
2. Select the add icon (+) at the top of the table to add a new product group.
3. Enter a name and ID for your new product group.
4. Select the confirm icon (✓) to save changes.

24.1.2 Configure Unit of Measure

As an administrator, you can define new trading units of measurement and maintain the predefined units in the application.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Product](#) ► [Units of Measure](#) ►.
2. Select the add icon (+) at the top of the table to add a new unit of measure.
3. Enter a code, description, and decimals for your unit of measure.
Custom codes have to start with **Z** followed by two numbers. For example: Z12
4. Toggle the **Enabled** button to activate the unit of measurement.
You can delete a unit of measurement by selecting the delete icon () in the [Action](#) column.

i Note

Use the **Translate** toggle button to translate the description into any of the supported languages.

24.1.3 Configure Language

As an administrator, you can select the languages that are to be used in the Products worklist. The selected languages are used for translating the contents on the system and communication.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [General](#) ► [Language & Region](#) ►.
2. Select the [Language](#) tab.
3. Use the toggle button under [System Language](#) and [Communication Language](#) against each language to enable or disable.

24.1.4 Configure Replication of Products from S/4HANA

As an administrator, you can manually create products or upload the products data from SAP S/4HANA using the migration process found under [Settings](#).

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Inbound Configuration](#) ►.
2. Select the inbound configuration [Replicate Product from SAP S/4HANA](#).

24.1.5 Configure Replication of Product Groups from S/4HANA

As an administrator, you can manually create product groups or upload the product groups data from SAP S/4HANA using the migration process found under [Settings](#).

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Inbound Configuration](#) ►.
2. Select the inbound configuration [Replicate Product Group from SAP S/4HANA](#).

24.2 Create Products

You can create a product from the Products worklist.

In the Products worklist:

1. Click the create icon (+) to create a new product
The [Product Name](#), [Unit of Measure](#), and [Status](#) fields are mandatory.
2. Click [Save](#), the status of the product is set to [In Preparation](#) by default.
Based on your requirement, you can change the status of a product to [Active](#), [Blocked](#), or [Obsolete](#).

Note

Products that are manually created won't be exchanged with products that are migrated from external systems such as SAP S/4HANA

24.3 Manage Products

Explore all the options to manage your products and product groups in the solution.

You have multiple options such as filtering, searching, and adding products in the solution. Your administrator configures many of these options based on how your organization uses products.

24.3.1 Products Worklist

The *Products* worklist displays all products in the solution.

Choose *Products* from the app menu () to view all products that are currently maintained in the solution.

Use the search () option to search for a product by keyword, such as ID, External ID, or name. Filter all the maintained products either by *Status* or *Product Group* to view a specific set of products. You can filter the status as per:

- In Preparation
- Active
- Blocked
- Obsolete

24.3.1.1 Products Quick View

Products quick view allows you to access information about a product without having to leave the products worklist.

Click on the name of a product to launch the quick view and a new tile opens with details such as product description, sales information and change log. You can also edit the status, product group, unit of measure and other field values inline.

24.3.1.2 Products Detail View

Access a more detailed view of a product.

Click the *Open in Detailed View* () icon next to a product or in a product's quick view to launch the detailed view of the product. You can expand the header section by clicking the () icon to view information such as created on, created by, updated on, and updated by. You can also edit fields such as:

- Name
- Status
- Unit of Measure

- Product Group
- Description

In this view, you get a detailed insight into a product and all its attributes through the various tabs. The different tabs and their functions are as follows:

Product Description

In the [Product Description](#) tab, you can edit the description or add additional textual information to a product. The description maintained here can be part of the external communication with your customers, for example, in output forms. You can also save the information in any of the supported languages.

Click the add icon (+) to add a new description. You can edit or delete an existing description by clicking the more icon(⋮). Use the filter icon (🔍) to filter the information based on languages.

Languages

In the [Languages](#) tab, you can manage the translation of your product name in different supported languages. You can configure the system language in [Settings](#).

Click the add icon (+) to add a new language. You can edit or delete an existing language by clicking the delete icon(🗑️). Use the filter icon (🔍) to filter the product based on languages.

Attachments

In the [Attachments](#) tab, you can manage all the attachments of a product such as documents, images, and certificates. Based on the usage, you can upload an attachment type, for example, an image to represent the product in header section or a supporting document for the product.

Click the add icon (+) to upload an attachment as link or click [Browse files](#) to upload a file.

Changes

In the Changes tab, you can view the change history associated with the product in a chronological order. You can see various change details such as changed by, date and time, value changes and so on.

Click the icon (🔄) to refresh the list.

Mapping for Integration

In the [Mapping for Integration](#) tab, you can view the configuration between the external ID and business system ID. Use the icon (✓) under the column [Main](#) to activate or deactivate a mapping.

Sales

In the [Sales](#) tab, you can view the existing sales organization details or add a new sales organization with a distribution channel and sales unit of measure. Click the add icon (+) to add a new sales organization.

i Note

When the status of the product within a sales organization is active and if you change the status of the product to blocked in the header section, then the status of the product in all associated sales organization is also set to blocked. However, if you revert the status of the product to active in the header section, then status in the sales organization doesn't change. You need to set the status of the product to active in the sales organization, case by case.

If the status of a product is blocked in the header section, then you can't change the product's status for corresponding sales organization.

Quantity Conversion

In the [Quantity Conversion](#) tab, you can maintain multiple conversion rates for a product. Click the icon (+) to add a new entry. Enter the quantity and choose the unit from the dropdown and enter the corresponding quantity and choose the corresponding unit from the dropdown.

During pricing, the quantity conversion takes effect if the unit of measure requested in the document differs from the price unit maintained in the price list. As a prerequisite, the quantity conversions must be maintained in the product master data.

24.3.2 Adaptation of Products View

As an administrator, you can change the look and feel of the products worklist for all users by changing layout settings. You can add or remove columns based on your organization's requirement.

24.3.2.1 Customize Your Solution

As an administrator, you can customize the UI look and feel for all users.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#).

The system displays an orange line in the header to indicate that you are in adaptation mode.

2. Make your changes. Note that any change you make applies to only the main layout.
3. When your changes are complete, click [End Session](#) in the header area.

24.3.2.2 Manage Fields

Add, hide, and reorder fields.

Procedure

1. Navigate to your user profile on the top-right corner of your screen, and from the dropdown list click [Start Adaptation](#). The system opens in the adaptation mode and the section where the changes can be made is marked with a thin orange outline.
2. Hover over the desired section to view the editing icons.
3. Select the **Edit Section** icon (✎) to open the [Section Items](#) pop-up box.
 - a. To change the sequence of fields, drag the **Move** icon (☰) next to a field and move the fields around.
 - b. To hide fields, click the **Hide Field** icon (✕).
 - c. To add fields, select the **Add from list** icon (≡+) to bring up the fields available in the entity. You can select one or multiple fields at a time.
4. Click [Apply](#).

25 Output Management

Output management comprises all activities related to the output of documents. Form-based documents can be output on an ad hoc basis or as an integrated part of a business process.

25.1 Maintain Main Templates

As an administrator, you can create company-specific main templates for use in form templates.

Form main templates are used to define the logo, header, sender address, and language-dependent footer that are used in form-based business documents.

SAP provides one default form main template for your solution. If you require a different logo or sender address for different companies or org units in your organization, you can create additional form main templates. These can then be assigned to the relevant company/org unit.

If you do not want to use any form main template for a business document, you can deactivate the use of main templates in the form template. If you specify that a main template must not be used for a form template, then you can alternatively create a unique header, logo, and footer for the form template using Adobe® LiveCycle Designer.

When a business document is output, the system uses the following logic to determine which form main template to use:

- The system searches for a form main template assigned to the company/org unit in which the business document was created. For example, if an invoice is created in the invoicing unit *Sales US West Coast* and is to be sent by email, the system tries to find a main template assigned to the org unit *Sales US West Coast*.
- If no match is found, the system searches upwards in the organizational structure for a main template assigned to a higher-level org unit.
- If no match is found, the system uses the default main template.

25.1.1 Create New Form Main Templates

If you use templates, creating company-specific main templates makes all templates consistent, saving you time. You can define company-specific main templates to use as a basis for other forms.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Main Template](#) ►.

2. Show [Form Main Templates](#).
3. Click [New](#).
4. In the [Create Main Template View](#) tab, enter a name and assign org unit (if any) for your new main template.

Use the [Is Default](#) toggle button to mark the template as the default template.

5. In the [Design Main Template](#) tab, enter the header and footer information.
 - a. In the [Header](#) tab, upload a logo.

Adjust the size and alignment of the logo, as required. The file formats .gif, .bmp, .jpg, .png, and .tif are supported. The logo must be lesser than 40 mm in width and 20 mm in height. If you upload a graphic that is larger than 40 mm x 20 mm, it is resized automatically.

- b. Use the [Header Divider](#) toggle button to add a graphical divider on your form.
 - c. Enter your company's address as the sender address. The sender address is shown above the recipient's address in a letter.
 - d. In the [Footer](#) tab, choose the number of footer columns you want to display.

Enter the footer text exactly as you would like it to appear on the form. You can maintain up to four footer blocks with a maximum of 10 lines per footer.

Note that the width of each footer block is equally divided over the available space. If a line of text in the footer is too long for the current font size, the system automatically enters a line break. If this automatic line break does not meet your needs, you can enter the text after the line break as a separate line.

6. Preview your changes by refreshing the screen.
7. Save your changes.

You can now assign the main template to one or more companies, org units, or output channels.

25.1.2 Edit Form Main Templates

As an administrator, you can update the main template design or assign organizational units to the template.

Procedure

1. Go to your user profile and navigate to ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Main Template](#) ►.
2. Select any main template that you want to edit and click .
3. Update the name and org unit details in the [Create Main Template View](#) tab or the header and footer details in the [Design Main Template](#) tab.
4. Save your changes.

25.2 Maintain Form Templates

Use form templates to define the content and layout of documents that can be output from the system.

Form templates are in the form of print forms that are read-only portable document format (PDF) documents that are generated from data stored in the system. Print forms can be printed, sent as an email attachment, or faxed to business partners.

Variants

Variants of a form template are copies of the content displayed in other languages, or with information specific to that country/region. Language variants are used to provide translations of form templates into other languages.

When a document is output from the system, the system automatically determines which language version to use. Most form templates are intended for internal review. Therefore the system uses the form template matching the language selected by the user on the log on screen. Certain objects are intended for review by a business partner or customer, and the system selects a template based on the recipient language.

Language variants are determined by the user log-on language, or in certain cases, by the preferred language of the receiving business partner. For example, if a customer in a German company sends product information to a business partner in France, the system searches for a French language variant of the form template and uses this to generate the document. If a language variant is not available in the recipient's language, then the form template in the fallback language English is used.

You can use the appropriate buttons to delete form templates or variants.

! Restriction

It is not possible to delete a form template that has already been used to output a document because that template is referenced in a document's output history.

25.2.1 Create Form Templates

You can create custom form templates, but we recommend editing templates provided by SAP, which saves you setup time.

Context

→ Recommendation

While it's possible to create new form templates using this procedure, we recommend that you edit form templates provided by SAP or partners instead. Editing provided templates means that you don't have

to adapt any output settings. It is not possible to copy or edit print forms that contain legally-sensitive content.

Procedure

1. Go to your user profile and navigate to ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Choose a predelivered template, click  and select [Copy Template](#).
3. Enter a name and description and select a language.
4. If you are creating a form template specific to a language, select the language. If the template is not specific to language, leave this field blank.
5. Turn on the [Form Main Template](#) toggle button to use master templates to determine the layout of the template.
6. Save and publish your changes.

When a new version of a predelivered template is released, the new version of the predelivered template is automatically set to active. To avoid any changes to the existing version, create a copy of the predelivered template and use the custom template while selecting the form template. If you want to make changes in the newer predelivered templates, create a copy of the same and start using it. Else, continue using the existing custom template.

25.2.2 Edit Form Templates

As an administrator, you can update the name and description of a form template, upload a form template variant, or download a template and make changes using Adobe LiveCycle Designer.

Prerequisites

You have installed Adobe LiveCycle Designer.

Context

→ Recommendation

We recommend that you edit form templates provided by SAP or partners instead of creating new form templates. Editing provided templates means that you do not have to adapt any output settings and you can also manage all versions of the template. Form template versions that you customize are not overwritten by SAP or partner upgrades.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Click  and select [Edit](#).
3. Update the name and description of a form template, upload a form template variant, or download a template and make changes using Adobe LiveCycle Designer.

By default, the [Form Main Template](#) toggle is on. If you do not want to use company-specific main templates, turn off the toggle. Note that only .xdp files can be uploaded.

4. Save and publish your changes.

In the Adobe LiveCycle Designer, you can make changes such as the following:

- Change position of title, logo, and fields
 - Change alignment of title and logo (unless you define these with main templates)
 - Change title and template texts
 - Change sender address, number, and content of footers (unless you define these with main templates)
 - Change style and size of font
 - Change line style and add borders
 - Adjust visibility of fields, change field labels, and the sequence of fields
 - Adjust visibility of columns, change column order, and column headers
 - Add standard fields available in the form data structure
- If you want to add new fields that are not available in the form data structure, you can add extension fields. Other fields require support from the SAP Cloud Service Center.
- Add placeholders to free text blocks
 - Add new texts or captions
 - Show and hide page numbers or folder marks

Note

If you require complex changes to a print form, you can also contact the SAP Cloud Service Center. They can assist you with complex scripting such as dynamically displaying or hiding fields, sorting table rows, adding new or extension fields to a form, or creating a completely new form. For more information, see the [Business Center for Cloud Solutions from SAP](#) .

For more information about using Adobe LiveCycle Designer, see the Adobe product documentation.

25.2.3 Create Language Variants

Variants are copies of an existing form template that are defined in other languages. You can use variants to create translations of forms in other languages.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Click *** and select [Copy Variant](#).
3. In the dialog box that appears, select a language.
4. By default, the [Form Main Template](#) toggle is on. If you do not want to use company-specific main templates, turn off the toggle.
 - If you use a main template, some of the elements are predefined by the main template, which creates consistency among documents.
 - If you do not use a main template, then you can define the template's header, footers, logo, and sender address individually for this template.
5. Save and publish your changes.

25.2.4 Download Template Variants

Download a template variant either to edit it offline or to use it as sample data to preview a form template.

Procedure

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Maintain Form Template](#) ►.
2. Select the template you want and choose [Download](#).
3. Save the archive file using `.zip` as the file extension.

The ZIP file contains the following files:

- An XML data package (XDP) file containing the selected form template variant.
- An XML schema definition (XSD) file that defines the structure of the XML document. This is required for offline editing.
- If available, XML files containing sample data for previewing your changes.

i Note

If no sample data files are available, you can also create your own sample data files using the [View XML Data](#) button on a document's output history.

25.3 Select Form Templates

Administrators can create rules to determine which form template is used for the output of a business document. A default assignment of form templates to documents is provided. If you create new templates, you must create a form template rule to define when the templates are used.

You can either define a form template rule that applies to all instances of the selected business document, or you can use the parameters provided to create more complex rules. If you have created more than one rule for a document, you also need to define the sequence of the rules. The system processes the rules in the order in which they appear in the table until a rule is found for which the conditions are met. Therefore, enter more specific rules at the top of the list and generic rules at the bottom. If you create a rule and leave a parameter blank, then this is handled as a wildcard, and any value is considered as having fulfilled the condition.

i Note

In some cases, a business document represents a group of related document types. For example, the business document Customer Invoice includes invoices, credit memos, correction invoices, and down payments.

Create a Form Template Rule

1. Go to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Output Management](#) ► [Select Form Template](#) ►.
2. Select the [Entity](#) and [Type](#).
3. Click [Add](#).
4. From the [Form Template](#) drop-down list, select a form template.
5. Under [Parameter Settings](#), enter the parameters for which you want this rule to apply.
6. Use the [Rearrange](#) icon to adjust the sequence in which the form template rules are applied in order to ensure that the required rule is applied by the system.
7. Save your changes.

26 Data Import and Export

This document describes how you can use Data Import and Export to import data into your solution and export data from your solution.

You can use Data Import and Export to import the following:

- Data from a legacy system into SAP Service Cloud Version 2. For example, you can transfer all the existing data from any Cloud CRM system to SAP Service Cloud Version 2.
- Operational data from third-party systems into SAP Service Cloud Version 2.
- Data that is manually maintained in a CSV file (data file) into SAP Service Cloud Version 2. For example, customer and product information collected from a campaign and maintained in a data file.

As an administrator, you are expected to understand the following:

- Your business requirement and knowledge of data imported to SAP Service Cloud Version 2.
- Sequence in which the entities must be imported. For information on each of the entities, see entity documentation.

Typical tasks include:

- Downloading templates for various entities and nodes
- Preparing data to an accepted format
- Importing data

Prerequisites

Type	Task
Technical	Template files can be downloaded from the system. To check if the file is in an acceptable format, do the following: 1. Open the CSV file in Notepad. 2. Click  File > Save As >  3. In the Save As dialog, choose Encoding as UTF-8. Keep the file extension as .CSV. 4. Once the file is saved, check if the content is displayed correctly.
Functional	You understand the business requirement of the data to be imported. For each entity you want to import data into, you understand the dependency within the entity section of the respective entity document.

26.1 Download Template

You can download templates to prepare data for importing.

Procedure

1. Navigate to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Monitor](#) ►.
2. Select [Download Template](#).
3. Select the entity for which you want to export the data.
4. Select the [File Type](#) between [Comma Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).
5. Enable the [Complete Entity](#) toggle button if you want to import the root entity and all its subentities. Disable it if only the root entity or root entity with selected subentities is to be imported.

When [Complete Entity](#) is selected, the system downloads a zip file with multiple csv and xlsx template files (one file each for the root entity and individual subentities).

When [Complete Entity](#) is unchecked, you can optionally select subentities based on your import requirement along with root entity template file. considering only root entity is to be imported or root entity only with select child entities is to be imported.

- If no subentity is selected, a single csv/excel template file of the root entity is made available for download.
 - If certain subentities are selected, a zip file with multiple csv/xlsx template files (one file each for root entity and every selected subentity) is available for download.
6. Select [Download Template](#).
 7. Update the downloaded template with data to be imported.

26.2 Prepare Data

Data that is either downloaded from a legacy system, a third-party system, or manually maintained in a data file, it must be cleansed to ensure that it is in an acceptable format.

For example, you must ensure that all the reference IDs are maintained based on the dependency, and also check if data is in acceptable format.

Instructions

- For date fields, follow these instructions:
 - Use format yyyy-mm-dd. For example, 2015-11-25.
 - To import a timestamp field, use format yyyy-mm-ddThh:mm:ssZ. For example, 2021-12-31T23:59:59Z.

i Note

Enter UTC time.

- If you have a Boolean entry, ensure that the value is entered either as true or false in small case.

Accepted Values

- true
- True
- TRUE

- false
- False
- FALSE

i Note

Any blank boolean field is treated as undefined.

- Columns with numbers that are treated as text.
Data of a few columns that display numbers will be treated as text. Refer to the code list in the templates for fields that have a specific list of values.

→ Remember

Use a notepad application to modify the CSV data file before importing. This would avoid any format changes to the CSV data file.

When using any excel application to modify the CSV data file, make sure to import the **.csv** file using **UTF-8** character encoding.

- In addition to the import data fields, you must also provide the following:
 - An external_key in the root entity file
 - A parent_external_key in the root entity file in case of hierarchical entity structure
 - A parent_external_key in the subentity files
Add the parent record's external_key as parent_external_key for child records to define parent-child relationship between records.
- external_key must be unique for each record in the import file.
- The external_key and parent_external_key fields are string type fields used by data impex for processing the respective import tasks. It does not reflect in the entity records created in the system via import. Data impex does not store these field values as they are used only for a particular import task.

catalogId	parentId	displayId	imageUrl	name	description	language-Code	external_key	parent_external_key
-----------	----------	-----------	----------	------	-------------	---------------	--------------	---------------------

In this example, external_key is a unique non-null value.

- Even if subentities are not being used, you must include the template files for those subentities during import.

The primary key ID of the respective entity is different from the `external_key` and `parent_external_key` fields. The primary key ID of the entity is generated when the entity records imported are created in the tenant. Hence, when new records are created during import, this field will have to be left blank if it is present in the template.

i Note

To import only the root entity, use a single csv/excel file.

To import a complete entity or a root entity along with select subentities, use a .zip file that includes multiple csv/excel files

Do not create a zip file by zipping a folder that contains the entity files. To prepare a zip file, select all of the csv/excel files (except codelist files) to include in the .zip file, right-click, and create a zipped folder. Do not create a zipped folder directly from the parent folder that was downloaded.

Attachments present in the template cannot be imported via Data Impex. However, include the csv/excel blank template in the zip file to import the complete entity.

26.3 Import

Once you have prepared the data, you can import it.

You can only import data. You cannot update or delete the data during import. Templates are available by default for each entity type for administrators who have a business understanding of the fields that needs to be mapped between systems. This ensures consistency in mapping across all users.

You can import data to SAP Service Cloud Version 2 for a business entity. To import data to SAP Service Cloud Version 2, perform the following steps:

1. Navigate to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Monitor](#) ►.
2. Click the [Import](#) icon.
3. Select the entity for which you want to import the data.
4. Select the [File Type](#) between [Comma Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).
5. Enable the [Complete Entity](#) toggle button if you want to import the root entity and all its subentities. Disable it if only the root entity or root entity with selected subentities is to be imported.
6. Browse for the *.zip prepared data file you want to upload.

i Note

Maintaining external key is mandatory for import of the complete entity.

7. Click [Import](#).
The system displays a message on the status of the import. You can go to the [Monitor](#) view, and check the status of your import.

26.4 Export

You can export data from your solution.

Procedure

1. Navigate to your user profile and select ► [Settings](#) ► [All Settings](#) ► [Data Import and Export](#) ► [Monitor](#) ►.
2. Click the [Export](#) icon.
3. Select the entity for which you want to export the data.
4. Select the [File Type](#) between [Comma Separated Values \(.csv\)](#) and [Excel Workbook \(.xlsx\)](#).
5. Select [Apply Filter](#) to filter based on the values available for the respective entities.
 - a. Select [Define Conditions](#) to add specific filter conditions.
 - b. Select [Add Condition Group](#) to add more conditions.
6. Select entities that need to be downloaded as part of the export file.
7. Select [Export Data](#).

The system displays the status of the export. Select [Monitor](#) to check the status of your export.

26.5 Monitor

[Monitor](#) allows you to view the status of the import, export, and downloaded template tasks and download error and warning files for each task. You can also download templates and export files via the hyperlink.

Once an import task is triggered, it goes through multiple stages.

[New](#) indicates that a new task is triggered.

[In Process](#) indicates that the task is being processed.

[Data Processed](#) indicates that all records are processed. The tool starts to generate error and warning files, if any, and the next task moves to In Process.

[Completed](#) indicates that the task is processed without any errors or warnings.

[Completed With Errors](#) indicates that the task is completed but with errors or warnings. Error and warning files are generated and the task moves to Finished. Errors or warning generated, if any, are available for download. If an import task's status is Completed with Errors but there is corresponding error file, it might be due to an incorrect file format or invalid data format for any of field in the import file. Verify the file and import it again.

Note

Tasks older than 30 days are automatically removed from Monitor, irrespective of task status.

In cases where you want to retain information beyond one month (such as external keys used for an entity during import), we recommend you to maintain local copies of such information.

You can view all tasks triggered in your tenant.

26.6 Available Entities

Data Import and Export supports the following entities:

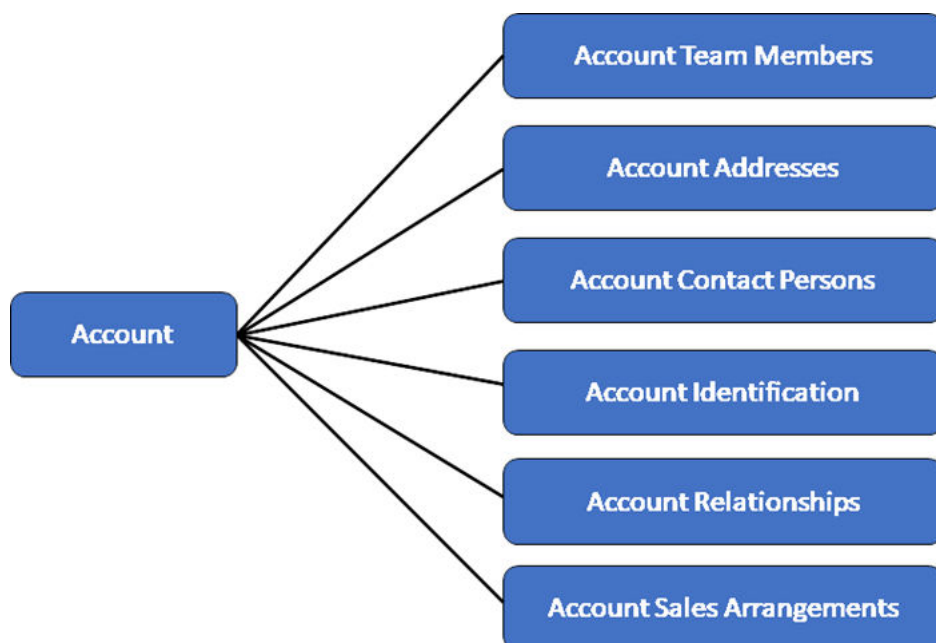
- Accounts
- Contacts
- Employees
- Exchange Rate
- Individual Customers
- Organizational Unit
- Product
- Service Catalog
- Service Category
- Warranty

26.7 Entity-Specific Documentation

26.7.1 Account

Account management capabilities offer a holistic view of the customer, and allow you to capture, monitor, store and track all critical business information about customers, prospects and partners. Use this information to

focus on your most profitable customers, maintain satisfaction and loyalty and consistent interaction across all channels.



26.7.1.1 Account Dependencies

Dependencies on Other Entities

Entity	Description
Contact Person	Used in Account Contact Persons to assign a Contact to Corporate account(s)
Employee	Used in Account Team Members to assign internal and external employees to the account
Organizational Unit	Used in Account Team to assign the employees to a specific Sales Organization to Corporate account(s) Used in Account Sales Data to assign a Sales Organization to the account
Account Relationships	The related Business Partner, and in case it is provided, the related Sales Organization must exist before creating a relationship to the account

Dependencies within the Entity

Entity	Description
Account Addresses	The default address can either be maintained in Account or Account Addresses but not in both places simultaneously
Account Contact Persons	Default address maintained at Account or Account Addresses must exist to fill workplace address fields

26.7.1.2 Account Fields

Download the template from [Data Import and Export](#) for the field level information.

26.7.1.2.1 Account

The Account header entity contains all central information of a corporate account.

It contains information about the account name, main address with communication data like email, ABC classification and blocking information. Use the address fields in the Account header entity in case only the default account address is to be maintained. It is highly recommended to fill the Country field even if you do not provide full address details.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Account ID	STRING	If this is not provided a number from a number range is drawn. Display ID of Account
Prospect	BOOLEAN	A prospect is someone who may become a customer.
Role	STRING	This field is mandatory Use CRM000 for customer Or BUP002 for prospect Or any specific Z-code
ABC Classification	STRING	ABC classification of a customer
Nielsen ID	STRING	
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.

Template Header	Data Type	Notes
Natural Person	BOOLEAN	
Source Lead Technical ID	STRING	
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	
Name	STRING	This field is mandatory Name of the account holder
Additional Name	STRING	
Additional Name 2	STRING	
Additional Name 3	STRING	
Industry	STRING	Primary business activity of the company
Legal Form	STRING	
Contact Permission	STRING	
D-U-N-S	STRING	
Owner Technical ID	STRING	Technical ID of the Account Owner (Employee) part of the Account Team
Main Address_Country/Region	STRING	The country of the account main address
Main Address_State_State	STRING	The region (federal state, county) of the address
Main Address_Postal Code	STRING	Postal code for the street address
Main Address_City	STRING	City of the address
Main Address_Street	STRING	Street
Main Address_House Number	STRING	
Main Address_County	STRING	
Main Address_District	STRING	District of the address
Main Address_Address Line 1	STRING	
Main Address_Address Line 2	STRING	
Main Address_Address Line 4	STRING	
Main Address_Address Line 5	STRING	
Main Address_P.O. Box	STRING	
Main Address_P.O. Box Address	BOOLEAN	
Main Address_Latitude	NUMBER	
Main Address_Longitude	NUMBER	

Template Header	Data Type	Notes
Main Address_Internal Geo Location	BOOLEAN	
Main Address_Language	STRING	
Main Address_Best Reached By	STRING	
Main Communication Data_Email	STRING	Email address
Main Communication Data_Website	STRING	
Main Communication Data_Fax	STRING	
Main Communication Data_Phone	STRING	
Main Communication Data_Mobile	STRING	

26.7.1.2.2 Account Team Members

The Account Team entity allows access to the account team. Here you can assign employees with a specific role to the corporate account.

Template Header	Data Type	Notes
parent_external_key	STRING	
Employee Technical ID	STRING	This field is mandatory Technical ID of Employee
Employee ID	STRING	Employee display ID can be used to reference an employee alternatively to Employee Technical ID. (Either of the two can be used.)
Role	STRING	This field is mandatory The type of responsibility the employee has for the account.
Sales Organization Technical ID	STRING	Technical ID of the Sales Organization. It is required for Sales Area dependent Account Team.
Distribution Channel	STRING	
division	STRING	
Main	BOOLEAN	Main indicator for given party role, sales area and validity period
Valid From	DATETIME	Start date of the validity of the account
Valid To	DATETIME	End date of the validity of the account

26.7.1.2.3 Account Addresses

The Account Addresses entity offers access and maintenance of the corporate accounts addresses.

The address includes postal address details as well as communication data details like email or phone number. You can specify for an individual address whether it should be the main address. Use the Account Addresses entity only in case of multiple addresses. If you want to maintain only the main address, then use the Account entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Main	BOOLEAN	Specification of main address
Country/Region	STRING	The country/region of the address
State_State	STRING	The region (federal state, county) of the address
City	STRING	City or district of the address
Street	STRING	Street of the address
House Number	STRING	
Postal Code	STRING	Postal code for the street address
County	STRING	
District	STRING	District of the address
Address Line 1	STRING	
Address Line 2	STRING	
Address Line 4	STRING	
Address Line 5	STRING	
P.O. Box	STRING	
P.O. Box Address	BOOLEAN	
Latitude	NUMBER	
Longitude	NUMBER	
Internal Geo Location	BOOLEAN	
Language	STRING	
Best Reached By	STRING	
Email	STRING	Email address
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

26.7.1.2.4 Account Contacts Persons

The Account Contact Persons entity offers access to the corporate accounts contact data.

With this entity you can assign contacts to an account and add business address details like communication data email, phone or mobile number. The relationship to the account will always be established based on the main address of the account.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Contact Technical ID	STRING	This field is mandatory
Contact ID	STRING	The Contact display ID can be used to reference an contact person alternatively to Contact Technical ID. (Either of the two can be used.)
Main	STRING	Specifies that the assigned Contact person is the main contact for this Account.
Function	STRING	The role of the contact in the company.
Department	STRING	The department to which the contact belongs.
VIP Contact	STRING	
Department From Business Card	STRING	The department on the contact's business card.
Job Title	STRING	Description of the contact's job.
Building	STRING	Company's building name
Floor	STRING	The company floor number of the contact.
Room	STRING	The company room number of the contact.
Phone	STRING	Business number of the contact
Mobile	STRING	
Email	STRING	Specify the business address email.
Best Reached By	STRING	

26.7.1.2.5 Account Identification

The Account Identification entity offers access to the corporate accounts identification data.

Template Header	Data Type	Notes
parent_external_key	STRING	
ID Type	STRING	This field is mandatory A code representation of the type of an identification number
ID Number	STRING	This field is mandatory The identification number itself
Valid From	DATETIME	
Valid To	DATETIME	
Institution Responsible	STRING	Name of the institution (government agency, registry office), company (Dun & Bradstreet), an organization (UN) that issued the identification number
Entry Date	DATETIME	
Country/Region	STRING	
State_State	STRING	

26.7.1.2.6 Account Relationships

The Account Relationships entity offers access to the corporate accounts relationship data.

Relationships from the account can be established to other business partners by providing the Business Partner Technical ID and the Relationship Role, which classifies the relationship.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Business Partner Relationship Role	STRING	This field is mandatory A coded representation of the type of Relationship.
Business Partner Technical ID	STRING	This field is mandatory
Business Partner ID	STRING	The Business Partner ID can be used to reference an Business Partner alternatively to Business Partner Technical ID. (Either of the two can be used.)

Template Header	Data Type	Notes
Sales Organization Technical ID	STRING	For Sales Area dependent relationships the Sales Organization Technical ID must be provided.
Distribution Channel	STRING	
Division	STRING	
Main Relationship	BOOLEAN	Indicates whether the relationship for the provided Business Partner Relationship Role and the Sales Area is the main relationship.

26.7.1.2.7 Account Sales Arrangements

The Account Sales Data entity allows access to the corporate accounts sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	This field is mandatory Technical ID of the Sales Organization.
Sales Organization ID	STRING	
Distribution Channel	STRING	If you provide a distribution channel you need to provide Sales Organization ID
Division	STRING	If you provide a division you need to provide distribution channel and Sales Organization ID
Marked for Deletion	BOOLEAN	Use this to mark a Sales Area for deletion. Required in integrated environments where physical deletion of sales arrangements is not allowed
Sales Office Technical ID	STRING	
Sales Office ID	STRING	
Sales Group Technical ID	STRING	
Sales Group ID	STRING	
Delivery Priority	STRING	
Complete Delivery	BOOLEAN	
Currency	STRING	
Customer Group	STRING	
Price List	STRING	
Price Group	STRING	

Template Header	Data Type	Notes
Cash Discount Terms	STRING	
Incoterms_Incoterms	STRING	
Incoterms_Incoterms Location	STRING	
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	

26.7.2 Contact Person

Use contacts to represent the relationship between contacts and accounts. Contacts can be maintained only for corporate accounts.

Contact Persons are individuals you are having direct interactions with for your business to be successful. Contacts mostly have a relationship with a corporate account and they may be involved in business processes like activities, orders, opportunities, and so on. Contact Person Relationships can only be maintained for corporate accounts whereas in general a contact can also exist without any account assignment yet. With the contact entity and its dedicated sub-entities you can access and modify all central and related information of a contact (master data).



26.7.2.1 Contact Person Dependencies

Dependencies on Other Entities

Entity	Description
Account	Used in Contact Person (Root) and (Is) Contact Person For entity to assign a Contact to existing Corporate Account(s)

Entity	Description
Account Address	Used as basis and prerequisite for the contacts business address based communication and workplace data

Dependencies within the Entity

Entity	Description
(Is) Contact Person For	The contacts main account relationship data can either be maintained in Contact Person (Root) or (Is) Contact Person For entity not in both places simultaneously. Recommendation is to maintain this Business Address based details like communication data email, phone or mobile number for the main account at the Contact Person (Root) entity whereas non-main-account information needs to be handled via the specific sub-entity.

26.7.2.2 Contact Person Fields

Download the template from [Data Import and Export](#) for the field level information.

26.7.2.2.1 Contact Person

The Contact Person entity allows you access to all central information for a contact person such as Contact Name and assignment of the contact to the main account (including the business address details of that relationship). The assignment to an account is mandatory in case you want to maintain any workplace details like "Email", "Phone", "Mobile", "Fax", "Building", "Floor", "Room", etc. based on the business address. Use the address fields in the Contact Person header entity in case only the workplace address to the main account is to be maintained.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Contact ID	STRING	Human-readable identifier of a contact person. The Contact ID is internally determined by the system via number range if not explicitly specified.

Template Header	Data Type	Notes
Last Name	STRING	This field is mandatory. Family name of a contact person
First Name	STRING	Given name of a contact person
Middle Name	STRING	
Date of Birth	DATETIME	
Gender	STRING	
Title	STRING	
Academic Title	STRING	
Language	STRING	
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.
Account Technical ID	STRING	UUID of the main account to which the contact is assigned to
Account ID	STRING	Display ID of the main account to which the contact is assigned to (Read Only)
Department	STRING	
Department From Business Card	STRING	
Job Title	STRING	Description of the function of the Contact Person within the main accounts company
Function	STRING	
VIP Contact	STRING	
Building	STRING	Building maintained for the assignment to the main account
Floor	STRING	
Room	STRING	
Phone	STRING	Conventional phone number for the contact person's workplace with main account
Mobile	STRING	Mobile phone number for the contact person's workplace with main account
Email	STRING	Email address for the contact person's workplace with main account
Best Reached By	STRING	
Contact Permission	STRING	

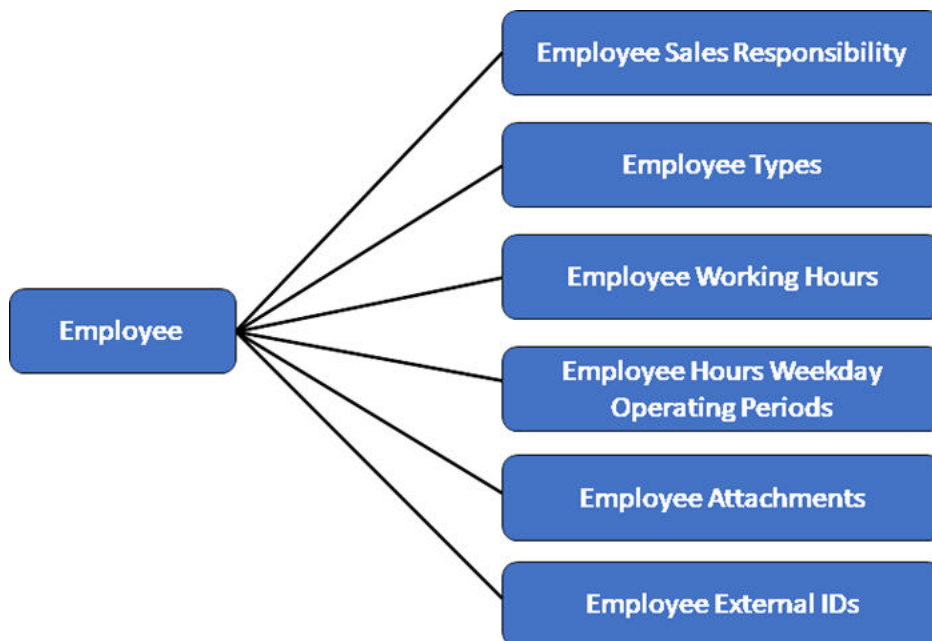
26.7.2.2.2 Contact Person for

The 'Contact Person Is Contact Person For' sub-entity offers access to the contact account relationship data. With this entity you can assign contacts to accounts and add business address details like communication data email, phone or mobile number. The relationship to the account will always be established based on the main address of the account. This specific entity has to be used in case a contact person serves as contact for multiple accounts and the referred account is not the main one. The related information for the contacts' main account can also be maintained at the Contact Person (root) entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Account Technical ID	STRING	This field is mandatory for the sub-entity UUID of the account to which the contact is assigned to
Account ID	STRING	
Main Account	BOOLEAN	Indicates whether the contact person is the account's main contact
Main Contact	BOOLEAN	Indicates whether the account is the contact person's main account (in case of multiple account assignments for the contact)
Function	STRING	
Department	STRING	
VIP Contact	STRING	
Department From Business Card	STRING	
Job Title	STRING	Description of the function of the contact person within the assigned accounts company
Building	STRING	Building maintained for the assignment to the assigned account
Floor	STRING	
Room	STRING	
Phone	STRING	Conventional phone number for the contact person's workplace with assigned account
Mobile	STRING	Mobile phone number for the contact person's workplace with main account
Email	STRING	Email address for the contact person's workplace with main account
Best Reached By	STRING	

26.7.3 Employee

Employee Master Data capabilities offer a holistic view of internal employees (a business partner person with valid employee type "INTERNAL"). It allows you to maintain all relevant data for your internal employees such as basic data (name, workplace address, communication data), organizational data (manager assignment, department, job), sales responsibility data, working hours information, provider skills and so on.



26.7.3.1 Employee Dependencies

Dependencies on Other Entities

Entity	Description
Organizational Unit	Used in Employee Sales Responsibility to assign the employees to a specific Sales Organization and OrgUnit

Dependencies within the Entity

Entity	Description
Employee Type	The employee type address can be maintained in Employee
Employee Working Hours Weekday Operating Period	The Employee Working Hours Weekday Operating Period can be maintained in Employee

26.7.3.2 Employee Fields

Download the template from [Data Import and Export](#) for the field level information.

26.7.3.2.1 Employee

The Employee header entity contains all central information of a employee. It contains information about the account name and workplace address with communication data like email.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Business Partner ID	STRING	A proprietary identifier for a business partner; alternative identifier for the employee
Employee ID	STRING	This field is mandatory An identifier for an Employee
Title	STRING	
Academic Title	STRING	
First Name	STRING	Given name of a person
Middle Name	STRING	
Last Name	STRING	This field is mandatory Family name of a person
Additional Last Name	STRING	
Date of Birth	DATETIME	The date on which the employee was born.
Nationality	STRING	
Marital Status	STRING	
Gender	STRING	

Template Header	Data Type	Notes
Language	STRING	
Workplace Address_Street	STRING	Street of the address
Workplace Address_House Number	STRING	House number of the address
Workplace Address_City	STRING	City of the address
Workplace Address_Postal Code	STRING	Postal code for the street address
Workplace Address_Country/Region	STRING	The country/region of the address
Workplace Address_State_State	STRING	The state of the address
Workplace Address_District	STRING	The district of the address
Workplace Address_Latitude	NUMBER	
Workplace Address_Longitude	NUMBER	
Workplace Address_Building	STRING	
Workplace Address_Floor	STRING	
Workplace Address_Room	STRING	
Workplace Address_Phone	STRING	
Workplace Address_Mobile	STRING	
Workplace Address_Email	STRING	Email address
Workplace Address_Address Line 1	STRING	
Workplace Address_Address Line 2	STRING	
Workplace Address_Address Line 4	STRING	
Workplace Address_Address Line 5	STRING	

26.7.3.2.2 Employee Sales Responsibility

The Employee Sales Responsibility entity allows the employee to maintain the sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	
Distribution Channel	STRING	A coded representation of a distribution channel
Division	STRING	A coded representation of a division
Default Sales Responsibility	BOOLEAN	An indicator to mark one Employee Sales Responsibility-record as default for the employee

26.7.3.2.3 Employee Types

The Employee Types entity offers the possibility to maintain the type of the employee (internal or external) and the validity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Internal Employee	BOOLEAN	This field is mandatory
Valid From	DATETIME	This field is mandatory
Valid To	DATETIME	This field is mandatory

26.7.3.2.4 Employee External IDs

The Employee External IDs entity displays the external ids for employee records.

Template Header	Data Type	Notes
parent_external_key	STRING	
External ID	STRING	
Business System Technical ID	STRING	
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

26.7.3.2.5 Employee Working Hours

The Employee Working Hours entity offers access to the working hours data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Type	STRING	A coded representation of a working hours type
Valid From	DATETIME	The period from which the operating hours are valid
Valid To	DATETIME	The period till which the operating hours are valid

Template Header	Data Type	Notes
Time Zone	STRING	Time zone in which the operating hours are specified
Working Day Calendar	STRING	Calendar with general working days in a week and public holidays

26.7.3.2.6 Employee Working Hours Operating Periods

The Employee Working Hours Operating Periods entity offers access to the operating period data. With this entity you can assign working hours operating period to working hours.

Template Header	Data Type	Notes
parent_external_key	STRING	
Weekday	STRING	
Start Time	DATETIME	An active time period of a day in a day program
End Time	DATETIME	An active time period of a day in a day program

26.7.4 Organizational Unit

Organizational unit containing a hierarchy that represents the hierarchical structure of the organization.

26.7.4.1 Organizational Unit Dependencies

Dependencies on Other Entities

Entity	Description
Organizational Unit	Assign a parent org unit
Employee	You can assign an employee to the org unit

26.7.4.2 Organizational Unit Fields

Download the template from [Data Import and Export](#) for the field level information.

26.7.4.2.1 Organizational Unit

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	If not provided, defaulted to current date
Valid To	DATETIME	If not provided, defaulted to the maximum end date, 31-December-9999
Organizational Unit ID	STRING	This field is mandatory
Name	STRING	
Status	STRING	
Current Functions_Company	BOOLEAN	
Current Functions_Sales Organization	BOOLEAN	
Current Functions_Service Organization	BOOLEAN	
Current Functions_Sales Office	BOOLEAN	
Current Functions_Sales Group	BOOLEAN	
Current Functions_Sales	BOOLEAN	
Current Functions_Service	BOOLEAN	
Current Functions_Currency	STRING	
Manager_Manager Technical ID	STRING	
Parent Unit_Parent Unit Technical ID	STRING	
Address_Country/Region	STRING	
Address_State	STRING	
Address_State	STRING	
Address_Postal Code	STRING	
Address_City	STRING	
Address_Street	STRING	
Address_House Number	STRING	
Address_County	STRING	

Template Header	Data Type	Notes
Address_District	STRING	
Address_Address Line 1	STRING	
Address_Address Line 2	STRING	
Address_Address Line 4	STRING	
Address_Address Line 5	STRING	
Address_Latitude	NUMBER	
Address_Longitude	NUMBER	
Communication Data_Email	STRING	
Communication Data_Website	STRING	
Communication Data_Fax	STRING	
Communication Data_Phone	STRING	
Communication Data_Mobile	STRING	

26.7.4.2.2 Organizational Unit Employee Assignments

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Employee Technical ID	STRING	The employee should be present in the system. This field or the employee display ID is mandatory. The employee must be valid during the period given in the Valid From and Valid To.
Role	STRING	

26.7.4.2.3 Organizational Unit External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
External ID	STRING	
Business System Technical ID	STRING	
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

26.7.4.2.4 Organizational Unit Functions

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Company	BOOLEAN	
Sales Organization	BOOLEAN	
Service Organization	BOOLEAN	
Sales Office	BOOLEAN	
Sales Group	BOOLEAN	
Sales	BOOLEAN	
Service	BOOLEAN	
Currency	STRING	Recommended to maintain a currency either for the company or for the Sales Organization

26.7.4.2.5 Organizational Unit Name and Address

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Name	STRING	Recommended to maintain this field
Address Technical ID	STRING	
Country/Region	STRING	
_Country/Region	STRING	
_State	STRING	
City	STRING	
Street	STRING	
House Number	STRING	
Postal Code	STRING	
County	STRING	
District	STRING	
Address Line 1	STRING	
Address Line 2	STRING	
Address Line 4	STRING	
Address Line 5	STRING	
Latitude	NUMBER	
Longitude	NUMBER	
Email	STRING	
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

26.7.4.2.6 Parent Organizational Unit

Template Header	Data Types	Notes
parent_external_key	STRING	
Technical ID	STRING	
Valid From	DATETIME	
Valid To	DATETIME	
Parent Unit Technical ID	STRING	Recommended to maintain a valid Parent Unit Technical ID or a Parent Unit Display ID
Reporting Line	BOOLEAN	There can be only one reporting line parent org unit at the same point in time.

26.7.4.2.7 Organizational Unit Sales Area Assignments

Template Header	Data Types	Notes
parent_external_key	STRING	
Technical ID	STRING	
Sales Organization ID	STRING	Maintain a valid Sales Organization Technical ID.
Distribution Channel	STRING	
Division	STRING	

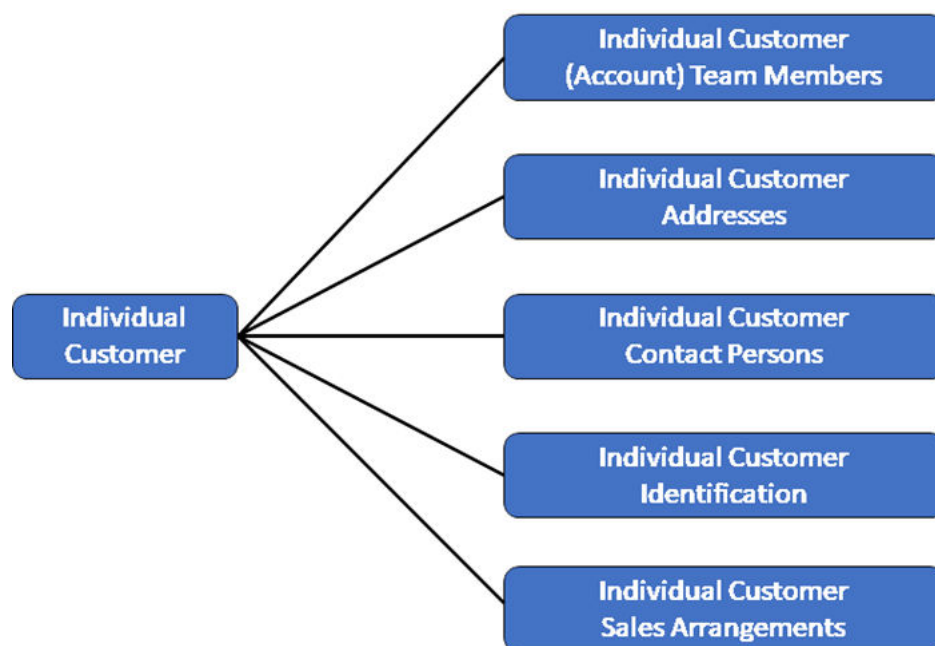
26.7.4.2.8 Organizational Unit Sales Area

Template Header	Data Types	Notes
parent_external_key	STRING	
Technical ID	STRING	
Distribution Channel	STRING	
Division	STRING	

26.7.5 Individual Customer

Individual Customers represent persons with whom business is done.

Similar to account management capabilities for accounts with this entity a holistic view of the Individual Customer is offered, that allows you to capture, monitor, store and track all critical business information about customers and prospects. The Individual Customer thereby directly represents a person with whom your business is done, whereas an account is always an organization. Use this information to focus on your most profitable individual customers, maintain satisfaction and loyalty and consistent interaction across all channels.



26.7.5.1 Individual Customer Dependencies

Dependencies on Other Entities

Entity	Description
Contact Person	Used in Individual Customer Contact Persons entity to assign a Contact to individual customers
Employee	Used in Individual Customer (Root) and Individual Customer Account Team Members to assign internal and external employees e.g. as owner to the Individual Customer

Entity	Description
Organizational Unit	Used in Individual Customer Account Team Members and Individual Customer Sales Data to assign data relevant for a specific Sales Organization to the Individual Customer

Dependencies within the Entity

Entity	Description
Individual Customer Addresses	The default address can either be maintained in the Individual Customer (Root) or Individual Customer Addresses entity but not in both places simultaneously.
Individual Customer Account Team Members	The owner for the individual customer can either be maintained in the Individual Customer (Root) or Individual Customer Account Team Members entity but not in both places simultaneously.
Individual Customer Contact Persons	A default address maintained at Individual Customer (Root) or Individual Customer Addresses must exist to fill work-place address fields (e.g. communication data) for the business address here.

26.7.5.2 Individual Customer Fields

Download the template from [Data Import and Export](#) for the field level information.

26.7.5.2.1 Individual Customer

The Individual Customer header entity contains all central information of an individual customer. It contains information about the customers name, main address with communication data like email, ABC classification and blocking information. Use the address fields in the individual customer header entity in case only the default address is to be maintained. It is recommended to fill the Country field even if you do not provide full address details.

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	

Template Header	Data Type	Notes
Customer ID	STRING	Human-readable identifier of an Individual Customer. The Customer ID is internally determined by the system via number range if not explicitly specified
Prospect	BOOLEAN	A prospect is someone who may become a customer.
Role	STRING	This field is mandatory Use CRM000 for customer Or BUP002 for prospect Or any specific Z-code
ABC Classification	STRING	ABC classification of a customer
Status	STRING	ACTIVE will be determined by the system if not explicitly specified.
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	
Date of Birth	DATETIME	The date of birth
Language	STRING	
Nationality	STRING	Identifies the nation to which the customer belongs
Gender	STRING	
Title	STRING	A form of address, for example, Mr. or Ms.
Academic Title	STRING	Identifies the academic degree
First Name	STRING	Given name of an individual customer
Middle Name	STRING	
Last Name	STRING	This field is mandatory Family name of an individual customer
Contact Permission	STRING	An indicator that defines whether it is allowed to contact the customer
Owner Technical ID	STRING	Technical ID of the Account Owner (Employee) as part of the Individual Customer (Account) Team
Main Address_Country/Region	STRING	The country of the customer's address
Main Address_State_State	STRING	The region (federal state, county) of the customer's address

Template Header	Data Type	Notes
Main Address_Postal Code	STRING	The postal code for the customer's street address
Main Address_City	STRING	The city of the customer's address
Main Address_Street	STRING	The street of the customer's address
Main Address_House Number	STRING	The house number of the customer's address
Main Address_County	STRING	
Main Address_District	STRING	District of the address
Main Address_Address Line 1	STRING	
Main Address_Address Line 2	STRING	
Main Address_Address Line 4	STRING	
Main Address_Address Line 5	STRING	
Main Address_Latitude	NUMBER	
Main Address_Longitude	NUMBER	
Main Address_Best Reached By	STRING	
Main Communication Data_Email	STRING	
Main Communication Data_Website	STRING	
Main Communication Data_Fax	STRING	
Main Communication Data_Phone	STRING	Phone number of the customer
Main Communication Data_Mobile	STRING	

26.7.5.2.2 Individual Customer Account Team Members

The Individual Customer (Account) Team Members entity allows access to the Individual Customer Team. Here you can assign employees with a specific party role to the individual customer. These assignments can even be done sales organization dependent.

Template Header	Data Type	Notes
parent_external_key	STRING	
Employee Technical ID	STRING	This field is mandatory Technical ID of the assigned Employee
Employee ID	STRING	
Role	STRING	This field is mandatory The type of responsibility the employee has for the business partner.

Template Header	Data Type	Notes
Sales Organization Technical ID	STRING	Technical ID of the Sales Organization. It is required for Sales Area dependent Individual Customer Team.
Distribution Channel	STRING	
Division	STRING	
Main	BOOLEAN	Main indicator for given party role, sales area and validity period
Valid From	DATETIME	Start date of the validity of the customer team assignment of the employee
Valid To	DATETIME	End date of the validity of the customer team assignment of the employee

26.7.5.2.3 Individual Customer Addresses

The Individual Customer Addresses entity offers access and maintenance of the individual customer addresses. The address includes postal address details as well as communication data details like email or phone number. You can specify for a dedicated address whether it should be the main address for the individual customer. Use the Individual Customer Addresses entity in case of multiple addresses. If you want to maintain/change only the main address, then use the Individual Customer header entity.

Template Header	Data Type	Notes
parent_external_key	STRING	
Main	BOOLEAN	Specification of main address
Country/Region	STRING	Country/region of the customer's address
State_State	STRING	The region (federal state, county) of the customer's address
City	STRING	City of customer's address
Street	STRING	Street of customer's address
House Number	STRING	House number of customer's address
Postal Code	STRING	Postal code of the customer's street address
County	STRING	
District	STRING	District of customer's address
Address Line 1	STRING	
Address Line 2	STRING	
Address Line 4	STRING	

Template Header	Data Type	Notes
Address Line 5	STRING	
Latitude	NUMBER	
Longitude	NUMBER	
Best Reached By	STRING	
E-mail	STRING	Email address
Website	STRING	
Fax	STRING	
Phone	STRING	
Mobile	STRING	

26.7.5.2.4 Individual Customer Contact Persons

The Individual Customer Contact Persons entity offers access to the individual customers contact data. With this entity you can assign contacts to the individual customer and add business address details like communication data email, phone or mobile number. The relationship to the contact will always be established based on the main address of the individual customer.

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Contact Technical ID	STRING	This field is mandatory Technical ID of the Contact person
Main	BOOLEAN	Specifies that the assigned Contact person is the main contact for this customer.
Function	STRING	The role of the contact in the company.
Department	STRING	The department to which the contact belongs.
VIP Contact	STRING	
Department	STRING	Department From Business Card - The department on the contact's business card.
Building	STRING	Company's building name
Floor	STRING	The company floor number of the contact.
Room	STRING	The company room number of the contact.

Template Header	Data Type	Notes
Phone	STRING	Business phone number of the contact (conventional)
Mobile	STRING	Business mobile number of the contact
Email	STRING	Business email address of the contact
Best Reached By	STRING	

26.7.5.2.5 Individual Customer Identifications

The Individual Customer Identification entity offers access to the individual customers identification data.

Template Header	Data Type	Notes
parent_external_key	STRING	
ID Type	STRING	This field is mandatory A code representation of the type of an identification number
ID Number	STRING	This field is mandatory The identification number itself
Valid From	DATETIME	
Valid To	DATETIME	
Institution Responsible	STRING	Name of the institution (government agency, registry office), company (Dun & Bradstreet), an organization (UN) that issued the identification number
Entry Date	DATETIME	
Country/Region	STRING	
State_State	STRING	

26.7.5.2.6 Individual Customer Sales Arrangements

The Individual Customer Sales Data entity allows access to the individual customers sales data.

Template Header	Data Type	Notes
parent_external_key	STRING	
Sales Organization Technical ID	STRING	This field is mandatory Technical ID of the Sales Organization.

Template Header	Data Type	Notes
Sales Organization ID	STRING	
Distribution Channel	STRING	
Division	STRING	
Marked for Deletion	BOOLEAN	Use this to mark a Sales Area for deletion. Required in integrated environments where physical deletion of sales arrangements is not allowed
Sales Office Technical ID	STRING	
Sales Office ID	STRING	
Sales Group Technical ID	STRING	
Sales Group ID	STRING	
Delivery Priority	STRING	
Complete Delivery	BOOLEAN	
Currency	STRING	
Customer Group	STRING	
Price List	STRING	
Price Group	STRING	
Cash Discount Terms	STRING	
Incoterms_Incoterms	STRING	
Incoterms_Incoterms Location	STRING	
Blocking Reasons_Billing Block	STRING	
Blocking Reasons_Delivery Block	STRING	
Blocking Reasons_Order Block	STRING	
Blocking Reasons_Sales Support Block	BOOLEAN	

26.7.6 Product

A product is the item offered for sale and it can be a service or a physical item. A product can be of real or virtual form. Each product is made at a cost and sold at a price and the price depends on the quality, market, the marketing, and the targeted segment.

26.7.6.1 Product Dependencies

Dependencies on Other Entities

Entity	Description
Product Group	Used in product root to assign a product group to product(s)
Organizational Unit	Used in product sales distribution chains to assign sales data to product(s)

26.7.6.2 Product Fields

Download the template from Data Import and Export for the field level information.

26.7.6.2.1 Product

Template Header	Data Type	Notes
external_key	STRING	
Technical ID	STRING	
One Domain Model ID	STRING	One Domain Model Identifier or ODM ID is used as object ID and in references for ODM objects in all cross application use cases.
Product ID	STRING	Mandatory field
Status Code	STRING	Mandatory field. If no value is provided, then the default will be In_PREPARATION
Unit of Measure Code	STRING	Mandatory field
Name	STRING	Mandatory field
Description	STRING	
Product Group Technical ID	STRING	Technical ID of the product group created in the system.
Product Group ID	STRING	Display ID of the product group created in the system.

26.7.6.2.2 Product Descriptions

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Description	STRING	A user-defined language-dependent text that provides more information about the product. Maximum length: 5000
Language Code	STRING	

26.7.6.2.3 Product External IDs

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
External ID	STRING	
Business System Technical ID	STRING	
Business System ID	STRING	
ID Type	STRING	
Main	BOOLEAN	

26.7.6.2.4 Product Names

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Name	STRING	Definition of product names in multiple languages. Maximum length: 255
Language Code	STRING	

26.7.6.2.5 Product Sales Distribution Chains

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Sales Organization Technical ID	STRING	Organizational Unit Technical ID which is created in the system. This is a mandatory field.
Distribution Channel Code	STRING	This is a mandatory field.
Sales Unit Of Measure Code	STRING	
Minimum Order Quantity_Quantity	NUMBER	
Minimum Order Quantity_Unit of Measure Code	STRING	
Sales Status Code	STRING	Sales status code that is defined in administrator settings > sales status schema

26.7.6.2.6 Product Unit of Measure Conversion

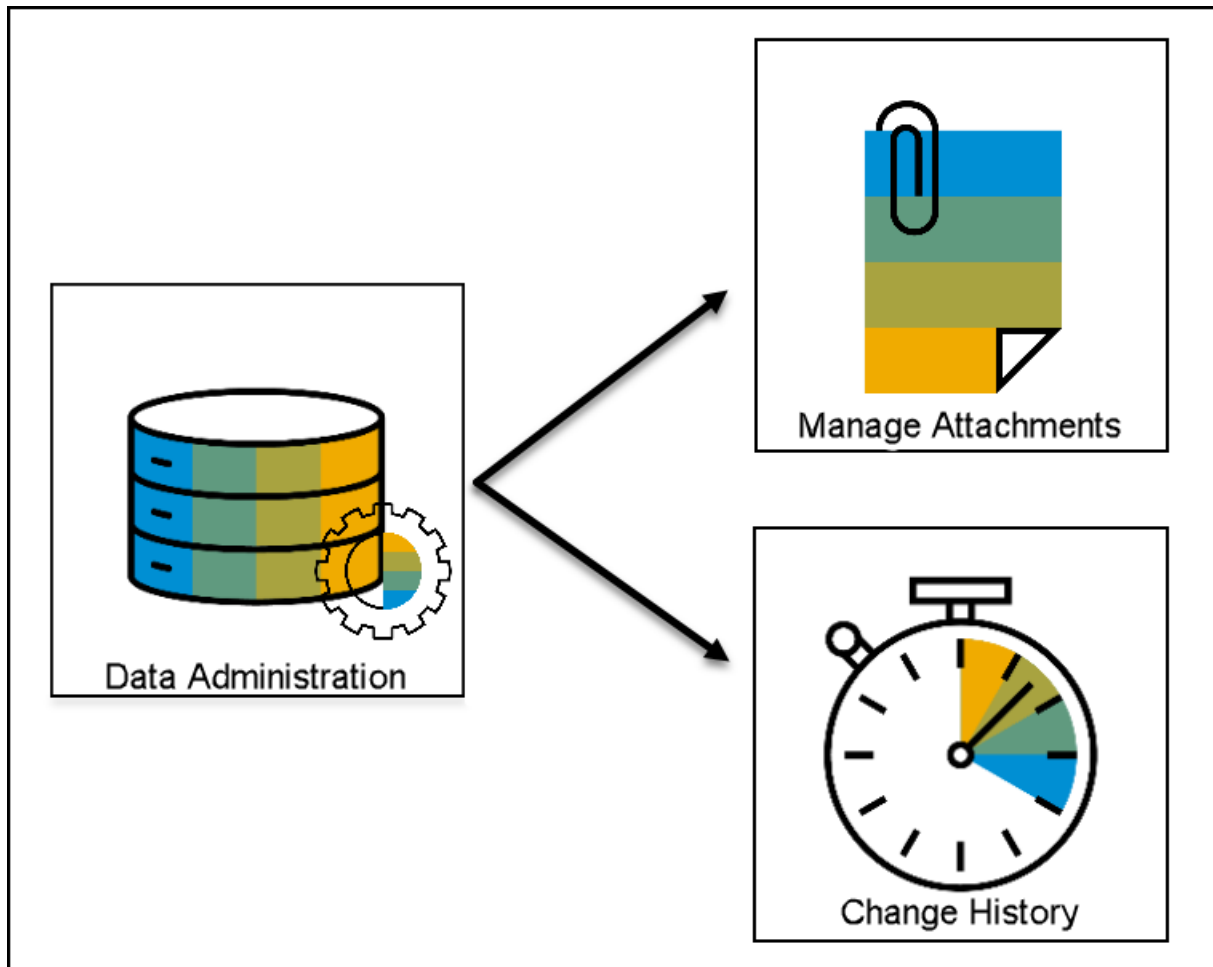
Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Quantity	NUMBER	
Unit of Measure Code	STRING	
Corresponding Quantity	NUMBER	
Corresponding Unit of Measure Code	STRING	

26.7.6.2.7 Product Unit of Measure

Template Header	Data Type	Notes
parent_external_key	STRING	
Technical ID	STRING	
Unit of Measure Code	STRING	

27 Data Administration

As an administrator, you can maintain and manage the files attached to an entity as additional information and check when and by whom an entity was modified.



27.1 Manage Attachment Types

Define and manage attachment types in the system.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Data Administration](#) ► [Manage Attachment Types](#) ►.
4. Click the add icon (+) in the [Customer Defined Document Types](#) section, enter a document code and a description, and choose the accept icon (✓) under [Actions](#) column to save the document type.

Note

Ensure that when you define the document type, your document code starts with the capital alphabet Z.

Results

You've now successfully defined an attachment type. You can further assign these attachment types to different business services using the [Assign Attachment Types](#) app.

27.2 Assign Attachment Types

Assign attachment types to business services.

Select a business service from the list of business services on the left side, assign it to one or more than one document types on the right, and then click [Save](#).

27.3 Change History

Learn how to view all the changes made in the solution.

Procedure

1. Go to your user profile, navigate to ► [Settings](#) ► [All Settings](#) ► [Change History](#) ►.
2. Select the business entity from the drop down.
3. You can view all the changes made to the entity here. The details of the change, changed by, changed date, time, attribute, type of change, value changed from, and value changed to, are also available.
4. You can search for changes made using the search icon (🔍) in the change history screen.
 - When an entity is deleted all change history entries for this entity are also deleted.
 - When a data subject gets depersonalized their corresponding change history records are also deleted.

27.4 Define MIME Types for Attachments

As an administrator, you can define additional MIME types for attachments for your users in your organization.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Data Administration](#) ► [MIME Types for Attachments](#) ►.
4. Click the [add](#) icon (+) enter the MIME Type, and add a description for your MIME type.
5. Select or unselect the [Allowed](#) switch, and then click the accept icon (✓) to save the MIME type.
6. You can further choose add activity icon (✓) to add extension types to your MIME type.

Results

You've now defined MIME types for your attachments.

27.4.1 Predelivered MIME Types for Attachments

Lists the various predelivered MIME types for attachments in the system.

The following table lists the predelivered MIME types available for attachments.

Predelivered MIME Types for Attachments

MIME Type	Description
application/pdf	Portable Document Format
application/pdi	Portable Database Image
application/dwf	Design Web Format
application/dxf	Drawing Exchange Format
application/json	JavaScript Object Notation
application/mif	FrameMaker Interchange Format
application/msonenote	Microsoft OneNote
application/msword	Microsoft Office Word Document
application/octet-stream	Octet Stream
application/oda	Open Document Architecture
application/pdf	Adobe Portable Document Format Document
application/pkcs7-mime	Single PKCS #7 object of type signedData
application/pkcs7-signature	PKCS#7 Signature
application/postscript	Postscript
application/rtf	Rich Text Format
application/vnd.adobe.xdp+xml	XML Data Package
application/vnd.hp-PCL	Printer Command Language
application/vnd.hp-PCLXL	Printer Command Language XL
application/vnd.ms-excel.addin.macroEnabled.12	Microsoft Office Excel Add-In
application/vnd.ms-excel.sheet.binary.macroEnabled.12	Microsoft Office Excel-Binary Worksheet
application/vnd.ms-excel.sheet.macroEnabled.12	Microsoft Office Excel-Macro
application/vnd.ms-excel.template.macroEnabled.12	Microsoft Office Excel-Macro Template
application/vnd.ms-mediapackage	Microsoft Media Package
application/vnd.ms-officetheme	Microsoft Office Theme
application/vnd.ms-package.relationships+xml	Microsoft Package Relationships
application/vnd.ms-pki.certstore	Microsoft PKI Certification Store
application/vnd.ms-pki.pko	Microsoft PKI PKO
application/vnd.ms-pki.seccat	Microsoft PKI Cat
application/vnd.ms-pki.stl	Microsoft PKI STL
application/vnd.ms-powerpoint	Microsoft Office PowerPoint

MIME Type	Description
application/vnd.ms-powerpoint	Microsoft Office PowerPoint Slide
application/vnd.ms-powerpoint.addin.macroEnabled.12	Microsoft Office PowerPoint Add-In
application/vnd.ms-powerpoint.presentation.macroEnabled.12	Microsoft Office PowerPoint Presentation
application/vnd.ms-powerpoint.slide.macroEnabled.12	Microsoft Office PowerPoint Presentation
application/vnd.ms-powerpoint.slideshow.macroEnabled.12	Microsoft Office PowerPoint Presentation
application/vnd.ms-powerpoint.template.macroEnabled.12	Microsoft Office PowerPoint Presentation
application/vnd.ms-project	Microsoft Projects
application/vnd.ms-publisher	Microsoft Office Publisher Document
application/vnd.ms-visio.viewer	Microsoft Office Visio Drawing
application/vnd.ms-word.document.macroEnabled.12	Microsoft Office Word Document
application/vnd.ms-word.template.macroEnabled.12	Microsoft Office Word Document
application/vnd.ms-wpl	Windows Media Playlist
application/vnd.ms-xpsdocument	XPS Document
application/vnd.msaccess	Microsoft Access Database File
application/vnd.oasis.opendocument.formula	OpenDocument Formula
application/vnd.oasis.opendocument.graphics	OpenDocument Drawing
application/vnd.oasis.opendocument.presentation	OpenDocument Presentation
application/vnd.oasis.opendocument.spreadsheet	OpenDocument Spreadsheet
application/vnd.oasis.opendocument.text	OpenDocument Text
application/vnd.oasis.opendocument.text-master	OpenDocument Master Document
application/vnd.oasis.opendocument.text-web	HTML Document Template
application/vnd.openofficeorg.extension	OpenOffice.org Extension
application/vnd.openxmlformats-officedocument.presentationml.presentation	Microsoft Office PowerPoint Presentation
application/vnd.openxmlformats-officedocument.presentationml.slide	Microsoft Office PowerPoint Slide
application/vnd.openxmlformats-officedocument.presentationml.slideshow	Microsoft Office PowerPoint Presentation
application/vnd.openxmlformats-officedocument.presentationml.template	Microsoft Office PowerPoint Template
application/vnd.openxmlformats-officedocument.spreadsheetml.sheet	Microsoft Office Excel Sheet
application/vnd.openxmlformats-officedocument.spreadsheetml.template	Microsoft Office Excel Template
application/vnd.openxmlformats-officedocument.wordprocessingml.document	Microsoft Office Word Document

MIME Type	Description
application/vnd.openxmlformats-officedocument.wordprocessingml.template	Microsoft Office Word Template
application/vnd.sap_kw.itutor	SAP ITutor
application/vnd.visio	Microsoft Office Visio Drawing
application/winhelp	Help File
application/wordperfect5.1	Word Perfect
application/x-bat	Batch File
application/x-director	Adobe Shockwave Player Video
application/x-dvi	DVI Movie
application/x-futuresplash	Shockwave Flash Object
application/x-gzip	ZIP Archive
application/x-javascript	Javascript File
application/x-latex	LaTeX-File
application/x-msmetafile	Open APK File
application/x-perl	Perl Script
application/x-rar-compressed	RAR Archive
application/x-sapshortcut	SAP Shortcut
application/x-screencam	RAR Archive
application/x-shockwave-flash	Shockwave Flash Object
application/x-stuffit	Stuffit Expander
application/x-tar	Tape Archive
application/x-tex	TeX File
application/x-x509-ca-cert	Private Keys and Certificates
application/x-zip-compressed	ZIP Archive
application/xhtml+xml	XHTML
application/xml	XML File
application/xslt+xml	XSLT
application/zip	Zip Archive
audio/basic	Basic Audio
audio/mid	MIDI Sequence
audio/mp4	MP4
audio/mpeg	MPEG Audio
audio/x-aiff	AIFF Audio
audio/x-pn-realaudio	RealAudio
audio/x-wav	WAV File

MIME Type	Description
flup	Flip
image/bmp	Bitmap Image
image/cgm	Computer Graphics Metafile
image/gif	GIF Image
image/ico	Icon
image/ief	IEF Image
image/jpeg	JPEG Image
image/jpg	JPEG Image
image/pjpeg	Progressive JPEG
image/png	PNG Image
image/prs.btif	BTIF Image
image/svg+xml	SVG Image
image/tiff	TIFF Image
image/vnd.fpx	FlashPix
image/vnd.xiff	eXtended Image File
image/x-dwg	AutoCAD Drawing
image/x-rgb	eXtended RGB
image/x-xbitmap	eXtended Bitmap
message/rfc822	E-mail
model/iges	IGESModel
model/mesh	Finite Element Model
model/vrml	VRML Model
text/asp	Active Server Page
text/calendar	Calender
text/css	CSS Style Sheet
text/csv	Comma-Separated Values
text/html	HTML Document
text/html	HTML Text
text/plain	Text Document
text/richtext	Rich Text Format
text/thtml	HTML Document
text/vnd.ms-mediapackage	Microsoft Media Package
text/vnd.wap.sli	WAP File Extension
text/vnd.wap.wbmp	Wireless Bitmap

MIME Type	Description
text/vnd.wap.wml	Wireless Markup Language
text/vnd.wap.wmlscript	Wireless Markup Language Script
text/x-datev	Datev File
text/x-vcalendar	vCalender
text/x-vcard	vCard
text/xml	XML File
video/avi	AVI Video Clip
video/mppeg	MPEG Video Clip
video/quicktime	Quicktime Movie
video/vdo	VDO Movie
video/vnd.vivo	Vivo Video
video/x-flv	Flash Video
video/x-ms-asf	Windows Media Audio/Video
video/x-ms-wmv	Windows Media Video

27.5 Attachment Configuration

As an administrator, you can configure file size limit for your attachments.

Configure the file size limit for attachments by entering the file size in the [Enter File Size Limit](#) field, and clicking [Save](#).

i Note

The maximum file size of your attachments must be up to 200 MB.

You can also enable uploading password-protected files by enabling the [Enable Uploading Password-Protected Files](#) switch.

i Note

SAP doesn't recommend setting upload password-protected files as default. However, you can do so at your own discretion, by activating the [Enable Uploading Password-Protected Files](#) switch.

28 Analytics

Analytics is integrated in the solution to support and monitor business processes, helping you to make informed decisions.

By default, the system is available with a prepackaged integration with SAP Analytics Cloud (SAC), offering an embedded SAC to all users. With this prepackaged embedded SAC, you can use all features of embedded SAC in the system.

Features

The following are the SAP Analytics Cloud, Embedded Edition features available in the system:

- *Query Designer*: This feature is an SAC OEM-specific feature available for applications Embedding SAC with extended HANA live type of connectivity.
- *Linked Analysis*: This feature can be used to drill through hierarchical data or create filters that simultaneously update multiple charts in a story.
- *Restricted Measures*: Measures are numerical values on which you can use mathematical functions. Restricted measures feature helps you to compare a value to a set of other values within the same chart or a table.

Limitations

The following are the limitations of Analytics, available in the system:

- You can build stories only on standard analytical models. Also, you can't create stories from data sources from any other SAP products apart from SAP Service Cloud Version 2.
- Embedded stories aren't available for mobile applications.
- The following features aren't supported:
 - Embedding tiles on the home page or other UIs on the system.
 - Viewing hierarchical data.
 - Reusing models across multiple stories.
 - Transporting stories across tenants.
 - Individual widget support for embedded tiles.
 - Intent-based navigation.
 - Lifecycle management scenarios such as tenant copy, tenant refresh, and so on.
 - Story metadata translations.
 - Support for extension fields.
 - User timezone conversion.
 - Calculate date or time difference.

- ID-based sorting in SAC stories.
- Currency and unit conversion.
- Support for different themes.

Related Information

[Query Designer](#)

[Creating a Linked Analysis](#)

[Launching the Explorer from a Story Canvas](#)

[Running a Forecast in a Time Series or Line Chart](#)

[Create Restricted Measures](#)

28.1 Configure Single Sign-On for SAP Analytics Cloud

Administrators can configure single sign-on for the system to ensure that your users can seamlessly connect to SAP Analytics Cloud (SAC) to access, create, and manage stories.

Enable Single Sign-On for SAC

To enable Single Sign-On for SAC, you've to connect the identity provider to the SAC tenant. To do so, start by downloading the service provider metadata of the SAC tenant from the system and then upload the metadata XML file to your identity provider. Later, download the configured metadata XML from your identity provider and upload this XML to the system to start using SAC seamlessly.

Prerequisites:

- Ensure your admin user has the **analyticsSACSPSetupservice** business service assigned and the **sap.crm.service.analyticsSACSPSetupservice** app enabled to configure single sign-on for SAP Analytics Cloud.
- Identity provider is configured on your system and is in active status. Only active identity providers can be selected while linking IdP metadata to the system.

Download System Metadata

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to **All Settings** > **Analytics** > **SSO Configuration for SAP Analytics Cloud**.
4. Click [Download Metadata](#). A zip file containing two (BTP_account_sp_metadata.xml and SAC_sp_metadata.xml) XML files is downloaded to your local drive.

Configure Metadata in Your Identity Provider (IdP)

Note

Use this procedure to configure the IdP provided by SAP Identity Authentication Service (IAS). You can follow similar steps to configure external IdPs as well.

1. Log on to [SAP Identity Authentication Service](#) as an administrator. You can also use an external identity provider.
2. On the IAS screen, navigate to ► [Applications & Resources](#) ► [Applications](#) ►, and click [Add](#) to create a new application for the SAC service provider.
3. In the [Add Application](#) pop-up window, enter a name and click [Save](#). A new application page opens.
4. Under ► [Trust](#) ► [Single Sign-On](#) ►, configure the following settings:
 - Click [Type](#), and select SAML 2.0
 - Click [SAML 2.0 Configuration](#), and upload the SAC metadata XML from your local drive.
 - Click [Subject Name Identifier](#), and configure the attribute, which the application uses to identify the users.
Ensure that you use the same attribute that is maintained in your system. For example, if your system-specific application in the IdP is configured as **Login Name**, then use the same value in SAC-specific application as well.
 - Click ► [Assertion Attributes](#) ► [+ Add](#) ►, maintain the following user attributes and assertion attributes, and then click [Save](#).

User Attribute	Assertion Attribute
Language	preferredLanguage
Display Name	displayName
Last Name	familyName
First Name	givenName
E-mail	email

5. In the [Add Application](#) pop-up window, enter a name and click [Save](#). A new application page opens.
6. Under ► [Trust](#) ► [Single Sign-On](#) ►, configure the following settings:
 - Click [Type](#), and select SAML 2.0
 - Click [SAML 2.0 Configuration](#), and upload the BTP metadata XML from your local drive.
 - Click [Subject Name Identifier](#), and configure the attribute, which the application uses to identify the users.
Ensure that you use the same attribute that is maintained in your system. For example, if your system-specific application in the IdP is configured as **Login Name**, then use the same value in BTP-specific application as well.
7. Navigate to [Tenant Settings](#), and click [SAML 2.0 Configuration](#) to open a new screen.
8. Click [Download Metadata File](#) to download the configured IdP metadata file for both SAC and BTP configured metadata XML files.

Link IdP Metadata to the System

1. Log in to the system as an administrator.

2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to [All Settings](#) > [Analytics](#) > [SSO Configuration for SAP Analytics Cloud](#).
4. Click **+**, and select the IdP that needs to be linked. The metadata and the subject name identifier are populated from the system IdP metadata.
5. Use the refresh icon  to view the status of IdP metadata linking. The status gets updated once both the IdP and users synchronization is successful.

i Note

Don't link any further IdP entries when the existing IdP entry is in progress.

You've now configured Single Sign-On to the SAC system, where your users can design stories seamlessly.

Related Information

[Configure Identity Provider \[page 67\]](#)

28.2 Design Stories

Create and design stories on the SAC user interface.

Prerequisites

Ensure **analyticUserStoriesService** business service and the corresponding **sap.crm.analyticUserStoriesService** app are assigned to your users to design stories on the interface.

Context

A story is a presentation-style document that uses charts, visualizations, text, images, and pictograms to describe data. Once you create or open a story, you can add and edit pages, sections, and elements.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.

3. Go to [All Settings](#) > [Analytics](#) > [Design Stories](#) .

Note

A popup appears once you choose [Design Stories](#). If you've pop-up blockers enabled on your web browser, allow the popup to successfully log in to SAC.

4. Click [New](#) to create a new story. The query designer loads in the SAP Analytics Cloud interface.
5. In the right pane, under [Available Data](#), select an analytical model and drag and drop it to add it to the query designer.
6. Click the table icon () to select fields from the data source that you want to include in the model.
7. Optional: Click the data source node to see the list of master data attributes.
8. Optional: Click [Preview Query](#) to view the preview of your story.
9. Click [Finish](#). In the popup window, enter a name for your new SAC model. The system opens the story designer where you can add charts, tables, and so on.
10. Click the save icon (). In the popup window, enter a name and description for your new story. The story is now available on the [Design Stories](#) page.
11. Navigate to the [Design Stories](#) page, select a story, click the [Assign/Unassign Roles](#) button, search for a role, select a role and choose [Save](#) to assign a story to users with business roles. The users assigned to these business roles can now view these stories in the [Analytical Stories](#) view in the left navigation.

Note

- [Assign/Unassign Roles](#) functionality is supported only for one story at a time, and multiple story selections at a time isn't supported.
- You can access these stories only through the system. Also, accessing the SAP Analytics Cloud tenant independently isn't possible.

12. You can edit or delete stories by selecting them and then choosing [Edit](#) and [Delete](#) buttons respectively on the [Design Stories](#) page.

Related Information

[Creating a New Story \(All the features discussed here may not be applicable for in-built Analytics\)](#)

[View Stories \[page 266\]](#)

[Getting started with Embedded analytics in SAP Service Cloud Version 2](#) 

28.3 View Stories

View analytical stories assigned to your user.

Prerequisites

To view the stories assigned to your user, ensure that you've **analyticBusinessUserStoriesService** business service and the corresponding application **sap.crm.analyticBusinessUserStoriesService** app assigned to your business role.

View the Assigned Stories

You can view all the stories assigned to your user by clicking the hamburger icon in the left navigation, and then selecting Analytical Stories.

i Note

A popup appears once you choose [Analytical Stories](#). If you've pop-up blockers enabled on your web browser, allow the popup to successfully log in to SAC.

i Note

If you're unable to view any of the stories, verify role level assignments at a story level and also verify if that specific role is assigned to your user.

28.4 SAP Analytics Cloud FAQ

This section answers commonly asked questions about SAP Analytics Cloud.

1. **Issue:** The user account not active in the popup when accessing Design Stories or Analytical Stories.
Solution:
 1. Check if the subject name identifier set in IdP and the one set in the system are the same.
 2. If your subject name identifier is User ID, verify if the user's User ID in the system and the Login Name in IdP are the same (including the case).
 3. If your subject name identifier is Email, verify if the user's email in the system and the email maintained in IdP are the same (including the case).
 4. If the Email of the user is already used by some other user, then change the user's email in the existing user. Ensure that a unique email ID is updated for each of your users.
2. **Issue:** The user account is inactive for the new user created after IdP upload.

Solution: This scenario can occur if the email of the existing user was previously used or changed in any other user. For example: If a user1 is created with email test101@test.com, later is changed to test102@test.com, and now user2 is created with test101@test.com then for user2 the status displays as user inactive. Update Assertion Attributes for the SAC tenant in the IdP system and update the email ID of user1 in IdP to test102@test.com and now the user1 must log in to the system and access any of the existing stories. After a successful logon by the user1, create a user2 with test101@test.com and activate user2.

29 Microsoft Teams Integration

An embedded in-app integration of the solution with Microsoft Teams.

With this integration, you can share workspaces with Microsoft Teams. The real-time and automated data updates ensure transparency, consistency, and collaboration of the cases for the service teams.

→ Remember

The solution is best displayed at a screen resolution of 1920 x 1024 and 100% scale.

29.1 Configure Microsoft Teams Integration

Your administrators can enable Microsoft Teams integration.

Prerequisites

Ensure that you've the Microsoft Office 365 exchange online license.

Procedure

1. Add *SAP Service Cloud Version 2* app in the Teams Apps.
 - a. Log in to Microsoft Teams admin center.
 - b. From the home page, expand the navigation menu and go to ► *Teams apps* ► *Manage apps* ►.
 - c. Search and select the *SAP Service Cloud Version 2* app from the list.
 - d. Click *Allow*. A new window opens.
 - e. Click *Allow* again.
2. Grant consent for Microsoft Teams integration
 - a. Go to your user profile, navigate to ► *Settings* ► *All Settings* ►.
 - b. Under *Communication & Information Exchange*, select *Microsoft Teams Integration*.
A new window opens.
 - c. Ensure that the following options are enabled:
 - *Share Workspace*
 - Optional: *Create Teams*
 - d. Click *Start Consent* to open the sign in page.

- e. Sign in with your Microsoft 365 username and password.

A new window opens for consent.

- f. Click [Accept](#).

29.2 Use Microsoft Teams Integration

Learn how you can use the features using MS Teams integration.

i Note

Ensure that your administrator enables Microsoft Teams integration by following the steps in [Configure Microsoft Teams Integration \[page 268\]](#) to use the features.

[Create Appointments \[page 269\]](#)

Create an appointment with Microsoft Teams collaboration.

[Share Workspaces \[page 270\]](#)

Share any list or detail view of your supported entities on Microsoft Teams.

[Make Outbound Calls \[page 272\]](#)

Use Microsoft Teams to make your outbound calls.

29.2.1 Create Appointments

Create an appointment with Microsoft Teams collaboration.

Procedure

1. Navigate to [Appointments](#).
2. Click the plus (+) icon to create a new appointment.
3. Update the necessary fields such as [Subject](#), [Start Date/Time](#), and so on.
4. Check whether the Microsoft Teams collaboration is enabled or not. If not, turn on the [Teams Meeting](#) toggle button.
5. Go to [Attendees](#) and click the add (+) icon to search and select the required attendees.

i Note

You must add at least one attendee to create a Microsoft Teams meeting.

6. Click [Save](#).

i Note

You can't remove Microsoft Teams collaboration from the appointment once it is saved.

29.2.2 Share Workspaces

Share any list or detail view of your supported entities on Microsoft Teams.

The supported objects are [Cases](#), [Leads](#), and [Guided Selling](#).

i Note

For [Leads](#), we support only detail view from the solution. From Microsoft Teams, we support both list view and detail view.

29.2.2.1 Share Workspace from Solution

You can share any list or detail view of your supported entities on Microsoft Teams from your solution.

Procedure

1. Navigate to any of the supported entities that you want to share on Microsoft Teams.
2. Click the  (Share Workspace) icon.
3. If the workspace is already shared, the system shows a window with a list of existing teams and channels where you can join directly if you have not already joined, or you can click [Create a Team](#) and proceed.
4. If the workspace is not shared, a new window opens and click [Next](#).
5. In the [Team](#) tab, you can either select an existing team from the dropdown or select [Create a team](#) and enter a name. The system generates the [Tab Name](#) depending on the object that you select. You can edit it if required.
6. Click [Next](#).
7. In the [Channel tab](#), you can either select an existing channel from the dropdown or select [Create a channel](#) and edit the name if required.

You can't use the existing channels if you have created a new team.

8. Click [Next](#) to move to the next tab.
9. To add members to the team, enter their email ID and press [Enter](#).

i Note

If you are adding guest users, it may take up to 24 hours to reflect on Microsoft Teams.

10. Click [Next](#) to review the updated information.
11. Once you finish, click [Save](#).

The system generates a link.

12. Click the link to open the existing or new team on Microsoft Teams.

29.2.2.2 Share Workspace from Microsoft Teams

You can share any list or detail view of your supported entities directly from Microsoft Teams.

Procedure

1. Sign in to Microsoft Teams.
2. Click [Apps](#) and select SAP Service Cloud Version 2 application.
3. Choose one of the following options:
 - Add to a team
 - Add to a chat
4. Search and select a team or chat from the dropdown based on the previous selection.
5. Click [Set up a tab](#).
6. If required, you can edit the default [Tab Name](#) from [SAP Service Cloud](#).
7. Enter the [Tenant URL](#) based on the following table:

Type	URL Format	Example	Re- marks
List View	<Tenant URL>go/list/<Entity>	https://*crm.cloud.sap/go/list/case	You can replace the entity with case, lead, or guided selling in the URL which is case sensitive.
Detail View	<Tenant URL>go/detail/<Entity>?id=<uuid>	https://*crm.cloud.sap/go/detail/case?id=b977f162-ae68-11ec-9733-bd67a002f51f	

8. Click [Save](#) to move to the sign in page.
9. Click [Sign in](#) to view the entity from your solution.

! Restriction

If you try to include users outside the team to a private channel of Microsoft Teams, they may not be added.

29.2.3 Make Outbound Calls

Use Microsoft Teams to make your outbound calls.

i Note

Ensure that you've the required Microsoft Teams license for outbound calls.

To use this feature, you must disable the following options:

- Computer Telephony Integration (CTI) if it's configured in your solution.
- Pop-up blocker in your browser settings.

29.3 Microsoft Teams Integration FAQ

This section answers commonly asked questions about Microsoft Teams integration in your solution.

29.3.1 What licenses are required for Teams integration?

You should have a Microsoft Office 365 Exchange Online license for Teams integration.

29.3.2 Does the Service tab work in Teams mobile app?

No, it doesn't work.

30 Surveys

Surveys are an integral feature in retail business, service scenarios, and lead nurturing phases.

You can design surveys that sales representatives will use to capture business related data. The information captured in the surveys offers you an insightful direction for your next set of business actions. The survey offering includes comprehensive survey design, execution and reporting capabilities.

Surveys can be consumed only if they are assigned to a business object such as visit, service ticket, lead, opportunity and so on.

General surveys, also known as checklist surveys are supported. This survey type contains a collection of questions in a list format.

Survey Question Types

Use [Question Types](#) to specify the format of the answer options.

The question types offered are as follows:

- **Numeric:** Capture numeric responses pertaining to:
 - [Amount](#): Collect information in the currency that you specify. Validations are performed against the lower or upper limits that you have set.
 - [Quantity](#): Collect information in the unit of measure that you specify.
 - [Numeric Value](#): Collect numeric responses that do not specify **Unit of Measurement** or **Currency** as a mandatory field.

For information on deriving calculations, see [Derive Calculations for Numeric Questions \[page 277\]](#).

- **Date:** Choose required dates from the calendar as [Date](#) and [Date Range](#).
- [Matrix](#): Gather more details at once using a matrix of rows and columns.
- **Multiple Choice:** Choose one or more predefined responses from:
 - [Multiple Choice Single Select](#): Select one option from predefined response options using radio buttons, toggle buttons or dropdown list.
 - [Multiple Choice Multiple Select](#): Respondents can select multiple options from predefined options using checkboxes or dropdown lists.
- [Text](#): Enter any additional information you want to give your survey respondents.
- [Signature](#): Ask for signature upon completion of a survey.

Structure your content using [Sections](#) and [Page Breaks](#).

Survey Status

Once you start creating a survey, you'll need to pay attention to its status. The different statuses for surveys are as follows:

- **In Preparation** (initial status): In this status you can add, change or delete questions and answers. Surveys with status **In Preparation** cannot be assigned to an activity plan, visit, playbook, lead, or opportunity. From **In Preparation** you can switch to **Active**.
- **Active**: Indicates that the survey is ready to be used. You can use the survey for plan, visit, playbook, lead, or opportunity. From **Active** you can switch to **In Preparation** and **Obsolete**.
- **Obsolete**: The survey can no longer be assigned to an activity plan, visit, playbook, lead, or opportunity. If the survey is already used for open, or in process store visits, the survey can still be completed during the visit.
- **Inactive**: The survey can no longer be edited or reused on changing the status. You cannot change the status from **Inactive** to **Obsolete**: If a new version of a survey is created and activated, the previous one becomes inactive.

30.1 Configure Surveys

As an administrator, you can configure surveys in [Settings](#).

You must create business roles and assign business users to services for using surveys.

- Add the business service [sap.crm.service.surveyDesignService](#) to design surveys.
- Add the business service [sap.crm.service.surveyService](#) for answering surveys.

Related Information

[Create Business Roles and Assign a Business User \[page 63\]](#)

30.1.1 Configure Survey Categories

As an administrator, you can configure survey categories for a specific survey type. This allows the user to create and analyze surveys for various categories.

By default, **Checklist** and **Call Script** categories are provided.

To define more categories, follow the steps below:

1. Go to your user profile, choose [Settings](#) > [Survey](#) > [Survey Categories](#) .
2. Choose the **Add** icon (+).

3. Enter the *Category Code*, *Category Name*, and *Category Typecode*.
4. Choose *Create*.

30.1.2 Configure Question and Answer Categories

When creating survey questions and answers, you can assign a category to help classify them in a larger business context. You can then create reports based on specific categories.

Context

As an administrator, you can configure categories for questions and answers.

Procedure

1. Go to your user profile, choose **Settings** > **Survey** > **Question/ Answer Categories**.
2. Choose the *Add* icon (+).
3. Enter the *Question/ Answer Category* and *Category Name*.
4. Choose *Create*.

30.2 Create Surveys

As a survey designer, you can create, edit and modify surveys to help businesses run better. You can use a variety of advanced features such as branching, scoring, and translating.

Context

Create a survey using various question types based on business needs and share it with your end users to get clear insights.

For an overview, watch the following video:

Procedure

1. After logging into the system, select [Surveys](#) from the navigation menu.
2. Select the [Create Survey](#) icon (+).
3. Enter the mandatory details for your survey.
4. Choose [Add Questions](#).
5. Add questions from the [Builder](#) or from the [Question Bank](#).
6. Add all the required details for your questions depending on the question type.
7. Optional: Add exceptions or follow-ups.
8. Optional: Preview the survey to understand the final look of the survey.
9. Choose [Save and Activate](#). Choose [Save](#) to save a draft version, so you can edit later.

30.2.1 Edit Surveys

You can edit a survey that is in the in-preparation status without any survey version changes. When you click a question that you want to edit, an in-line edit option is provided.

To edit an inactive or obsolete survey, you must change the status to in preparation first.

You can create new versions of an active survey that is already being consumed in an application. Implicit versioning is enabled, which means the system decides if a new version is required.

i Note

- You can have multiple versions of a survey to meet your changing business needs.
- Only one version of any survey can be in the active state and in preparation state at a given time.

30.2.2 Add Questions to Question Bank

Store survey questions in the question bank for easy reuse in multiple surveys.

Context

The [Question Bank](#) stores frequently used questions that you can add to multiple surveys. You can add questions to the question bank when you create questions for a survey or create questions in the bank directly for later use.

Procedure

1. Log in to the system.
2. From the navigation menu, choose [Question Bank](#).
3. Choose the [Create](#) icon (+).
4. Add the required question type from the [Question Types](#) pane.
5. Enter the required details.
6. Choose [Save](#) if you want to make further changes, or choose [Save & Activate](#) to use the question in the survey.

You can also add questions to the question bank while creating them in your surveys. Use the [Add From Question Bank](#) icon () beside the question that you want to store.

To use this question in your survey, the question must be active, and not in-preparation or obsolete.

30.2.3 Derive Calculations for Numeric Questions

Quick calculated response values are available based on responses, thus helping you in deriving easy inferences.

Context

In a calculation, you can use various operands, groups, reference questions, choose a response from an existing question, or enter a constant numeric value as the operand. Calculations can be derived only for numeric question types that includes amount, quantity, and numeric value.

Procedure

1. Log in to the system and choose [Surveys](#) from the navigation menu.
2. Choose the [Create](#) icon (+).
3. From the question pane, select an amount, quantity, or numeric value question type.
4. Select the [Derive By Calculation](#) checkbox.

i Note

You can see the [Calculations](#) tab instead of the [Logic](#) tab.

5. In the [Calculations](#) tab, choose [Add](#).
6. Enter the calculation logic.
7. Choose [Save](#).

You cannot edit calculated responses during runtime.

If you select the *Derive by Calculation* option, the *Set default answer*, *Upper Limit*, and *Lower Limit* fields are disabled.

30.2.4 Add Scores to Questions

Use scores in your surveys to help in better analysis of results and get clearer insights on responses.

Context

Add scores to questions and responses based on question types.

Procedure

- 1. When you add a question to a survey, go to the *Options* tab and select *Show More*.
- 2. Enter the required value in the *Question Weight* field.
- 3. Based on the question type, choose one of the following:

Procedure for different question types

Question Types	Procedure
Multiple Choice	Enter a score for each response option.
Numeric	In the <i>Score Ranges</i> tab, enter the minimum and maximum values to create answer ranges that correspond to a score. These values must align with the Upper Limit and Lower Limit values you have set for the response field in the <i>Options</i> tab.

i Note

The range includes the minimum and maximum values you set, so you cannot set the same value for the maximum of one range and the minimum of the next range.

Question Types	Procedure
Matrix	Enter scores for each column in the Columns tab. Use the dropdown for Score order to score the columns in ascending or descending order, or set the same score for all columns. In the Rows tab, choose Maintain Score to select a score order. For unique values for rows, enter scores against each entry. Choose Clear Score to clear all the scores in the matrix.
Date and Text	You can assign a question weight. You cannot assign scores for answers. Based on whether the question is responded to, the score you assign is considered for calculation.

- Click the question to save the scores you assigned.

After you create a survey, you can preview it in the simulation mode. In this mode, you can answer questions, but cannot submit the survey.

To preview scores, select the  ([View Score](#)) icon.

If scoring is enabled, you can see the following scores as you answer the survey:

- [Question Weight](#): Score assigned by the administrator while designing the survey. For multiple choice multiple select questions, based on the responses, the score is calculated as a sum of all the response scores.
- [Response Score](#): For multiple choice multiple select, score is calculated based on the response provided and the score (or score ranges) assigned by the administrator while designing the survey.
- [Final Score](#): Score calculated in real-time as a product of response score and question score based on the responses.

i Note

- You can preview scores in the full view mode only.
- [Survey Score](#) is calculated as a sum of all the question weights, and is displayed against the survey name in the business process which has consumed the survey.

30.2.5 Survey Score Calculations

Calculate your minimum and maximum scores for multiple choice questions, matrix questions and surveys overall.

In a multiple choice survey, the values you set during design time determine the [Minimum Score](#), [Maximum Score](#), and [Final Score](#) for each question.

i Note

The scores are not shown when the employee fills out the survey.

Question Type	Minimum Score	Maximum Score	Final Score
Multiple Choice Single Select	The score is calculated by multiplying the lowest response score by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Question 1 Minimum Score: 1*1=1	The score is calculated by multiplying the highest response score by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Question 1 Maximum Score: 3*1=3	This score is calculated by multiplying the response chosen by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Response Chosen = A Question 1 Final score: 1*1=3
Multiple Choice Multiple Select	The score is calculated by summing up the lowest reachable response scores and multiplying it by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Minimum Score: 1*1=1	The score is calculated by summing up the highest reachable response scores and multiplying it by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Maximum Score: (2+3+1)*1=6	This score is calculated by summing up the chosen response scores and multiplying it by the question weight. ♣ Example Question 1 Question Weight: 1 - Response A: 1 - Response B: 2 - Response C: 3 Responses Chosen = A and B Question 1 Final score: (1+2)*1=3

Matrix (only column score enabled)

i Note

Mandatory conditions are not set.

For checkbox and radio button responses, the score is calculated by multiplying the least score in the columns by the question weight.

❖ Example

Question 1

Question Weight: 2

- Column 1: 5
- Column 2: 3
- Column 3: 2

Minimum Score: $2 \times 2 = 4$

For radio button response, the score is calculated by multiplying the highest score in the columns by the question weight.

❖ Example

Question 1

Question Weight: 2

- Column 1: 5
- Column 2: 3
- Column 3: 2

Maximum Score: $5 \times 2 = 10$

For checkbox response, the score is calculated as a sum of all column scores multiplied by the question weight that is, Maximum Score = $[(5 \times 1) + (3 \times 1) + (2 \times 1)] \times 2 = 20$

This score is calculated by multiplying the column score of response chosen by the question weight.

❖ Example

Question 1

Question Weight: 2

- Column 1: 5
- Column 2: 3
- Column 3: 2

The response chosen is from column 2

Question 1 Final score: $3 \times 2 = 6$

Matrix (both row and column score enabled)

Note

Mandatory conditions are not set.

For checklist and radio button responses, the score is calculated by summing up the lowest reachable response scores and multiplying it by the question weight.

❖ Example

Question 1

Question Weight: 2

- Row 1: 1
- Row 2: 2
- Row 3: 3
- Column 1: 1
- Column 2: 2
- Column 3: 3

Minimum Score: $(1*1)*2=2$

For radio button responses, the score is calculated by summing up the highest reachable response scores and multiplying it by the question weights.

❖ Example

Question 1

Question Weight: 2

- Row 1: 1
- Row 2: 2
- Row 3: 3
- Column 1: 1
- Column 2: 2
- Column 3: 3

Maximum Score:

$$[(1*3)+(2*3)+(3*3)]*2=36$$

For checkbox response, the score is calculated as a sum of all element scores multiplied by the question weight that is, Maximum

$$\text{Score} = \{[(1*1)+(1*2)+(1*3)] + [(2*1)+(2*2)+(2*3)] + [(3*1)+(3*2)+(3*3)]\}*2=72$$

This score is calculated by multiplying the response's column score, row score and question weight.

❖ Example

Question 1

Question Weight: 2

- Row 1: 1
- Row 2: 2
- Row 3: 3
- Column 1: 1
- Column 2: 2
- Column 3: 3

For radio buttons, if the responses chosen are:

- The second entry in first row.
- The first entry in second row.
- The third entry in third row.

Final Score:

$$[(2*1)+(2*1)+(3*3)]*2=26$$

Similarly, for checklists, this score is calculated by multiplying all the response's column score, row score and question weight.

Overall Survey Score

This total is shown in the *Details* view of the survey as *Score Min/Max*.

The overall survey minimum score is the sum of all minimum scores in the survey.

The overall survey maximum score is the sum of all maximum scores in the survey.

The overall survey score is the sum of all final scores in a survey.

30.2.6 Add Logic

You can apply different types of logic based on your question type.

Following are the logic types:

- Conditional logic
- Branching logic
- Hide logic

30.2.6.1 Add Conditional Logic to a Survey Question

Conditional logic allows you to select the responses that you see in subsequent questions based on the response to the source question.

Context

This logic is supported in multiple choice question types only. To enable conditional logic for a question, follow the steps below:

Procedure

1. In the survey question, go to the [Logic](#) tab and choose [Add](#).
2. In the condition, define whether the question should be [answered](#) or [unanswered](#), from the dropdown list.
3. Select the response type and the value. There can be a specific response, or any response.
4. For defining the logic, select [show](#) from the dropdown list.
5. Select the question and responses that have to be displayed based on whether the condition is met or not. You can set multiple questions and responses.
6. Save your entries.

30.2.6.2 Add Branching Logic to a Survey Question

Based on how respondents answer the current question, they can skip questions that are no longer applicable, and jump to a later question in the survey sequence. Define survey branching when you design the question.

Context

This feature allows you to branch respondents to different questions in the survey sequence based on their responses.

→ Tip

It's recommended that you define all the questions in the survey before you set the branching logic.

Procedure

1. In the question for which you want to define the branching logic, go to the [Logic](#) tab and choose [Add](#).
2. In the condition, define whether the question must be [answered](#) or [unanswered](#) from the dropdown list. Also select the required options for the response, and operator.

i Note

Branching only allows forward progress within a survey. You can only select a future question as the branching target.

If branching isn't defined, respondents are taken to the next question in your survey sequence.

3. If you choose [unanswered](#) as the response type, define what the next question is when no answer is submitted.

If the field isn't defined and no answer is submitted for a question with branching, respondents are taken to the next question in your survey sequence.

4. When you finish, save your entries. You can create multiple logic groups. In case of contradicting logic groups, the first one is executed.

Next Steps

Preview your survey to see if the questions are logically displayed as expected.

30.2.6.3 Add Hide Logic to a Survey Question

Based on the response to a question, the system can hide the next set of questions.

Context

During runtime of a survey, you can hide multiple choice, numeric, date, text, and numeric questions.

Procedure

1. In the survey question, go to the *Logic* tab and choose *Add*.
2. In the condition, define whether the question must be *answered* or *unanswered*, from the dropdown list.
3. Select the response type and the value.
4. For defining the logic, select *hide* from the dropdown list.
5. Select the question that has to be hidden when the condition you defined is met.

Note

Multiple questions can be hidden in a single condition.

6. Save your entries. You can create multiple logic groups. In case of contradicting logic groups, the first one is executed.

30.2.7 Add Exceptions

You can use condition based validations to inform sales reps of inappropriate survey responses, or issues found during audits.

Context

Exceptions can help identify potential errors such as entering a price too low or too high for an item, which allows users to correct their entries before submitting the survey and hence create more accurate data.

Exceptions can also alert sales users of specific problems detected at survey runtime. For example, an exception can be used to remind a sales rep of talking to a store manager when the quantity is less than the minimum value.

Procedure

1. Add all the survey questions.
2. Go to the [Exceptions](#) tab.
3. Click [Add Exception](#).
4. Enter the exception message.

The users see this message when the exception condition is met during survey runtime.

5. From the side pane, choose the questions to which you want to apply the exception.
6. Define the exception condition. Set the operators and values for your exception.
 - For multiple choice question types, select the answers that must be selected for triggering the exception.
 - For question types such as amount, date (excluding date range), numeric value, and quantity, set the operator and the value for your exception.

Note

You can create multiple validations for a rule. You can choose if any or all of the validations need to be applied during survey execution.

30.2.8 Add Translations to Surveys

Create a survey and add descriptions, questions, and responses in additional languages. In the [Translations](#) tab of the survey, you can see an overview of surveys in multiple languages, and modify the text based on your requirement.

Procedure

1. Select the survey that you want to translate and go to the [Translations](#) tab.
2. Below the globe's image, make a selection from the [Language](#) dropdown.
3. Choose [Translate](#). You can now see a section for the language you selected beside the survey in the primary language. Primary language is the language that you first create the survey in.
4. In the section for the language you chose, select the question that you want to translate.
5. Optional: Use the [Preview](#) button to simulate the survey and individual questions in the translated language.
6. After translating the questions into various languages, choose [Save & Exit](#). Similarly, you can translate more questions into multiple languages.

Caution

If you delete a language, all corresponding text fields within your survey will be deleted. Any existing text that you have entered for questions, descriptions, and answers in the language you deleted will also be deleted.

- Survey fields fallback to the primary language if the translation is not provided.
- If a question is not translated in a particular language, it will be shown in the primary language.

30.3 Answer Surveys

You can answer surveys and help collect important information for deriving insights.

Procedure

1. In the business process that you're working in, go to the [Surveys](#) section.
2. Choose the survey that you want to answer.
3. Enter all your responses to the questions.

i Note

If the administrator has enabled the [Derive by Calculation](#) option, the calculated fields are shown when the responses are provided.

4. Choose [Submit](#). You can also save the survey as a draft.

i Note

The system validates surveys only when you submit them.

5. The system validates the survey and displays the errors and exceptions that exist.
6. You can either rectify exceptions or proceed to submit as is.

To get an overview of the survey and see the related entities, use the expand button beside the survey. You can also add notes here.

30.4 Survey Reporting and Analysis

Create reports using the data collected from your surveys to better understand your current performance and make improvements.

As an administrator, to view reporting details of your survey, proceed as follows:

1. Log in to the system.
2. Choose [Surveys](#) from the navigation menu.
3. Select the survey that you want to analyse.

4. Go to the [Results](#) tab. Click the [Full View](#) icon () to get a complete insight to the survey. You can also view sentiment analysis here.

31 Library

Use [Library](#) to store, organize, and share files and folders.

[Library](#) offers you the following:

- Efficient management of files, such as documents and multimedia objects, for your daily business activities
- Searching, viewing, uploading, and downloading files, creating folder structures to organize them, and controlling access to them
- Provision of attaching files to sales processes such as sales quotes
- Quick and safe way to share and exchange files for collaboration among sales team members
- Public and private folders. You can set access control for private files and folders.
- Table view and tile view of all the files and folders in the right pane
- Collapsible tree view that shows folder structures in the left pane

31.1 Configure Library

Administrators can create file type filter categories and manage access restrictions for [Library](#).

31.1.1 Create File Type Filter Category

Administrators must create file type categories for users to filter files.

Procedure

1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Library](#) ► [File Type Filter Configuration](#) ►.
2. Click + ([Create](#)).

i Note

The standard SAP system delivers file type filter categories without file types added to them. You can modify them with the required file types or delete them.

To add file types to a standard filter category, allow the file types at ► [Settings](#) ► [Data Administration](#) ► [MIME Types for Attachments](#) ► if they are not allowed. For more information, see [Define MIME Types for Attachments](#) [page 255].

3. In the [Create File Type Filter Category](#) pane, enter *Title*.
4. Click the [Search and Add](#) icon to add the supported file types.
5. Click [Save](#).
 - The new file type filter category gets added in [File Type Filter Categories](#).
 - To delete a file type configuration, click  ([Delete](#)) in [File Type Filter Categories](#).
 - You can also modify file types added to an existing file type filter category.
 - Switch off file type filter categories if you do not want them to appear in the [File Type](#) filter in [Library](#).

31.1.2 Manage Access Restrictions

Administrators can provide users with restricted access to the library.

Context

With this access context, read and write access to folders and files can be restricted based on sales area and employee.

Sub folders inherit access restrictions of their parent folder. You can also set sub folders' access to [Private](#) and set restrictions that apply to only sub folders.

This access restriction also applies to attachments added from the library. Files added to the library and referenced in a business object, such as account, are visible to all under the [Attachments](#) section. However, unauthorized users cannot navigate to further details. Restricted documents are not listed in [Attachments](#). Attachments from outside of the library are not subject to access restriction.

Procedure

- Create a business role and add the business service [sap.crm.service.libraryService](#) and enable the apps [sap.crm.libraryservice.uiapp.libraryAdminApp](#) and [sap.crm.libraryservice.uiapp.libraryApp](#).

Related Information

[Create Business Roles and Assign a Business User \[page 63\]](#)

31.2 Organize Files and Folders

You can organize files and folders based on your organizational and business needs. This helps you find the right files quickly for your business processes.

Library offers you various operations to keep your files and folders organized.

i Note

You can perform these operations only if you are authorized.

31.2.1 Create Folders

You can create folders and folder structures to collect, group, and organize files and folders.

Context

Organizing files in folders helps you easily and quickly find and share them.

A navigation path that appears above the contents of a folder allows you to keep track of your current location and easily navigate back to an upper level folder in the hierarchy.

i Note

- You can build only five levels of folders.
- If you add a file within a folder, the file inherits the permissions of the parent folder.

Procedure

1. Click **+** ([Create](#)).
2. Select [Create Folder](#).
3. Enter [Name](#).

i Note

To rename the folder after creating it, click **⋮** ([More](#)) and then [✎](#) ([Rename](#)).

4. Do one of the following:
 - To provide access to all users, go to [step 5 \[page 292\]](#).
 - To provide access to the restricted sales area or employee or both, do the following:

1. Switch off [Public](#).
2. Do the following as required:
 - To restrict access to the required sales area, click [Sales Area](#) and **+** ([Add](#)) the required [Sales Org](#), [Distribution Channel](#), and [Division](#).
 - To restrict access to the required employees, click [Employee](#) and the [Search and Add](#) icon. To provide administration rights to the employee, switch on [Owner](#).
5. Click [Save](#).

Results

The folder is created, and it appears in [Library](#).

31.2.2 Upload Files

You can upload files from your local drive to [Library](#) and organize them in folders. You can also select multiple files for mass upload.

Procedure

1. Click **+** ([Create](#)) at the required folder level.
2. Click [Add Files](#).
3. In [Upload Files](#), do one of the following:
 - Drag and drop the files
 - Browse the files and click [Open](#).

Note

- The number of files in mass upload is restricted to six.
- You cannot upload a document if another document has the same name at the same level. We recommend that you upload the document with a different name and edit the title later.
- The size limit for files is as configured by your administrator. Contact your administrator to change the limit.

4. Do one of the following:
 - To provide access to all users, go to [step 5 \[page 293\]](#).
 - To provide access to the restricted sales area or employee or both, do the following:
 1. Switch off [Public](#).
 2. Do the following as required:
 - To restrict access to the required sales area, click [Sales Area](#) and **+** ([Add](#)) the required [Sales Org](#), [Distribution Channel](#), and [Division](#).

- To restrict access to the required employees, click [Employee](#) and the [Search and Add](#) icon. To provide administration rights to the employee, switch on [Owner](#).

5. Click [Save](#)

Results

The files are uploaded, and they appear in [Library](#).

i Note

To rename the uploaded file, click  ([More](#)) and then  ([Rename](#)).

Related Information

[Attachment Configuration](#) [page 260]

31.2.3 Delete Files and Folders

You can delete files and folders that are no longer valid or those you uploaded by mistake.

Procedure

1. Click  ([More](#)) on the folder or file.
2. Click  ([Delete](#)).

i Note

You cannot delete folders if they have contents.

Results

The files are deleted from [Library](#).

31.2.4 Download Files

You can download files on to your local drive from [Library](#).

Procedure

1. Select the required files and click *** ([More](#)).

i Note

You can download multiple files simultaneously. The files are first consolidated into a single zip folder and then downloaded to your local drive.

2. Click  ([Download](#)).

Results

The files are downloaded to your local drive.

31.2.5 Move Files

When you organize your files, you can move them to folders at any level.

Procedure

1. In the table view or the tile view, select the required files and click the [Move](#) icon on the right.

i Note

- If you are moving a single file, you can also drag and drop the file to the required folder.
- You cannot drag and drop multiple files. You can drag and drop only one file at a time.
- You cannot drag and drop files in the table view. You can drag and drop files only in the tile view.
- You can drag and drop files from tile view to tree view.
- You cannot drag and drop files within the tree view.
- You can move only files into a folder. You cannot move folders into a folder.
- You can only select multiple files that are on the same folder level.
- If you move a file into a folder, the access restrictions on the file change and it inherits the access restrictions of the new parent folder.

2. In the [Move Files](#) dialog, select the required folder.
3. Click [Move](#).

Results

The file is moved to the required folder.

31.2.6 Search and Filter Files and Folders

You can look for specific files and folders using the search option. Filters help you narrow down the displayed files and folders.

Procedure

1. Click  ([Search](#)).
2. Enter the keyword.

Files and folders with your keyword in their names are displayed.

Note

- The system searches both the parent folders and subfolders to display results.
- Search results are based on access control.

3. To filter the search results, select the required option from the standard SAP system views, [File Types](#), [Upload Date](#), [Uploaded By](#), or [Access](#).

- **! Restriction**

The standard SAP system views are only used to filter top-level documents and folders in the library. For example, if you switch the view from [Created by Me](#) to [Changed by Me](#) within a folder, you will be directed back to the top level in the library and a list of top-level documents and folders that have been changed by you are displayed.

- You can filter files and folders without searching for a particular keyword too, but the scope is restricted to the current folder.

31.3 Manage Access to Files and Folders

If you have unrestricted edit authorization or if you are an owner, you can define whether the file or folder is public or private. Access to a private file or folder can be restricted to specific sales areas and employees.

Context

You can authorize other employees to access the file or the folder in [Manage Access](#).

Users that have access to a parent folder can access all the subfolders and files within the folder. Users with such [Inherited Access](#) cannot be edited or deleted.

i Note

- You cannot change the administration right (the [Owner](#) switch in [Manage Access](#)) for an employee with inherited access. In a case like this, you must add the employee explicitly for editing.
- You can configure access to files and folders when you upload or create them respectively. For more information, see [Create Folders \[page 291\]](#) and [Upload Files \[page 292\]](#).

Inherited access can also be granted due to access control that has been configured by your administrator.

Procedure

1. Click **...** ([More](#)) on the required file or the folder.
2. Click [Manage Access](#).
3. Do one of the following:
 - To provide unrestricted access, switch on [Public](#)
 - To provide access to the restricted sales area or employee or both, do the following:
 1. Switch off [Public](#).
 2. Click  ([Edit](#)).
 3. Do the following as required:
 - To restrict access to the required sales area, click [Sales Area](#) and **+** ([Add](#)) the required [Sales Org](#), [Distribution Channel](#), and [Division](#).
 - To restrict access to the required employees, click [Employee](#) and the [Search and Add](#) icon. To provide administration rights to the employee, switch on [Owner](#).
4. Click [Save](#).

31.4 Share Links to Files and Folders

You can share links to files or folders.

Procedure

1. Click  ([More](#)) on the folder or the file.
2. Click  ([Link](#)).

Results

The URL is copied to the clipboard to be shared.

When you open a shared link to a private folder or document, you must have permission to access the folder or the file to see the content.

Note

- When you open a shared link to a folder, you can view the content within the shared folder displayed under your default view and the available actions on the right.
- When you open a shared link to a file, you can view the preview of the file along with the details of the file and available actions above it.

32 Integration

Configure settings for integrating SAP Service Cloud Version 2 with other SAP solutions.

Many SAP Service Cloud Version 2 business applications have requirements to integrate with other business applications. Therefore, these applications must be connected to run business processes end-to-end. To enable cloud native applications developed in SAP Service Cloud Version 2 to connect with other business applications (within SAP or outside of SAP), you must make technical settings and monitor data messages on a regular basis. To implement integration, see the integration set up guides.

Related Information

[SAP Service Cloud Version 2 Integration Guides for SAP S/4HANA](#) 

32.1 Configure User Roles for Integration

Create users with the roles and services that are necessary to perform integration relevant tasks and settings.

Procedure

1. Create the business role and add the following business services:

Business Service ID	Name
sap.crm.service.adminService	adminService
sap.crm.service.dataConnectorService	dataConnectorService
sap.crm.service.homepageService	homepageService
sap.crm.service.valueMappingService	valueMappingService

Note

You must switch on the [Admin](#) toggle switch.

2. Enable the following apps:

Business Service ID	App Name
sap.crm.service.adminService	sap.crm.adminservice.uiapp.admin
sap.crm.service.adminService	sap.crm.adminservice.uiapp.adminconsole
sap.crm.service.dataConnectorService	sap.crm.dataconnectorservice.uiapp.communicationSystemsApp
sap.crm.service.dataConnectorService	sap.crm.dataconnectorservice.uiapp.dataConnectorConfigApp
sap.crm.service.dataConnectorService	sap.crm.dataconnectorservice.uiapp.monitoringApp
sap.crm.service.homepageService	sap.crm.homepageservice.uiapp.homepage
sap.crm.service.valueMappingService	sap.crm.valuemappingservice.uiapp.valuemappingApp

Note

The status of the business role must be *Active*.

3. Create the employee.
4. Create the business user with the employee and the business role.

Related Information

[Create Employees \[page 39\]](#)

[Create Business Users and Assign a Business Role \[page 61\]](#)

[Create Business Roles and Assign a Business User \[page 63\]](#)

32.2 Monitor messages

Monitor all the incoming and outgoing messages and check their status with *Message Monitoring*.

Procedure

- Go to **User Menu** > **Settings** > **All Settings** > **Integration** > **Message Monitoring**.
 - Use the search option to look for specific messages.
 - Use the filter option to narrow down the displayed messages.
 - If the status of the message is *Error*, you can also click *Retry* to retry the message processing. The filter and search functionalities help you find the specific messages with errors.

32.3 Event Monitoring

Learn how to monitor the events you've subscribed for.

Navigate to ► [User Profile](#) ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Event Monitoring](#) ►.

You can see notifications for failed and successful events on this page. You can further filter the events records for the events that have occurred in the past hour, last 24 hours, last week, or last month. You can also filter events based on the status of events.

You can also resend event notifications from this page. Select one or more events on the page, and click [Resend](#). Later, refresh the list to see the latest notifications.

32.4 Communication Systems

A communication system represents an external system that is used for application integration. Communication systems can be, for example, an external time recording or master data system.

Context

On the [Communication Systems](#) page, you can filter the communication systems by status and authentication type. You can further create and edit communication systems to exchange data across systems electronically.

Procedure

1. Log in to the system as an administrator.
2. Navigate to user profile on the right-top corner, and access the [Settings](#) page.
3. Go to ► [All Settings](#) ► [Integration](#) ► [Communication Systems](#) ►.
4. Click the add icon (+) on the [Communication Systems](#) page, enter the display ID and description for the communication system.

i Note

Display ID must be the Business System ID of your external system.

5. Click [+ Create](#) button in the [Inbound](#) tab, enter a password or upload a certificate file for authentication, and click [Save and Activate](#) to create the inbound communication system.
6. Click [+ Create](#) button in the [Outbound](#) tab, choose a protocol, enter a host name, select an authentication method from the dropdown, and click [Save and Activate](#) to create the outbound communication system.

Results

You've now successfully created a communication system, which you can further use for application integration.

32.5 Configure Communication

Configure communication between SAP Service Cloud Version 2 and the application you are integrating it with.

Context

Communication configuration comprises all the required information for integrating SAP Service Cloud Version 2 with other applications. It contains the communication system, mapping of code values, and all inbound and outbound communications that must be executed for integration.

Procedure

1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Communication Configuration](#) ►.
2. Click the required communication configuration template.

You can also click **+** ([Create](#)) to create a communication configuration for your requirements.

3. Click  ([Copy](#)).
4. Go to the [Communication Configuration](#) page and click the copied communication configuration.
5. Edit [Communication System](#) and select the name of the communication system.
6. Edit [Value Mapping Group](#) and select the value mapping group.
7. In [Inbound Configuration](#), use the search icon to select the inbound configurations relevant to your scope.
8. In [Asynchronous Outbound Configuration](#), edit [Target Message Entity API Path](#) for the respective outbound communication scenarios.

i Note

The [Target Message Entity API Path](#) must match the respective iflow endpoints.

32.6 View Inbound Configuration

View preconfigured inbound communications.

Context

The inbound configurations are the standard SAP configurations for inbound messages and APIs of a business entity.

Inbound Configuration is a repository of inbound configurations.

Procedure

- Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Inbound Configuration](#) ►.

32.7 View Outbound Configuration

View preconfigured outbound communications.

Context

The outbound configurations are the standard SAP configurations for outbound messages and APIs of a business entity.

Outbound Configuration is a repository of outbound configurations.

Procedure

- Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Outbound Configuration](#) ►.

32.8 Create Value Mapping

Create value mapping to map code lists and their values between SAP Service Cloud Version 2 and the application you are integrating with.

Context

Value mapping matches code values of SAP Service Cloud Version 2 to the values of the external system to ensure smooth data transfer.

Procedure

1. Go to ► [User Menu](#) ► [Settings](#) ► [All Settings](#) ► [Integration](#) ► [Value Mapping](#) ►.

2. In [Mapping](#), click **+** ([Create](#)).

3. In [Create Mapping Definitions](#), do the following:

- a. From [Mapping Group](#), select a mapping group.

To add a new mapping group, click [Add New Mapping Group](#).

- b. Enter [Description](#).

- c. **Optional:** From [Parent Mapping Group](#), select the parent mapping group.

Parent mapping groups have code mappings that are common to most integration scenarios, and they are used if code mappings are not added to mapping groups.

If you must use deviations from the common code mappings, add them to the mapping group.

- d. From [Code List Name](#), select the data type name.

i Note

For the [Region](#) data type name, select the country from [Local Context](#).

- e. From [Rule for Missing Mapping](#), select the rule that SAP Service Cloud Version 2 must follow if code mappings are not added.

The choices are:

- [Pass-Through](#): SAP Service Cloud Version 2 considers the available internal or external value for integration and shows no error messages.
- [Raise Error](#): SAP Service Cloud Version 2 considers only explicit code mappings for integration and shows the error message.

4. Click [Save](#).

5. In the [Code Mapping](#) pane, enter the required values and click **✓** (Accept).

Important Disclaimers and Legal Information

Hyperlinks

Some links are classified by an icon and/or a mouseover text. These links provide additional information.

About the icons:

- Links with the icon  : You are entering a Web site that is not hosted by SAP. By using such links, you agree (unless expressly stated otherwise in your agreements with SAP) to this:
 - The content of the linked-to site is not SAP documentation. You may not infer any product claims against SAP based on this information.
 - SAP does not agree or disagree with the content on the linked-to site, nor does SAP warrant the availability and correctness. SAP shall not be liable for any damages caused by the use of such content unless damages have been caused by SAP's gross negligence or willful misconduct.
- Links with the icon  : You are leaving the documentation for that particular SAP product or service and are entering an SAP-hosted Web site. By using such links, you agree that (unless expressly stated otherwise in your agreements with SAP) you may not infer any product claims against SAP based on this information.

Videos Hosted on External Platforms

Some videos may point to third-party video hosting platforms. SAP cannot guarantee the future availability of videos stored on these platforms. Furthermore, any advertisements or other content hosted on these platforms (for example, suggested videos or by navigating to other videos hosted on the same site), are not within the control or responsibility of SAP.

Beta and Other Experimental Features

Experimental features are not part of the officially delivered scope that SAP guarantees for future releases. This means that experimental features may be changed by SAP at any time for any reason without notice. Experimental features are not for productive use. You may not demonstrate, test, examine, evaluate or otherwise use the experimental features in a live operating environment or with data that has not been sufficiently backed up.

The purpose of experimental features is to get feedback early on, allowing customers and partners to influence the future product accordingly. By providing your feedback (e.g. in the SAP Community), you accept that intellectual property rights of the contributions or derivative works shall remain the exclusive property of SAP.

Example Code

Any software coding and/or code snippets are examples. They are not for productive use. The example code is only intended to better explain and visualize the syntax and phrasing rules. SAP does not warrant the correctness and completeness of the example code. SAP shall not be liable for errors or damages caused by the use of example code unless damages have been caused by SAP's gross negligence or willful misconduct.

Bias-Free Language

SAP supports a culture of diversity and inclusion. Whenever possible, we use unbiased language in our documentation to refer to people of all cultures, ethnicities, genders, and abilities.

© 2023 SAP SE or an SAP affiliate company. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or for any purpose without the express permission of SAP SE or an SAP affiliate company. The information contained herein may be changed without prior notice.

Some software products marketed by SAP SE and its distributors contain proprietary software components of other software vendors. National product specifications may vary.

These materials are provided by SAP SE or an SAP affiliate company for informational purposes only, without representation or warranty of any kind, and SAP or its affiliated companies shall not be liable for errors or omissions with respect to the materials. The only warranties for SAP or SAP affiliate company products and services are those that are set forth in the express warranty statements accompanying such products and services, if any. Nothing herein should be construed as constituting an additional warranty.

SAP and other SAP products and services mentioned herein as well as their respective logos are trademarks or registered trademarks of SAP SE (or an SAP affiliate company) in Germany and other countries. All other product and service names mentioned are the trademarks of their respective companies.

Please see <https://www.sap.com/about/legal/trademark.html> for additional trademark information and notices.